

# Chapter 7

## EXTERNAL FORCED CONVECTION

### Drag Force and Heat Transfer in External Flow

**7-1C** The velocity of the fluid relative to the immersed solid body sufficiently far away from a body is called the *free-stream velocity*,  $\mathbf{V}_\infty$ . The *upstream* (or *approach*) *velocity*  $\mathbf{V}$  is the velocity of the approaching fluid far ahead of the body. These two velocities are equal if the flow is uniform and the body is small relative to the scale of the free-stream flow.

**7-2C** A body is said to be *streamlined* if a conscious effort is made to align its shape with the anticipated streamlines in the flow. Otherwise, a body tends to block the flow, and is said to be *blunt*. A tennis ball is a blunt body (unless the velocity is very low and we have “creeping flow”).

**7-3C** The force a flowing fluid exerts on a body in the flow direction is called *drag*. Drag is caused by friction between the fluid and the solid surface, and the pressure difference between the front and back of the body. We try to minimize drag in order to reduce fuel consumption in vehicles, improve safety and durability of structures subjected to high winds, and to reduce noise and vibration.

**7-4C** The force a flowing fluid exerts on a body in the normal direction to flow that tend to move the body in that direction is called *lift*. It is caused by the components of the pressure and wall shear forces in the normal direction to flow. The wall shear also contributes to lift (unless the body is very slim), but its contribution is usually small.

**7-5C** When the drag force  $F_D$ , the upstream velocity  $\mathbf{V}$ , and the fluid density  $\rho$  are measured during flow over a body, the drag coefficient can be determined from

$$C_D = \frac{F_D}{\frac{1}{2} \rho \mathbf{V}^2 A}$$

where  $A$  is ordinarily the *frontal area* (the area projected on a plane normal to the direction of flow) of the body.

**7-6C** The *frontal area* of a body is the area seen by a person when looking from upstream. The frontal area is appropriate to use in drag and lift calculations for blunt bodies such as cars, cylinders, and spheres.

**7-7C** The part of drag that is due directly to wall shear stress  $\tau_w$  is called the *skin friction drag*  $F_{D, \text{friction}}$  since it is caused by frictional effects, and the part that is due directly to pressure  $P$  and depends strongly on the shape of the body is called the *pressure drag*  $F_{D, \text{pressure}}$ . For slender bodies such as airfoils, the friction drag is usually more significant.

**7-8C** The friction drag coefficient is independent of surface roughness in *laminar flow*, but is a strong function of surface roughness in *turbulent flow* due to surface roughness elements protruding further into the highly viscous laminar sublayer.

**7-9C** As a result of streamlining, (a) friction drag increases, (b) pressure drag decreases, and (c) total drag decreases at high Reynolds numbers (the general case), but increases at very low Reynolds numbers since the friction drag dominates at low Reynolds numbers.

**7-10C** At sufficiently high velocities, the fluid stream detaches itself from the surface of the body. This is called *separation*. It is caused by a fluid flowing over a curved surface at a high velocity (or technically, by adverse pressure gradient). Separation increases the drag coefficient drastically.

### Flow over Flat Plates

**7-11C** The friction coefficient represents the resistance to fluid flow over a flat plate. It is proportional to the drag force acting on the plate. The drag coefficient for a flat surface is equivalent to the mean friction coefficient.

**7-12C** The friction and the heat transfer coefficients change with position in laminar flow over a flat plate.

**7-13C** The average friction and heat transfer coefficients in flow over a flat plate are determined by integrating the local friction and heat transfer coefficients over the entire plate, and then dividing them by the length of the plate.

**7-14** Hot engine oil flows over a flat plate. The total drag force and the rate of heat transfer per unit width of the plate are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible.

**Properties** The properties of engine oil at the film temperature of  $(T_s + T_\infty)/2 = (80 + 30)/2 = 55^\circ\text{C} = 328\text{ K}$  are (Table A-13)

$$\rho = 867\text{ kg/m}^3 \quad \nu = 123 \times 10^{-6}\text{ m}^2/\text{s}$$

$$k = 0.141\text{ W/m}\cdot^\circ\text{C} \quad Pr = 1505$$

**Analysis** Noting that  $L = 6\text{ m}$ , the Reynolds number at the end of the plate is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(3\text{ m/s})(6\text{ m})}{123 \times 10^{-6}\text{ m}^2/\text{s}} = 1.46 \times 10^5$$

which is less than the critical Reynolds number. Thus we have laminar flow over the entire plate. The average friction coefficient and the drag force per unit width are determined from

$$C_f = 1.328 Re_L^{-0.5} = 1.328(1.46 \times 10^5)^{-0.5} = 0.00347$$

$$F_D = C_f A_s \frac{\rho V_\infty^2}{2} = (0.00347)(6 \times 1\text{ m}^2) \frac{(867\text{ kg/m}^3)(3\text{ m/s})^2}{2} = 81.3\text{ N}$$

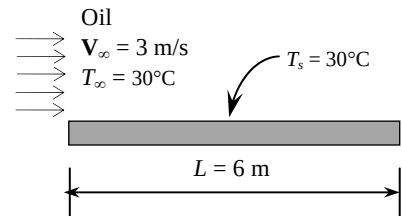
Similarly, the average Nusselt number and the heat transfer coefficient are determined using the laminar flow relations for a flat plate,

$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664(1.46 \times 10^5)^{0.5} (1505)^{1/3} = 2908$$

$$h = \frac{k}{L} Nu = \frac{0.141\text{ W/m}\cdot^\circ\text{C}}{6\text{ m}} (2908) = 68.3\text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat transfer is then determined from Newton's law of cooling to be

$$\dot{Q} = hA_s(T_\infty - T_s) = (68.3\text{ W/m}^2\cdot^\circ\text{C})(6 \times 1\text{ m}^2)(80 - 30)^\circ\text{C} = 2.05 \times 10^4\text{ W} = \mathbf{20.5\text{ kW}}$$

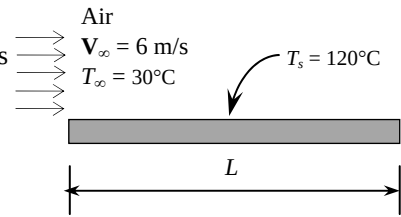


**7-15** The top surface of a hot block is to be cooled by forced air. The rate of heat transfer is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties.

**Properties** The atmospheric pressure in atm is

$$P = (83.4 \text{ kPa}) \frac{1 \text{ atm}}{101.325 \text{ kPa}} = 0.823 \text{ atm}$$



For an ideal gas, the thermal conductivity and the Prandtl number are independent of pressure, but the kinematic viscosity is inversely proportional to the pressure. With these considerations, the properties of air at 0.823 atm and at the film temperature of  $(120+30)/2=75^\circ\text{C}$  are (Table A-15)

$$k = 0.02917 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = \nu_{@1\text{atm}} / P_{\text{atm}} = (2.046 \times 10^{-5} \text{ m}^2/\text{s}) / 0.823 = 2.486 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7166$$

**Analysis** (a) If the air flows parallel to the 8 m side, the Reynolds number in this case becomes

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(6 \text{ m/s})(8 \text{ m})}{2.486 \times 10^{-5} \text{ m}^2/\text{s}} = 1.931 \times 10^6$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(1.931 \times 10^6)^{0.8} - 871](0.7166)^{1/3} = 2757$$

$$h = \frac{k}{L} Nu = \frac{0.02917 \text{ W/m}\cdot^\circ\text{C}}{8 \text{ m}} (2757) = 10.05 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (2.5 \text{ m})(8 \text{ m}) = 20 \text{ m}^2$$

$$\dot{Q} = hA_s(T_\infty - T_s) = (10.05 \text{ W/m}^2\cdot^\circ\text{C})(20 \text{ m}^2)(120 - 30)^\circ\text{C} = 18,096 \text{ W} = \mathbf{18.10 \text{ kW}}$$

(b) If the air flows parallel to the 2.5 m side, the Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(6 \text{ m/s})(2.5 \text{ m})}{2.486 \times 10^{-5} \text{ m}^2/\text{s}} = 6.034 \times 10^5$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(6.034 \times 10^5)^{0.8} - 871](0.7166)^{1/3} = 615.1$$

$$h = \frac{k}{L} Nu = \frac{0.02917 \text{ W/m}\cdot^\circ\text{C}}{2.5 \text{ m}} (615.1) = 7.177 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (8 \text{ m})(2.5 \text{ m}) = 20 \text{ m}^2$$

$$\dot{Q} = hA_s(T_\infty - T_s) = (7.177 \text{ W/m}^2\cdot^\circ\text{C})(20 \text{ m}^2)(120 - 30)^\circ\text{C} = 12,919 \text{ W} = \mathbf{12.92 \text{ kW}}$$

**7-16** Wind is blowing parallel to the wall of a house. The rate of heat loss from that wall is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties.

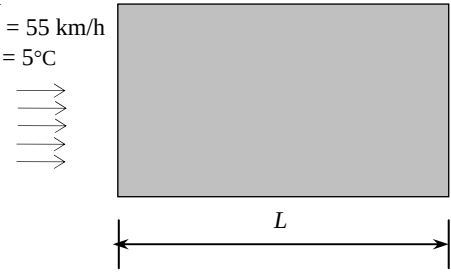
**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (12+5)/2 = 8.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02428 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.413 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7340$$

Air  
 $V_\infty = 55 \text{ km/h}$   
 $T_\infty = 5^\circ\text{C}$



**Analysis** Air flows parallel to the 10 m side:

The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(55 \times 1000 / 3600) \text{ m/s}](10 \text{ m})}{1.413 \times 10^{-5} \text{ m}^2/\text{s}} = 1.081 \times 10^7$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, heat transfer coefficient and then heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(1.081 \times 10^7)^{0.8} - 871](0.7340)^{1/3} = 1.336 \times 10^4$$

$$h = \frac{k}{L} Nu = \frac{0.02428 \text{ W/m}\cdot^\circ\text{C}}{10 \text{ m}} (1.336 \times 10^4) = 32.43 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (4 \text{ m})(10 \text{ m}) = 40 \text{ m}^2$$

$$\dot{Q} = hA_s (T_\infty - T_s) = (32.43 \text{ W/m}^2\cdot^\circ\text{C})(40 \text{ m}^2)(12 - 5)^\circ\text{C} = 9081 \text{ W} = \mathbf{9.08 \text{ kW}}$$

If the wind velocity is doubled:

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(110 \times 1000 / 3600) \text{ m/s}](10 \text{ m})}{1.413 \times 10^{-5} \text{ m}^2/\text{s}} = 2.163 \times 10^7$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(2.163 \times 10^7)^{0.8} - 871](0.7340)^{1/3} = 2.384 \times 10^4$$

$$h = \frac{k}{L} Nu = \frac{0.02428 \text{ W/m}\cdot^\circ\text{C}}{10 \text{ m}} (2.384 \times 10^4) = 57.88 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (10 \text{ m})(4 \text{ m}) = 40 \text{ m}^2$$

$$\dot{Q} = hA_s (T_\infty - T_s) = (57.88 \text{ W/m}^2\cdot^\circ\text{C})(40 \text{ m}^2)(12 - 5)^\circ\text{C} = 16,206 \text{ W} = \mathbf{16.21 \text{ kW}}$$

## 7-17 "PROBLEM 7-17"

## "GIVEN"

Vel=55 "[km/h], parameter to be varied"

height=4 "[m]"

L=10 "[m]"

"T\_infinity=5 [C], parameter to be varied"

T\_s=12 "[C]"

## "PROPERTIES"

Fluid\$='air'

k=Conductivity(Fluid\$, T=T\_film)

Pr=Prandtl(Fluid\$, T=T\_film)

rho=Density(Fluid\$, T=T\_film, P=101.3)

mu=Viscosity(Fluid\$, T=T\_film)

nu=mu/rho

T\_film=1/2\*(T\_s+T\_infinity)

## "ANALYSIS"

Re=(Vel\*Convert(km/h, m/s)\*L)/nu

"We use combined laminar and turbulent flow relation for Nusselt number"

Nusselt=(0.037\*Re<sup>0.8</sup>-871)\*Pr<sup>(1/3)</sup>

h=k/L\*Nusselt

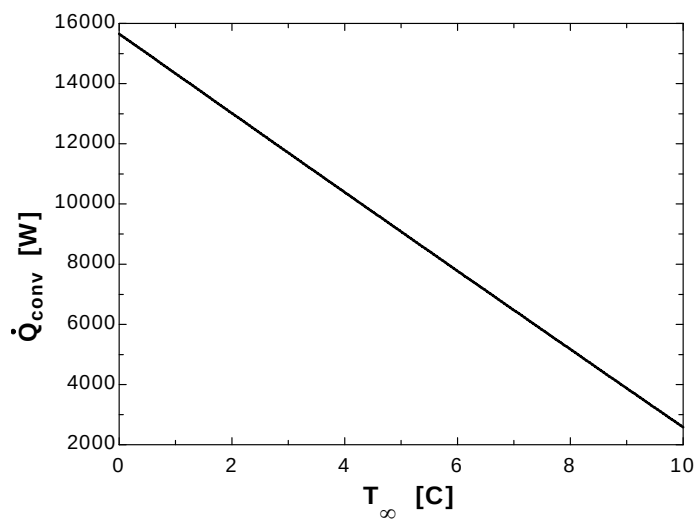
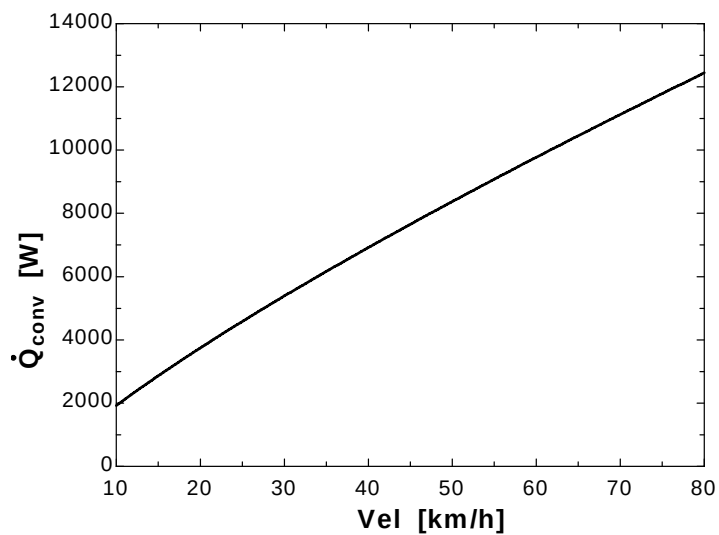
A=height\*L

Q\_dot\_conv=h\*A\*(T\_s-T\_infinity)

| Vel [km/h] | Q <sub>conv</sub> [W] |
|------------|-----------------------|
| 10         | 1924                  |
| 15         | 2866                  |
| 20         | 3746                  |
| 25         | 4583                  |
| 30         | 5386                  |
| 35         | 6163                  |
| 40         | 6918                  |
| 45         | 7655                  |
| 50         | 8375                  |
| 55         | 9081                  |
| 60         | 9774                  |
| 65         | 10455                 |
| 70         | 11126                 |
| 75         | 11788                 |
| 80         | 12441                 |

| T <sub>∞</sub> [C] | Q <sub>conv</sub> [W] |
|--------------------|-----------------------|
| 0                  | 15658                 |
| 0.5                | 14997                 |
| 1                  | 14336                 |
| 1.5                | 13677                 |
| 2                  | 13018                 |
| 2.5                | 12360                 |
| 3                  | 11702                 |
| 3.5                | 11046                 |
| 4                  | 10390                 |
| 4.5                | 9735                  |
| 5                  | 9081                  |
| 5.5                | 8427                  |
| 6                  | 7774                  |

|     |      |
|-----|------|
| 6.5 | 7122 |
| 7   | 6471 |
| 7.5 | 5821 |
| 8   | 5171 |
| 8.5 | 4522 |
| 9   | 3874 |
| 9.5 | 3226 |
| 10  | 2579 |



## Chapter 7 External Forced Convection

**7-18E** Air flows over a flat plate. The local friction and heat transfer coefficients at intervals of 1 ft are to be determined and plotted against the distance from the leading edge.

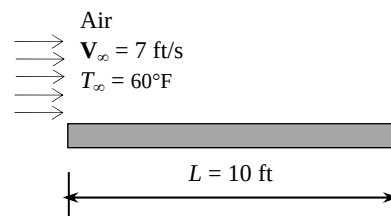
**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties.

**Properties** The properties of air at 1 atm and 60°F are (Table A-15E)

$$k = 0.01433 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.1588 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$Pr = 0.7321$$



**Analysis** For the first 1 ft interval, the Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(7 \text{ ft/s})(1 \text{ ft})}{0.1588 \times 10^{-3} \text{ ft}^2/\text{s}} = 4.407 \times 10^4$$

which is less than the critical value of  $5 \times 10^5$ . Therefore, the flow is laminar. The local Nusselt number is

$$Nu_x = \frac{hx}{k} = 0.332 Re_x^{0.5} Pr^{1/3} = 0.332(4.407 \times 10^4)^{0.5} (0.7321)^{1/3} = 62.82$$

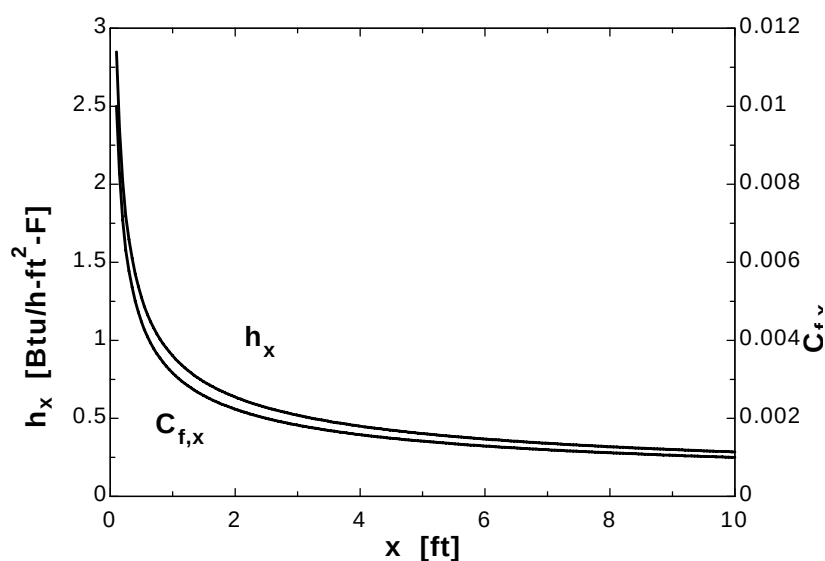
The local heat transfer and friction coefficients are

$$h_x = \frac{k}{x} Nu_x = \frac{0.01433 \text{ Btu/h.ft.}^\circ\text{F}}{1 \text{ ft}} (62.82) = 0.9002 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

$$C_{f,x} = \frac{0.664}{Re^{0.5}} = \frac{0.664}{(4.407 \times 10^4)^{0.5}} = 0.00316$$

We repeat calculations for all 1-ft intervals. The results are

| $x$ | $h_x$  | $C_{f,x}$ |
|-----|--------|-----------|
| 1   | 0.9005 | 0.003162  |
| 2   | 0.6367 | 0.002236  |
| 3   | 0.5199 | 0.001826  |
| 4   | 0.4502 | 0.001581  |
| 5   | 0.4027 | 0.001414  |
| 6   | 0.3676 | 0.001291  |
| 7   | 0.3404 | 0.001195  |
| 8   | 0.3184 | 0.001118  |
| 9   | 0.3002 | 0.001054  |
| 10  | 0.2848 | 0.001     |



## 7-19E "PROBLEM 7-19E"

## "GIVEN"

T<sub>air</sub>=60 "[F]"

"x=10 [ft], parameter to be varied"

Vel=7 "[ft/s]"

## "PROPERTIES"

Fluid\$='air'

k=Conductivity(Fluid\$, T=T<sub>air</sub>)Pr=Prandtl(Fluid\$, T=T<sub>air</sub>)rho=Density(Fluid\$, T=T<sub>air</sub>, P=14.7)mu=Viscosity(Fluid\$, T=T<sub>air</sub>)\*Convert(lbm/ft-h, lbm/ft-s)

nu=mu/rho

## "ANALYSIS"

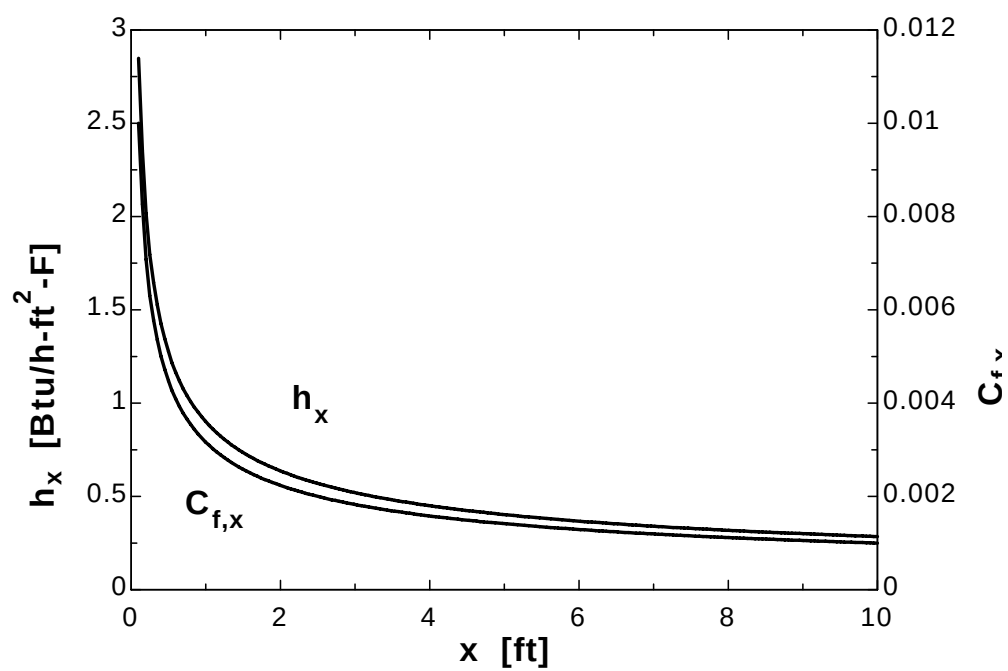
Re<sub>x</sub>=(Vel\*x)/nu

"Reynolds number is calculated to be smaller than the critical Re number. The flow is laminar."

Nusselt<sub>x</sub>=0.332\*Re<sub>x</sub><sup>0.5</sup>\*Pr<sup>(1/3)</sup>h<sub>x</sub>=k/x\*Nusselt<sub>x</sub>C<sub>f\_x</sub>=0.664/Re<sub>x</sub><sup>0.5</sup>

| x [ft] | h <sub>x</sub> [Btu/h.ft <sup>2</sup> .F] | C <sub>f_x</sub> |
|--------|-------------------------------------------|------------------|
| 0.1    | 2.848                                     | 0.01             |
| 0.2    | 2.014                                     | 0.007071         |
| 0.3    | 1.644                                     | 0.005774         |
| 0.4    | 1.424                                     | 0.005            |
| 0.5    | 1.273                                     | 0.004472         |
| 0.6    | 1.163                                     | 0.004083         |
| 0.7    | 1.076                                     | 0.00378          |
| 0.8    | 1.007                                     | 0.003536         |
| 0.9    | 0.9492                                    | 0.003333         |
| 1      | 0.9005                                    | 0.003162         |
| ...    | ...                                       | ...              |
| ...    | ...                                       | ...              |
| 9.1    | 0.2985                                    | 0.001048         |
| 9.2    | 0.2969                                    | 0.001043         |
| 9.3    | 0.2953                                    | 0.001037         |
| 9.4    | 0.2937                                    | 0.001031         |
| 9.5    | 0.2922                                    | 0.001026         |
| 9.6    | 0.2906                                    | 0.001021         |
| 9.7    | 0.2891                                    | 0.001015         |
| 9.8    | 0.2877                                    | 0.00101          |
| 9.9    | 0.2862                                    | 0.001005         |
| 10     | 0.2848                                    | 0.001            |





**7-20** A car travels at a velocity of 80 km/h. The rate of heat transfer from the bottom surface of the hot automotive engine block is to be determined.

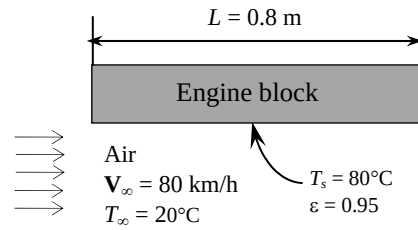
**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Air is an ideal gas with constant properties. 4 The flow is turbulent over the entire surface because of the constant agitation of the engine block.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (80 + 20)/2 = 50^\circ\text{C}$  are (Table A-15)

$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7228$$



**Analysis** Air flows parallel to the 0.4 m side. The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(80 \times 1000 / 3600) \text{ m/s}](0.8 \text{ m})}{1.798 \times 10^{-5} \text{ m}^2/\text{s}} = 9.888 \times 10^5$$

which is less than the critical Reynolds number. But the flow is assumed to be turbulent over the entire surface because of the constant agitation of the engine block. Using the proper relations, the Nusselt number, the heat transfer coefficient, and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = 0.037 Re_L^{0.8} Pr^{1/3} = 0.037(9.888 \times 10^5)^{0.8} (0.7228)^{1/3} = 2076$$

$$h = \frac{k}{L} Nu = \frac{0.02735 \text{ W/m}\cdot^\circ\text{C}}{0.8 \text{ m}} (2076) = 70.98 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (0.8 \text{ m})(0.4 \text{ m}) = 0.32 \text{ m}^2$$

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (70.98 \text{ W/m}^2\cdot^\circ\text{C})(0.32 \text{ m}^2)(80 - 20)^\circ\text{C} = \mathbf{1363 \text{ W}}$$

The radiation heat transfer from the same surface is

$$\begin{aligned} \dot{Q}_{rad} &= \epsilon A_s \sigma (T_s^4 - T_{surr}^4) = (0.95)(0.32 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(80 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4] \\ &= \mathbf{132 \text{ W}} \end{aligned}$$

Then the total rate of heat transfer from that surface becomes

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = (1363 + 132) \text{ W} = \mathbf{1495 \text{ W}}$$

**7-21** Air flows on both sides of a continuous sheet of plastic. The rate of heat transfer from the plastic sheet is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (90 + 30)/2 = 60^\circ\text{C}$  are (Table A-15)

$$\rho = 1.059 \text{ kg/m}^3$$

$$k = 0.02808 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.896 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7202$$

**Analysis** The width of the cooling section is first determined from

$$W = V\Delta t = [(15/60) \text{ m/s}](2 \text{ s}) = 0.5 \text{ m}$$

The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(3 \text{ m/s})(1.2 \text{ m})}{1.896 \times 10^{-5} \text{ m}^2/\text{s}} = 1.899 \times 10^5$$

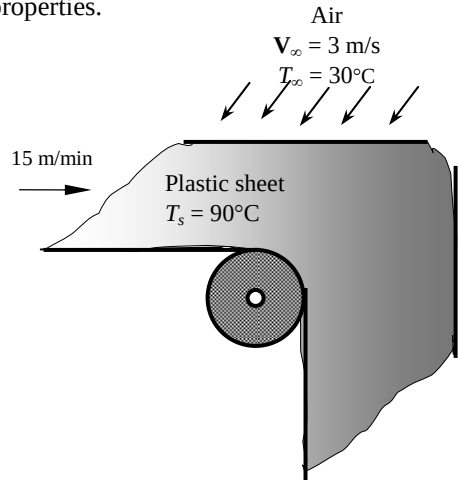
which is less than the critical Reynolds number. Thus the flow is laminar. Using the proper relation in laminar flow for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664(1.899 \times 10^5)^{0.5} (0.7202)^{1/3} = 259.7$$

$$h = \frac{k}{L} Nu = \frac{0.0282 \text{ W/m}\cdot^\circ\text{C}}{1.2 \text{ m}} (259.7) = 6.07 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = 2LW = 2(1.2 \text{ m})(0.5 \text{ m}) = 1.2 \text{ m}^2$$

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (6.07 \text{ W/m}^2\cdot^\circ\text{C})(1.2 \text{ m}^2)(90 - 30)^\circ\text{C} = \mathbf{437 \text{ W}}$$



**7-22** The top surface of the passenger car of a train in motion is absorbing solar radiation. The equilibrium temperature of the top surface is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation heat exchange with the surroundings is negligible. 4 Air is an ideal gas with constant properties.

**Properties** The properties of air at  $30^\circ\text{C}$  are (Table A-15)

$$k = 0.02588 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7282$$

**Analysis** The rate of convection heat transfer from the top surface of the car to the air must be equal to the solar radiation absorbed by the same surface in order to reach steady operation conditions. The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[70 \times 1000/3600] \text{ m/s}(8 \text{ m})}{1.608 \times 10^{-5} \text{ m}^2/\text{s}} = 9.674 \times 10^6$$

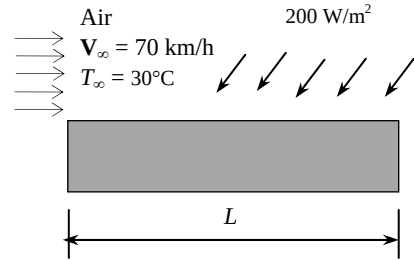
which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(9.674 \times 10^6)^{0.8} - 871](0.7282)^{1/3} = 1.212 \times 10^4$$

$$h = \frac{k}{L} Nu = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{8 \text{ m}} (1.212 \times 10^4) = 39.21 \text{ W/m}^2\cdot^\circ\text{C}$$

The equilibrium temperature of the top surface is then determined by taking convection and radiation heat fluxes to be equal to each other

$$\dot{q}_{rad} = \dot{q}_{conv} = h(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{q}_{conv}}{h} = 30^\circ\text{C} + \frac{200 \text{ W/m}^2}{39.21 \text{ W/m}^2\cdot^\circ\text{C}} = 35.1^\circ\text{C}$$



## 7-23 "PROBLEM 7-23"

**"GIVEN"**

Vel=70 "[km/h], parameter to be varied"

w=2.8 "[m]"

L=8 "[m]"

 $q_{\text{dot\_rad}}=200 \text{ [W/m}^2\text{]}, \text{ parameter to be varied}"$  $T_{\text{infinity}}=30 \text{ [}^\circ\text{C]}"$ **"PROPERTIES"**

Fluid\$='air'

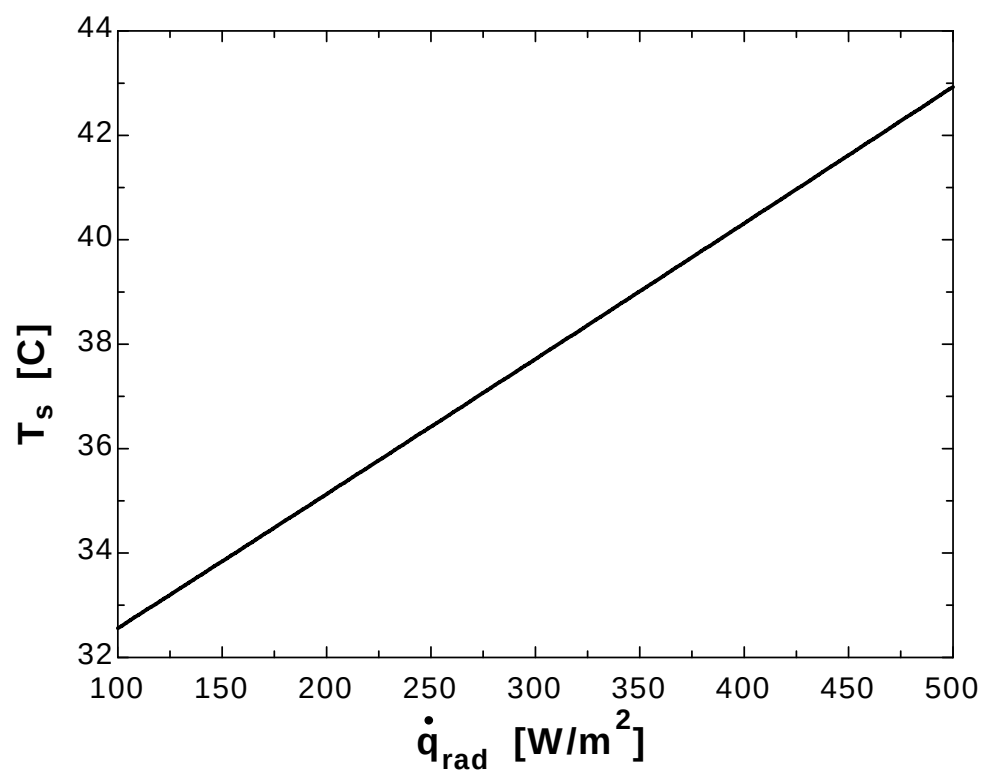
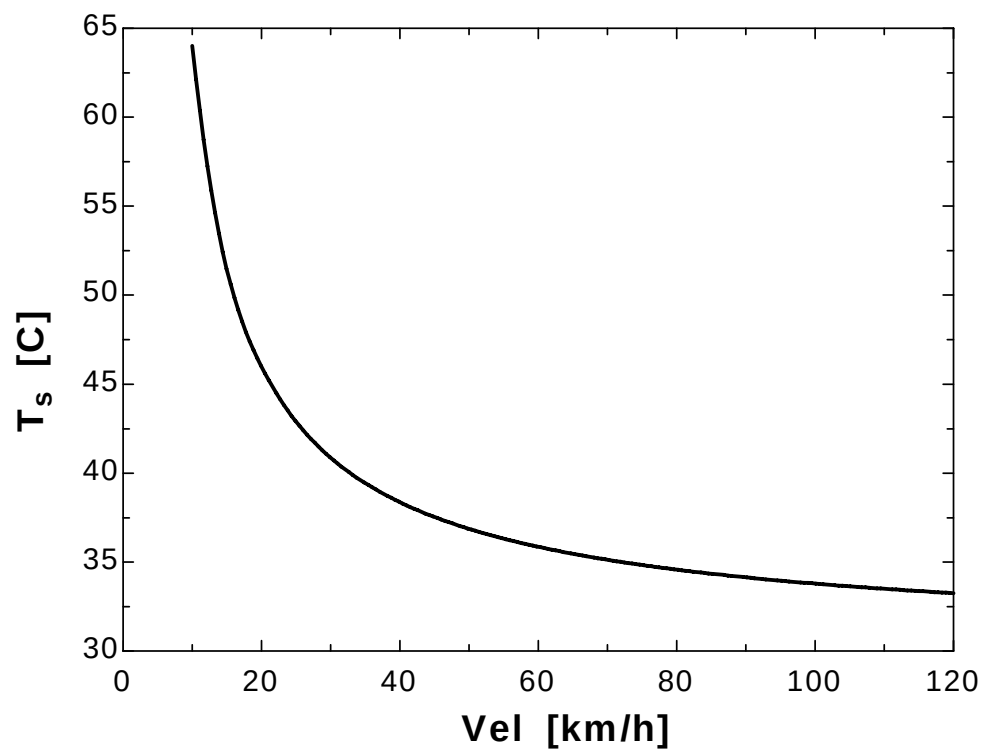
 $k=\text{Conductivity}(\text{Fluid}\$, T=T_{\text{film}})$  $\text{Pr}=\text{Prandtl}(\text{Fluid}\$, T=T_{\text{film}})$  $\rho=\text{Density}(\text{Fluid}\$, T=T_{\text{film}}, P=101.3)$  $\mu=\text{Viscosity}(\text{Fluid}\$, T=T_{\text{film}})$  $\nu=\mu/\rho$  $T_{\text{film}}=1/2*(T_s+T_{\text{infinity}})$ **"ANALYSIS"** $\text{Re}=(\text{Vel}*\text{Convert}(\text{km/h}, \text{m/s})*L)/\nu$ 

"Reynolds number is greater than the critical Reynolds number. We use combined laminar and turbulent flow relation for Nusselt number"

 $\text{Nusselt}=(0.037*\text{Re}^{0.8}-871)*\text{Pr}^{(1/3)}$  $h=k/L*\text{Nusselt}$  $q_{\text{dot\_conv}}=h*(T_s-T_{\text{infinity}})$  $q_{\text{dot\_conv}}=q_{\text{dot\_rad}}$ 

| Vel [km/h] | $T_s$ [C] |
|------------|-----------|
| 10         | 64.01     |
| 15         | 51.44     |
| 20         | 45.99     |
| 25         | 42.89     |
| 30         | 40.86     |
| 35         | 39.43     |
| 40         | 38.36     |
| 45         | 37.53     |
| 50         | 36.86     |
| 55         | 36.32     |
| 60         | 35.86     |
| 65         | 35.47     |
| 70         | 35.13     |
| 75         | 34.83     |
| 80         | 34.58     |
| 85         | 34.35     |
| 90         | 34.14     |
| 95         | 33.96     |
| 100        | 33.79     |
| 105        | 33.64     |
| 110        | 33.5      |
| 115        | 33.37     |
| 120        | 33.25     |

| $Q_{\text{rad}}$ [W/m <sup>2</sup> ] | $T_s$ [C] |
|--------------------------------------|-----------|
| 100                                  | 32.56     |
| 125                                  | 33.2      |
| 150                                  | 33.84     |
| 175                                  | 34.48     |
| 200                                  | 35.13     |
| 225                                  | 35.77     |
| 250                                  | 36.42     |
| 275                                  | 37.07     |
| 300                                  | 37.71     |
| 325                                  | 38.36     |
| 350                                  | 39.01     |
| 375                                  | 39.66     |
| 400                                  | 40.31     |
| 425                                  | 40.97     |
| 450                                  | 41.62     |
| 475                                  | 42.27     |
| 500                                  | 42.93     |



**7-24** A circuit board is cooled by air. The surface temperatures of the electronic components at the leading edge and the end of the board are to be determined.

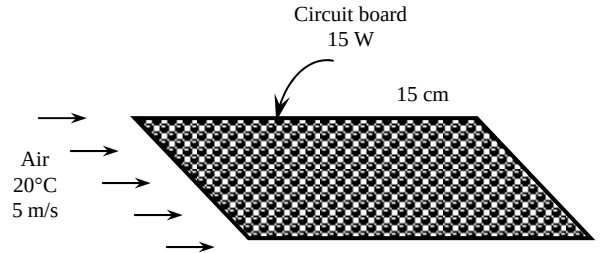
**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Any heat transfer from the back surface of the board is disregarded. 5 Air is an ideal gas with constant properties.

**Properties** Assuming the film temperature to be approximately  $35^\circ\text{C}$ , the properties of air are evaluated at this temperature to be (Table A-15)

$$k = 0.0265 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7268$$



**Analysis** (a) The convection heat transfer coefficient at the leading edge approaches infinity, and thus the surface temperature there must approach the air temperature, which is  $20^\circ\text{C}$ .

(b) The Reynolds number is

$$Re_x = \frac{V_\infty x}{\nu} = \frac{(5 \text{ m/s})(0.15 \text{ m})}{1.655 \times 10^{-5} \text{ m}^2/\text{s}} = 4.532 \times 10^4$$

which is less than the critical Reynolds number but we assume the flow to be turbulent since the electronic components are expected to act as turbulators. Using the Nusselt number uniform heat flux, the local heat transfer coefficient at the end of the board is determined to be

$$Nu_x = \frac{h_x x}{k} = 0.0308 Re_x^{0.8} Pr^{1/3} = 0.0308 (4.532 \times 10^4)^{0.8} (0.7268)^{1/3} = 147.0$$

$$h_x = \frac{k_x}{x} Nu_x = \frac{0.02625 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (147.0) = 25.73 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the surface temperature at the end of the board becomes

$$\dot{q} = h_x (T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{q}}{h_x} = 20^\circ\text{C} + \frac{(15 \text{ W})/(0.15 \text{ m})^2}{25.73 \text{ W/m}^2\cdot^\circ\text{C}} = 45.9^\circ\text{C}$$

**Discussion** The heat flux can also be determined approximately using the relation for isothermal surfaces,

$$Nu_x = \frac{h_x x}{k} = 0.0296 Re_x^{0.8} Pr^{1/3} = 0.0296 (45,320)^{0.8} (0.7268)^{1/3} = 141.3$$

$$h_x = \frac{k_x}{x} Nu_x = \frac{0.02625 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (141.3) = 24.73 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the surface temperature at the end of the board becomes

$$\dot{q} = h_x (T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{q}}{h_x} = 20^\circ\text{C} + \frac{(15 \text{ W})/(0.15 \text{ m})^2}{24.73 \text{ W/m}^2\cdot^\circ\text{C}} = 47.0^\circ\text{C}$$

Note that the two results are close to each other.



**7-25** Laminar flow of a fluid over a flat plate is considered. The change in the drag force and the rate of heat transfer are to be determined when the free-stream velocity of the fluid is doubled.

**Analysis** For the laminar flow of a fluid over a flat plate maintained at a constant temperature the drag force is given by

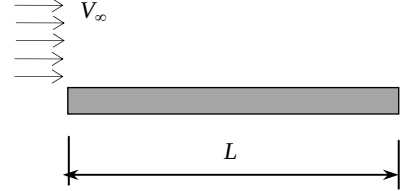
$$F_{D1} = C_f A_s \frac{\rho \mathbf{V}_\infty^2}{2} \quad \text{where} \quad C_f = \frac{1.328}{\text{Re}^{0.5}}$$

Therefore

$$F_{D1} = \frac{1.328}{\text{Re}^{0.5}} A_s \frac{\rho \mathbf{V}_\infty^2}{2}$$

Substituting Reynolds number relation, we get

$$F_{D1} = \frac{1.328}{\left(\frac{\mathbf{V}_\infty L}{\nu}\right)^{0.5}} A_s \frac{\rho \mathbf{V}_\infty^2}{2} = 0.664 \mathbf{V}_\infty^{3/2} A_s \frac{\nu^{0.5}}{L^{0.5}}$$



When the free-stream velocity of the fluid is doubled, the new value of the drag force on the plate becomes

$$F_{D2} = \frac{1.328}{\left(\frac{(2\mathbf{V}_\infty)L}{\nu}\right)^{0.5}} A_s \frac{\rho (2\mathbf{V}_\infty)^2}{2} = 0.664 (2\mathbf{V}_\infty)^{3/2} A_s \frac{\nu^{0.5}}{L^{0.5}}$$

The ratio of drag forces corresponding to  $\mathbf{V}_\infty$  and  $2\mathbf{V}_\infty$  is

$$\frac{F_{D2}}{F_{D1}} = \frac{(2\mathbf{V}_\infty)^{3/2}}{\mathbf{V}_\infty^{3/2}} = 2^{3/2}$$

We repeat similar calculations for heat transfer rate ratio corresponding to  $\mathbf{V}_\infty$  and  $2\mathbf{V}_\infty$

$$\begin{aligned} \dot{Q}_1 &= h A_s (T_s - T_\infty) = \left(\frac{k}{L} Nu\right) A_s (T_s - T_\infty) = \left(\frac{k}{L}\right) (0.664 \text{Re}^{0.5} \text{Pr}^{1/3}) A_s (T_s - T_\infty) \\ &= \frac{k}{L} 0.664 \left(\frac{\mathbf{V}_\infty L}{\nu}\right)^{0.5} \text{Pr}^{1/3} A_s (T_s - T_\infty) \\ &= 0.664 \mathbf{V}_\infty^{0.5} \frac{k}{L^{0.5} \nu^{0.5}} \text{Pr}^{1/3} A_s (T_s - T_\infty) \end{aligned}$$

When the free-stream velocity of the fluid is doubled, the new value of the heat transfer rate between the fluid and the plate becomes

$$\dot{Q}_2 = 0.664 (2\mathbf{V}_\infty)^{0.5} \frac{k}{L^{0.5} \nu^{0.5}} \text{Pr}^{1/3} A_s (T_s - T_\infty)$$

Then the ratio is

$$\frac{\dot{Q}_2}{\dot{Q}_1} = \frac{(2\mathbf{V}_\infty)^{0.5}}{\mathbf{V}_\infty^{0.5}} = 2^{0.5} = \sqrt{2}$$

**7-26E** A refrigeration truck is traveling at 55 mph. The average temperature of the outer surface of the refrigeration compartment of the truck is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties. 5 The local atmospheric pressure is 1 atm.

**Properties** Assuming the film temperature to be approximately  $80^\circ\text{F}$ , the properties of air at this temperature and 1 atm are (Table A-15E)

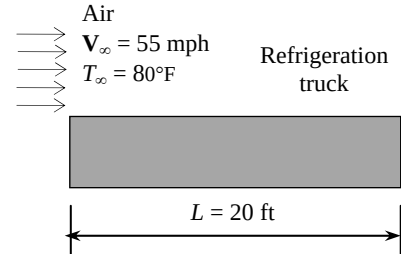
$$k = 0.01481 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.1697 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$Pr = 0.7290$$

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[55 \times 5280/3600] \text{ ft/s} (20 \text{ ft})}{0.1697 \times 10^{-3} \text{ ft}^2/\text{s}} = 9.506 \times 10^6$$



We assume the air flow over the entire outer surface to be turbulent. Therefore using the proper relation in turbulent flow for Nusselt number, the average heat transfer coefficient is determined to be

$$Nu = \frac{hL}{k} = 0.037 Re_L^{0.8} Pr^{1/3} = 0.037 (9.506 \times 10^6)^{0.8} (0.7290)^{1/3} = 1.273 \times 10^4$$

$$h = \frac{k}{L} Nu = \frac{0.01481 \text{ Btu/h.ft.}^\circ\text{F}}{20 \text{ ft}} (1.273 \times 10^4) = 9.427 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

Since the refrigeration system is operated at half the capacity, we will take half of the heat removal rate

$$\dot{Q} = \frac{(600 \times 60) \text{ Btu/h}}{2} = 18,000 \text{ Btu/h}$$

The total heat transfer surface area and the average surface temperature of the refrigeration compartment of the truck are determined from

$$A = 2[(20 \text{ ft})(9 \text{ ft}) + (20 \text{ ft})(8 \text{ ft}) + (9 \text{ ft})(8 \text{ ft})] = 824 \text{ ft}^2$$

$$\dot{Q} = hA_s (T_\infty - T_s) \longrightarrow T_s = T_\infty - \frac{\dot{Q}_{conv}}{hA_s} = 80^\circ\text{F} - \frac{18,000 \text{ Btu/h}}{(9.427 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(824 \text{ ft}^2)} = 77.7^\circ\text{F}$$

**7-27** Solar radiation is incident on the glass cover of a solar collector. The total rate of heat loss from the collector, the collector efficiency, and the temperature rise of water as it flows through the collector are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Heat exchange on the back surface of the absorber plate is negligible. 4 Air is an ideal gas with constant properties. 5 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at the film temperature of  $(35 + 25) / 2 = 30^\circ\text{C}$  are (Table A-15)

$$\begin{aligned} k &= 0.02588 \text{ W/m}\cdot^\circ\text{C} \\ \nu &= 1.608 \times 10^{-5} \text{ m}^2/\text{s} \\ Pr &= 0.7282 \end{aligned}$$

**Analysis** (a) Assuming wind flows across 2 m surface, the Reynolds number is determined from

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(30 \times 1000 / 3600) \text{ m/s}(2 \text{ m})}{1.608 \times 10^{-5} \text{ m}^2/\text{s}} = 1.036 \times 10^6$$

which is greater than the critical Reynolds number ( $5 \times 10^5$ ). Using the Nusselt number relation for combined laminar and turbulent flow, the average heat transfer coefficient is determined to be

$$\begin{aligned} Nu &= \frac{hL}{k} = (0.037 Re^{0.8} - 871) Pr^{1/3} = [0.037(1.036 \times 10^6)^{0.8} - 871](0.7282)^{1/3} = 1378 \\ h &= \frac{k}{L} Nu = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{2 \text{ m}} (1378) = 17.83 \text{ W/m}^2\cdot^\circ\text{C} \end{aligned}$$

Then the rate of heat loss from the collector by convection is

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (17.83 \text{ W/m}^2\cdot^\circ\text{C})(2 \times 1.2 \text{ m}^2)(35 - 25)^\circ\text{C} = 427.9 \text{ W}$$

The rate of heat loss from the collector by radiation is

$$\begin{aligned} \dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.90)(2 \times 1.2 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot^\circ\text{C})[(35 + 273 \text{ K})^4 - (-40 + 273 \text{ K})^4] \\ &= 741.2 \text{ W} \end{aligned}$$

and

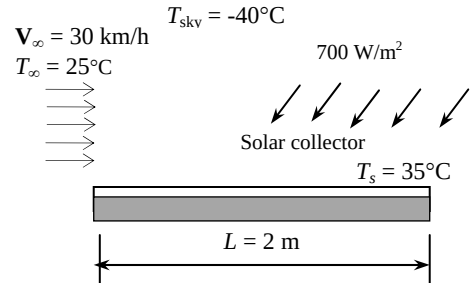
$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = 427.9 + 741.2 = \mathbf{1169 \text{ W}}$$

(b) The net rate of heat transferred to the water is

$$\begin{aligned} \dot{Q}_{net} &= \dot{Q}_{in} - \dot{Q}_{out} = \alpha AI - \dot{Q}_{out} \\ &= (0.88)(2 \times 1.2 \text{ m}^2)(700 \text{ W/m}^2) - 1169 \text{ W} \\ &= 1478 - 1169 = 309 \text{ W} \\ \eta_{collector} &= \frac{\dot{Q}_{net}}{\dot{Q}_{in}} = \frac{309 \text{ W}}{1478 \text{ W}} = \mathbf{0.209} \end{aligned}$$

(c) The temperature rise of water as it flows through the collector is

$$\dot{Q}_{net} = \dot{m} C_p \Delta T \longrightarrow \Delta T = \frac{\dot{Q}_{net}}{\dot{m} C_p} = \frac{309.4 \text{ W}}{(1/60 \text{ kg/s})(4180 \text{ J/kg}\cdot^\circ\text{C})} = \mathbf{4.44^\circ\text{C}}$$



**7-28** A fan blows air parallel to the passages between the fins of a heat sink attached to a transformer. The minimum free-stream velocity that the fan should provide to avoid overheating is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 The fins and the base plate are nearly isothermal (fin efficiency is equal to 1) 5 Air is an ideal gas with constant properties. 6 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (60 + 25)/2 = 42.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02681 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.726 \times 10^{-5} \text{ m}^2/\text{s}$$

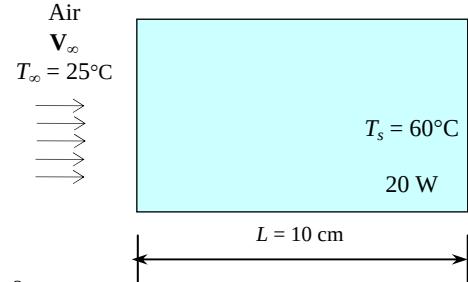
$$Pr = 0.7248$$

**Analysis** The total heat transfer surface area for this finned surface is

$$A_{s,\text{finned}} = (2 \times 7)(0.1 \text{ m})(0.005 \text{ m}) = 0.007 \text{ m}^2$$

$$A_{s,\text{unfinned}} = (0.1 \text{ m})(0.062 \text{ m}) - 7 \times (0.002 \text{ m})(0.1 \text{ m}) = 0.0048 \text{ m}^2$$

$$A_{s,\text{total}} = A_{s,\text{finned}} + A_{s,\text{unfinned}} = 0.007 \text{ m}^2 + 0.0048 \text{ m}^2 = 0.0118 \text{ m}^2$$



The convection heat transfer coefficient can be determined from Newton's law of cooling relation for a finned surface.

$$\dot{Q} = \eta h A_s (T_\infty - T_s) \longrightarrow h = \frac{\dot{Q}}{\eta A_s (T_\infty - T_s)} = \frac{20 \text{ W}}{(1)(0.0118 \text{ m}^2)(60 - 25)^\circ\text{C}} = 48.43 \text{ W/m}^2\cdot^\circ\text{C}$$

Starting from heat transfer coefficient, Nusselt number, Reynolds number and finally free-stream velocity will be determined. We assume the flow is laminar over the entire finned surface of the transformer.

$$Nu = \frac{hL}{k} = \frac{(48.43 \text{ W/m}^2\cdot^\circ\text{C})(0.1 \text{ m})}{0.02681 \text{ W/m}\cdot^\circ\text{C}} = 180.6$$

$$Nu = 0.664 Re_L^{0.5} Pr^{1/3} \longrightarrow Re_L = \frac{Nu^2}{0.664^2 Pr^{2/3}} = \frac{(180.6)^2}{(0.664)^2 (0.7248)^{2/3}} = 9.171 \times 10^4$$

$$Re_L = \frac{V_\infty L}{\nu} \longrightarrow V_\infty = \frac{Re_L \nu}{L} = \frac{(9.171 \times 10^4)(1.726 \times 10^{-5} \text{ m}^2/\text{s})}{0.1 \text{ m}} = \mathbf{15.83 \text{ m/s}}$$

**7-29** A fan blows air parallel to the passages between the fins of a heat sink attached to a transformer. The minimum free-stream velocity that the fan should provide to avoid overheating is to be determined.

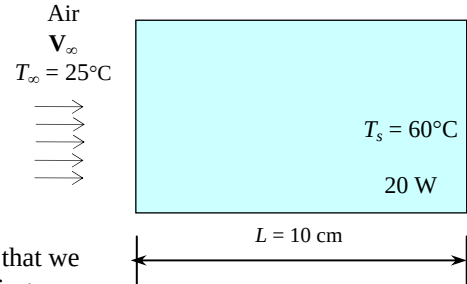
**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 The fins and the base plate are nearly isothermal (fin efficiency is equal to 1) 4 Air is an ideal gas with constant properties. 5 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at the film temperature of  $(T_s + T_\infty)/2 = (60+25)/2 = 42.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02681 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.726 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7248$$



**Analysis** We first need to determine radiation heat transfer rate. Note that we will use the base area and we assume the temperature of the surrounding surfaces are at the same temperature with the air ( $T_{surr} = 25^\circ\text{C}$ )

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.90)[(0.1\text{ m})(0.062\text{ m})](5.67 \times 10^{-8} \text{ W/m}^2 \cdot ^\circ\text{C})[(60 + 273\text{ K})^4 - (25 + 273\text{ K})^4] \\ &= 1.4 \text{ W}\end{aligned}$$

The heat transfer rate by convection will be 1.4 W less than total rate of heat transfer from the transformer. Therefore

$$\dot{Q}_{conv} = \dot{Q}_{total} - \dot{Q}_{rad} = 20 - 1.4 = 18.6 \text{ W}$$

The total heat transfer surface area for this finned surface is

$$A_{s,finned} = (2 \times 7)(0.1\text{ m})(0.005\text{ m}) = 0.007 \text{ m}^2$$

$$A_{s,unfinned} = (0.1\text{ m})(0.062\text{ m}) - 7 \times (0.002\text{ m})(0.1\text{ m}) = 0.0048 \text{ m}^2$$

$$A_{s,total} = A_{s,finned} + A_{s,unfinned} = 0.007 \text{ m}^2 + 0.0048 \text{ m}^2 = 0.0118 \text{ m}^2$$

The convection heat transfer coefficient can be determined from Newton's law of cooling relation for a finned surface.

$$\dot{Q}_{conv} = \eta h A_s (T_\infty - T_s) \longrightarrow h = \frac{\dot{Q}_{conv}}{\eta A_s (T_\infty - T_s)} = \frac{18.6 \text{ W}}{(1)(0.0118 \text{ m}^2)(60 - 25)^\circ\text{C}} = 45.04 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Starting from heat transfer coefficient, Nusselt number, Reynolds number and finally free-stream velocity will be determined. We assume the flow is laminar over the entire finned surface of the transformer.

$$Nu = \frac{hL}{k} = \frac{(45.04 \text{ W/m}^2 \cdot ^\circ\text{C})(0.1\text{ m})}{0.02681 \text{ W/m}\cdot^\circ\text{C}} = 168.0$$

$$Nu = 0.664 Re_L^{0.5} Pr^{1/3} \longrightarrow Re_L = \frac{Nu^2}{0.664^2 Pr^{2/3}} = \frac{(168.0)^2}{(0.664)^2 (0.7248)^{2/3}} = 7.932 \times 10^4$$

$$Re_L = \frac{V_\infty L}{\nu} \longrightarrow V_\infty = \frac{Re_L \nu}{L} = \frac{(7.932 \times 10^4)(1.726 \times 10^{-5} \text{ m}^2/\text{s})}{0.1\text{ m}} = \mathbf{13.7 \text{ m/s}}$$

**7-30** Air is blown over an aluminum plate mounted on an array of power transistors. The number of transistors that can be placed on this plate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible 4 Heat transfer from the back side of the plate is negligible. 5 Air is an ideal gas with constant properties. 6 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at the film temperature of  $(T_s + T_\infty)/2 = (65+35)/2 = 50^\circ\text{C}$  are (Table A-15)

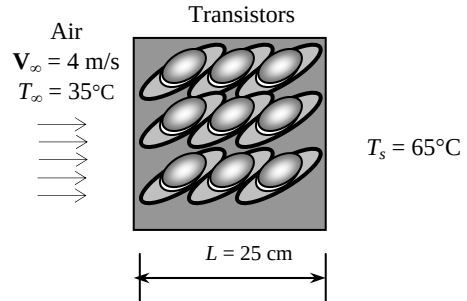
$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7228$$

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(4 \text{ m/s})(0.25 \text{ m})}{1.798 \times 10^{-5} \text{ m}^2/\text{s}} = 55,617$$



which is less than the critical Reynolds number ( $5 \times 10^5$ ). Thus the flow is laminar. Using the proper relation in laminar flow for Nusselt number, heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664(55,617)^{0.5} (0.7228)^{1/3} = 140.5$$

$$h = \frac{k}{L} Nu = \frac{0.02735 \text{ W/m}\cdot^\circ\text{C}}{0.25 \text{ m}} (140.5) = 15.37 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = wL = (0.25 \text{ m})(0.25 \text{ m}) = 0.0625 \text{ m}^2$$

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (15.37 \text{ W/m}^2\cdot^\circ\text{C})(0.0625 \text{ m}^2)(65 - 35)^\circ\text{C} = 28.83 \text{ W}$$

Considering that each transistor dissipates 3 W of power, the number of transistors that can be placed on this plate becomes

$$n = \frac{28.8 \text{ W}}{6 \text{ W}} = 4.8 \longrightarrow 4$$

This result is conservative since the transistors will cause the flow to be turbulent, and the rate of heat transfer to be higher.

**7-31** Air is blown over an aluminum plate mounted on an array of power transistors. The number of transistors that can be placed on this plate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Heat transfer from the backside of the plate is negligible. 5 Air is an ideal gas with constant properties. 6 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (65+35)/2 = 50^\circ\text{C}$  are (Table A-15)

$$k = 0.02735 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7228$$

Note that the atmospheric pressure will only affect the kinematic viscosity. The atmospheric pressure in atm is

$$P = (83.4 \text{ kPa}) \frac{1 \text{ atm}}{101.325 \text{ kPa}} = 0.823 \text{ atm}$$

The kinematic viscosity at this atmospheric pressure will be

$$\nu = (1.798 \times 10^{-5} \text{ m}^2/\text{s}) / 0.823 = 2.184 \times 10^{-5} \text{ m}^2/\text{s}$$

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(4 \text{ m/s})(0.25 \text{ m})}{2.184 \times 10^{-5} \text{ m}^2/\text{s}} = 4.579 \times 10^4$$

which is less than the critical Reynolds number ( $5 \times 10^5$ ). Thus the flow is laminar. Using the proper relation in laminar flow for Nusselt number, the average heat transfer coefficient and the heat transfer rate are determined to be

$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664 (4.579 \times 10^4)^{0.5} (0.7228)^{1/3} = 127.5$$

$$h = \frac{k}{L} Nu = \frac{0.02735 \text{ W/m}\cdot^\circ\text{C}}{0.25 \text{ m}} (127.5) = 13.95 \text{ W/m}^2\cdot^\circ\text{C}$$

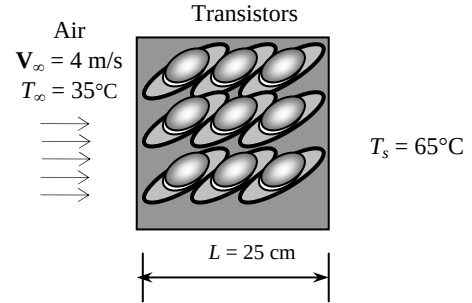
$$A_s = wL = (0.25 \text{ m})(0.25 \text{ m}) = 0.0625 \text{ m}^2$$

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (13.95 \text{ W/m}^2\cdot^\circ\text{C})(0.0625 \text{ m}^2)(65 - 35)^\circ\text{C} = 26.2 \text{ W}$$

Considering that each transistor dissipates 3 W of power, the number of transistors that can be placed on this plate becomes

$$n = \frac{26.2 \text{ W}}{6 \text{ W}} = 4.4 \longrightarrow 4$$

This result is conservative since the transistors will cause the flow to be turbulent, and the rate of heat transfer to be higher.



**7-32** Air is flowing over a long flat plate with a specified velocity. The distance from the leading edge of the plate where the flow becomes turbulent, and the thickness of the boundary layer at that location are to be determined.

**Assumptions** **1** The flow is steady and incompressible. **2** The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . **3** Air is an ideal gas. **4** The surface of the plate is smooth.

**Properties** The density and kinematic viscosity of air at 1 atm and 25°C are  $\rho = 1.184 \text{ kg/m}^3$  and  $\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$  (Table A-15).

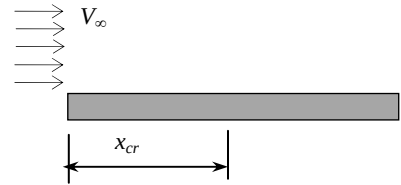
**Analysis** The critical Reynolds number is given to be  $Re_{cr} = 5 \times 10^5$ . The distance from the leading edge of the plate where the flow becomes turbulent is the distance  $x_{cr}$  where the Reynolds number becomes equal to the critical Reynolds number,

$$Re_{cr} = \frac{V_{\infty} x_{cr}}{\nu} \quad \rightarrow \quad x_{cr} = \frac{\nu Re_{cr}}{V_{\infty}} = \frac{(1.562 \times 10^{-5} \text{ m}^2/\text{s})(5 \times 10^5)}{8 \text{ m/s}} = 0.976 \text{ m}$$

The thickness of the boundary layer at that location is obtained by substituting this value of  $x$  into the laminar boundary layer thickness relation,

$$\delta_x = \frac{5x}{Re_x^{1/2}} \quad \rightarrow \quad \delta_{cr} = \frac{5x_{cr}}{Re_{cr}^{1/2}} = \frac{5(0.976 \text{ m})}{(5 \times 10^5)^{1/2}} = 0.006903 \text{ m} = \mathbf{0.69 \text{ cm}}$$

**Discussion** When the flow becomes turbulent, the boundary layer thickness starts to increase, and the value of its thickness can be determined from the boundary layer thickness relation for turbulent flow.





**7-33** Water is flowing over a long flat plate with a specified velocity. The distance from the leading edge of the plate where the flow becomes turbulent, and the thickness of the boundary layer at that location are to be determined.

**Assumptions** 1 The flow is steady and incompressible. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 The surface of the plate is smooth.

**Properties** The density and dynamic viscosity of water at 1 atm and 25°C are  $\rho = 997 \text{ kg/m}^3$  and  $\mu = 0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s}$  (Table A-9).

**Analysis** The critical Reynolds number is given to be  $Re_{cr} = 5 \times 10^5$ . The distance from the leading edge of the plate where the flow becomes turbulent is the distance  $x_{cr}$  where the Reynolds number becomes equal to the critical Reynolds number,

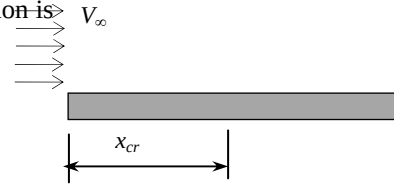
$$Re_{cr} = \frac{\rho V_{\infty} x_{cr}}{\mu} \rightarrow x_{cr} = \frac{\mu Re_{cr}}{\rho V_{\infty}} = \frac{(0.891 \times 10^{-3} \text{ kg/m} \cdot \text{s})(5 \times 10^5)}{(997 \text{ kg/m}^3)(8 \text{ m/s})} = 0.056 \text{ m} = \mathbf{5.6 \text{ cm}}$$

The thickness of the boundary layer at that location is obtained by substituting this value of  $x$  into the laminar boundary layer thickness relation,

$$\delta_{cr} = \frac{5x}{Re_x^{1/2}} \rightarrow \delta_{cr} = \frac{5x_{cr}}{Re_{cr}^{1/2}} = \frac{5(0.056 \text{ m})}{(5 \times 10^5)^{1/2}} = 0.00040 \text{ m} = \mathbf{0.4 \text{ mm}}$$

Therefore, the flow becomes turbulent after about 5 cm from the leading edge of the plate, and the thickness of the boundary layer at that location is 0.4 mm.

**Discussion** When the flow becomes turbulent, the boundary layer thickness starts to increase, and the value of its thickness can be determined from the boundary layer thickness relation for turbulent flow.



**7-34** The weight of a thin flat plate exposed to air flow on both sides is balanced by a counterweight. The mass of the counterweight that needs to be added in order to balance the plate is to be determined.

**Assumptions** **1** The flow is steady and incompressible. **2** The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . **3** Air is an ideal gas. **4** The surfaces of the plate are smooth.

**Properties** The density and kinematic viscosity of air at 1 atm and 25°C are  $\rho = 1.184 \text{ kg/m}^3$  and  $\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$  (Table A-15).

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{(10 \text{ m/s})(0.5 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2/\text{s}} = 3.201 \times 10^5$$

which is less than the critical Reynolds number of  $5 \times 10^5$ . Therefore the flow is laminar. The average friction coefficient, drag force and the corresponding mass are

$$C_f = \frac{1.328}{Re_L^{0.5}} = \frac{1.328}{(3.201 \times 10^5)^{0.5}} = 0.002347$$

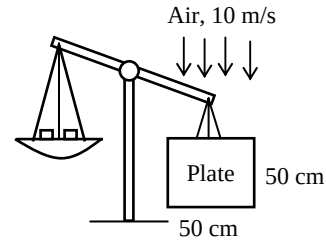
$$F_D = C_f A_s \frac{\rho V_\infty^2}{2} = (0.002347)[(2 \times 0.5 \times 0.5) \text{ m}^2] \frac{(1.184 \text{ kg/m}^3)(10 \text{ m/s})^2}{2} = 0.0695 \text{ kg} \cdot \text{m/s}^2 = 0.0695 \text{ N}$$

The mass whose weight is 0.069 N is

$$m = \frac{F_D}{g} = \frac{0.06915 \text{ kg} \cdot \text{m/s}^2}{9.81 \text{ m/s}^2} = \mathbf{0.00708 \text{ kg} = 7.08 \text{ g}}$$

Therefore, the mass of the counterweight must be 7 g to counteract the drag force acting on the plate.

**Discussion** Note that the apparatus described in this problem provides a convenient mechanism to measure drag force and thus drag coefficient.



## Flow Across Cylinders And Spheres

**7-35C** For the laminar flow, the heat transfer coefficient will be the highest at the stagnation point which corresponds to  $\theta \approx 0^\circ$ . In turbulent flow, on the other hand, it will be highest when  $\theta$  is between  $90^\circ$  and  $120^\circ$ .

**13-36C** Turbulence moves the fluid separation point further back on the rear of the body, reducing the size of the wake, and thus the magnitude of the pressure drag (which is the dominant mode of drag). As a result, the drag coefficient suddenly drops. In general, turbulence increases the drag coefficient for flat surfaces, but the drag coefficient usually remains constant at high Reynolds numbers when the flow is turbulent.

**13-37C** Friction drag is due to the shear stress at the surface whereas the pressure drag is due to the pressure differential between the front and back sides of the body when a wake is formed in the rear.

**13-38C** Flow separation in flow over a cylinder is delayed in turbulent flow because of the extra mixing due to random fluctuations and the transverse motion.

**7-39** A steam pipe is exposed to windy air. The rate of heat loss from the steam is to be determined. ✓

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (90+7)/2 = 48.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02724 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.784 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7232$$

$$\begin{aligned} \text{Air} \\ V_\infty &= 50 \text{ km/h} \\ T_\infty &= 7^\circ\text{C} \end{aligned}$$

$$\begin{aligned} \text{Pipe} \\ D &= 8 \text{ cm} \\ T_s &= 90^\circ\text{C} \end{aligned}$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(50 \text{ km/h})(1000 \text{ m/km})/(3600 \text{ s/h})](0.08 \text{ m})}{1.784 \times 10^{-5} \text{ m}^2/\text{s}} = 6.228 \times 10^4$$

The Nusselt number corresponding to this Reynolds number is

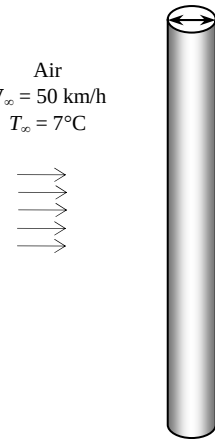
$$\begin{aligned} Nu &= \frac{hD}{k} = 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(6.228 \times 10^4)^{0.5} (0.7232)^{1/3}}{\left[1 + (0.4/0.7232)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{6.228 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 159.1 \end{aligned}$$

The heat transfer coefficient and the heat transfer rate become

$$h = \frac{k}{D} Nu = \frac{0.02724 \text{ W/m}\cdot^\circ\text{C}}{0.08 \text{ m}} (159.1) = 54.17 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.08 \text{ m})(1 \text{ m}) = 0.2513 \text{ m}^2$$

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (54.17 \text{ W/m}^2\cdot^\circ\text{C})(0.2513 \text{ m}^2)(90 - 7)^\circ\text{C} = \mathbf{1130 \text{ W}} \text{ (per m length)}$$

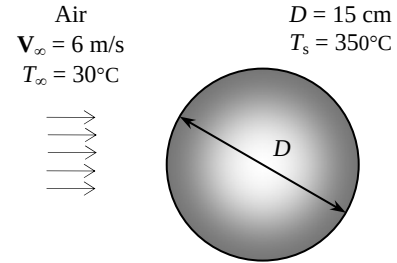


**7-40** A hot stainless steel ball is cooled by forced air. The average convection heat transfer coefficient and the cooling time are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The outer surface temperature of the ball is uniform at all times.

**Properties** The average surface temperature is  $(350+250)/2 = 300^\circ\text{C}$ , and the properties of air at 1 atm pressure and the free stream temperature of  $30^\circ\text{C}$  are (Table A-15)

$$\begin{aligned}k &= 0.02588 \text{ W/m}\cdot^\circ\text{C} \\ \nu &= 1.608 \times 10^{-5} \text{ m}^2/\text{s} \\ \mu_\infty &= 1.872 \times 10^{-5} \text{ kg/m}\cdot\text{s} \\ \mu_{s, @ 300^\circ\text{C}} &= 2.934 \times 10^{-5} \text{ kg/m}\cdot\text{s} \\ \text{Pr} &= 0.7282\end{aligned}$$



**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{(6 \text{ m/s})(0.15 \text{ m})}{1.608 \times 10^{-5} \text{ m}^2/\text{s}} = 5.597 \times 10^4$$

The Nusselt number corresponding this Reynolds number is determined to be

$$\begin{aligned}Nu &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(5.597 \times 10^4)^{0.5} + 0.06(5.597 \times 10^4)^{2/3} \right] (0.7282)^{0.4} \left( \frac{1.872 \times 10^{-5}}{2.934 \times 10^{-5}} \right)^{1/4} = 145.6\end{aligned}$$

Heat transfer coefficient is

$$h = \frac{k}{D} Nu = \frac{0.02588 \text{ W/m}\cdot^\circ\text{C}}{0.15 \text{ m}} (145.6) = \mathbf{25.12 \text{ W/m}^2\cdot^\circ\text{C}}$$

The average rate of heat transfer can be determined from Newton's law of cooling by using average surface temperature of the ball

$$\begin{aligned}A_s &= \pi D^2 = \pi (0.15 \text{ m})^2 = 0.07069 \text{ m}^2 \\ \dot{Q}_{ave} &= hA_s (T_s - T_\infty) = (25.12 \text{ W/m}^2\cdot^\circ\text{C})(0.07069 \text{ m}^2)(300 - 30)^\circ\text{C} = 479.5 \text{ W}\end{aligned}$$

Assuming the ball temperature to be nearly uniform, the total heat transferred from the ball during the cooling from  $350^\circ\text{C}$  to  $250^\circ\text{C}$  can be determined from

$$Q_{total} = mC_p(T_1 - T_2)$$

$$\text{where } m = \rho V = \rho \frac{\pi D^3}{6} = (8055 \text{ kg/m}^3) \frac{\pi (0.15 \text{ m})^3}{6} = 14.23 \text{ kg}$$

$$\text{Therefore, } Q_{total} = mC_p(T_1 - T_2) = (14.23 \text{ kg})(480 \text{ J/kg}\cdot^\circ\text{C})(350 - 250)^\circ\text{C} = 683,249 \text{ J}$$

Then the time of cooling becomes

$$\Delta t = \frac{Q}{\dot{Q}} = \frac{683,249 \text{ J}}{479.5 \text{ J/s}} = 1425 \text{ s} = \mathbf{23.75 \text{ min}}$$

## 7-41 "PROBLEM 7-41"

## "GIVEN"

D=0.15 "[m]"

T<sub>1</sub>=350 "[C]"T<sub>2</sub>=250 "[C]"T<sub>infinity</sub>=30 "[C]"

P=101.3 "[kPa]"

"Vel=6 [m/s], parameter to be varied"

rho<sub>ball</sub>=8055 "[kg/m<sup>3</sup>]"C<sub>p,ball</sub>=480 "[J/kg-C]"

## "PROPERTIES"

Fluid\$='air'

k=Conductivity(Fluid\$, T=T<sub>infinity</sub>)Pr=Prandtl(Fluid\$, T=T<sub>infinity</sub>)rho=Density(Fluid\$, T=T<sub>infinity</sub>, P=P)mu<sub>infinity</sub>=Viscosity(Fluid\$, T=T<sub>infinity</sub>)nu=mu<sub>infinity</sub>/rhomu<sub>s</sub>=Viscosity(Fluid\$, T=T<sub>s,ave</sub>)T<sub>s,ave</sub>=1/2\*(T<sub>1</sub>+T<sub>2</sub>)

## "ANALYSIS"

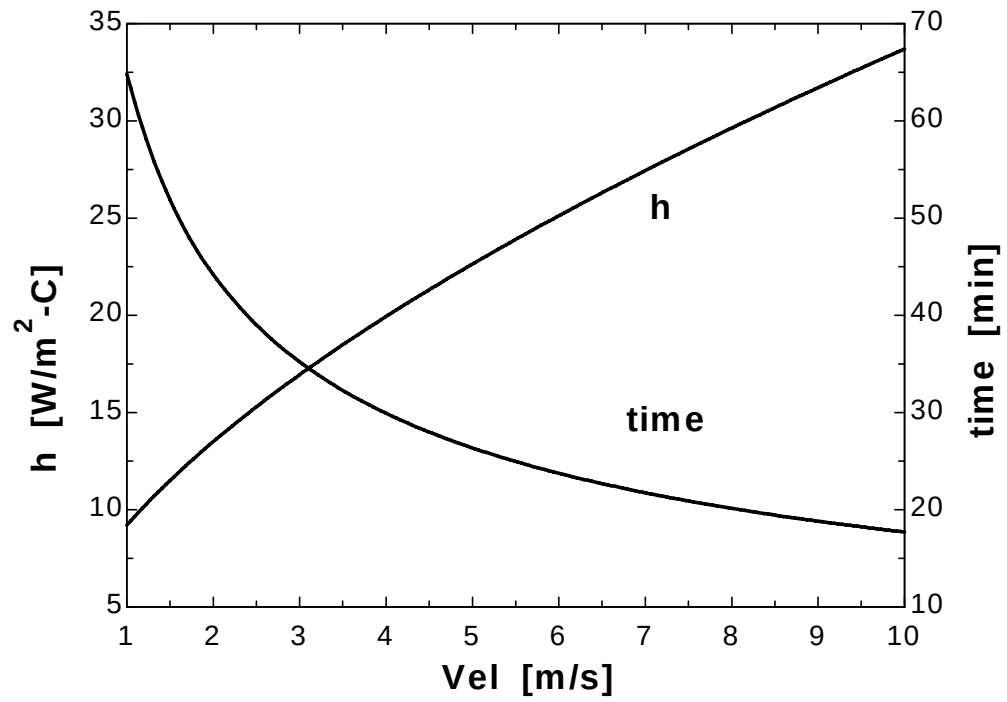
Re=(Vel\*D)/nu

Nusselt=2+(0.4\*Re<sup>0.5</sup>+0.06\*Re<sup>(2/3)</sup>)\*Pr<sup>0.4</sup>(mu<sub>infinity</sub>/mu<sub>s</sub>)<sup>0.25</sup>

h=k/D\*Nusselt

A=pi\*D<sup>2</sup>Q<sub>dot,ave</sub>=h\*A\*(T<sub>s,ave</sub>-T<sub>infinity</sub>)Q<sub>total</sub>=m<sub>ball</sub>\*C<sub>p,ball</sub>\*(T<sub>1</sub>-T<sub>2</sub>)m<sub>ball</sub>=rho<sub>ball</sub>\*V<sub>ball</sub>V<sub>ball</sub>=(pi\*D<sup>3</sup>)/6time=Q<sub>total</sub>/Q<sub>dot,ave</sub>\*Convert(s, min)

| Vel [m/s] | h [W/m <sup>2</sup> .C] | time [min] |
|-----------|-------------------------|------------|
| 1         | 9.204                   | 64.83      |
| 1.5       | 11.5                    | 51.86      |
| 2         | 13.5                    | 44.2       |
| 2.5       | 15.29                   | 39.01      |
| 3         | 16.95                   | 35.21      |
| 3.5       | 18.49                   | 32.27      |
| 4         | 19.94                   | 29.92      |
| 4.5       | 21.32                   | 27.99      |
| 5         | 22.64                   | 26.36      |
| 5.5       | 23.9                    | 24.96      |
| 6         | 25.12                   | 23.75      |
| 6.5       | 26.3                    | 22.69      |
| 7         | 27.44                   | 21.74      |
| 7.5       | 28.55                   | 20.9       |
| 8         | 29.63                   | 20.14      |
| 8.5       | 30.69                   | 19.44      |
| 9         | 31.71                   | 18.81      |
| 9.5       | 32.72                   | 18.24      |
| 10        | 33.7                    | 17.7       |



**7-42E** A person extends his uncovered arms into the windy air outside. The rate of heat loss from the arm is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The arm is treated as a 2-ft-long and 3-in.-diameter cylinder with insulated ends. 5 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (86+54)/2 = 70^\circ\text{F}$  are (Table A-15E)

$$k = 0.01457 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.1643 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$\text{Pr} = 0.7306$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(20 \times 5280/3600) \text{ ft/s}](3/12) \text{ ft}}{0.1643 \times 10^{-3} \text{ ft}^2/\text{s}} = 4.463 \times 10^4$$

The Nusselt number corresponding this Reynolds number is determined to be

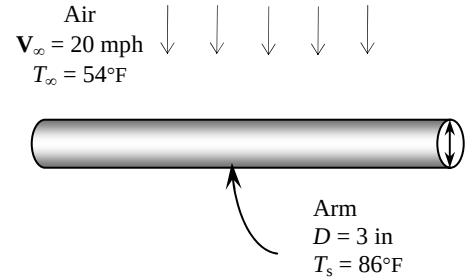
$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + \left(\frac{0.4}{\text{Pr}}\right)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(4.463 \times 10^4)^{0.5} (0.7306)^{1/3}}{\left[1 + \left(\frac{0.4}{0.7306}\right)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{4.463 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 129.6 \end{aligned}$$

Then the heat transfer coefficient and the heat transfer rate from the arm becomes

$$h = \frac{k}{D} \text{Nu} = \frac{0.01457 \text{ Btu/h.ft.}^\circ\text{F}}{(3/12) \text{ ft}} (129.6) = 7.557 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

$$A_s = \pi DL = \pi(3/12 \text{ ft})(2 \text{ ft}) = 1.571 \text{ ft}^2$$

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (7.557 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(1.571 \text{ ft}^2)(86 - 54)^\circ\text{F} = \mathbf{379.8 \text{ Btu/h}}$$



## 7-43E "PROBLEM 7-43E"

## "GIVEN"

$T_{\infty}=54$  "[F], parameter to be varied"

"Vel=20 [mph], parameter to be varied"

$T_s=86$  "[F]"

$L=2$  "[ft]"

$D=3/12$  "[ft]"

## "PROPERTIES"

Fluid\$='air'

$k=\text{Conductivity}(\text{Fluid}\$, T=T_{\text{film}})$

$\text{Pr}=\text{Prandtl}(\text{Fluid}\$, T=T_{\text{film}})$

$\rho=\text{Density}(\text{Fluid}\$, T=T_{\text{film}}, P=14.7)$

$\mu=\text{Viscosity}(\text{Fluid}\$, T=T_{\text{film}})*\text{Convert}(\text{lbf}/\text{ft}\cdot\text{s}, \text{lbf}/\text{ft}\cdot\text{s})$

$\nu=\mu/\rho$

$T_{\text{film}}=1/2*(T_s+T_{\infty})$

## "ANALYSIS"

$\text{Re}=(\text{Vel}*\text{Convert}(\text{mph}, \text{ft}/\text{s})*D)/\nu$

$\text{Nusselt}=0.3+(0.62*\text{Re}^{0.5}*\text{Pr}^{(1/3)})/(1+(0.4/\text{Pr})^{(2/3)})^{0.25}*(1+(\text{Re}/282000)^{(5/8)})^{(4/5)}$

$h=k/D*\text{Nusselt}$

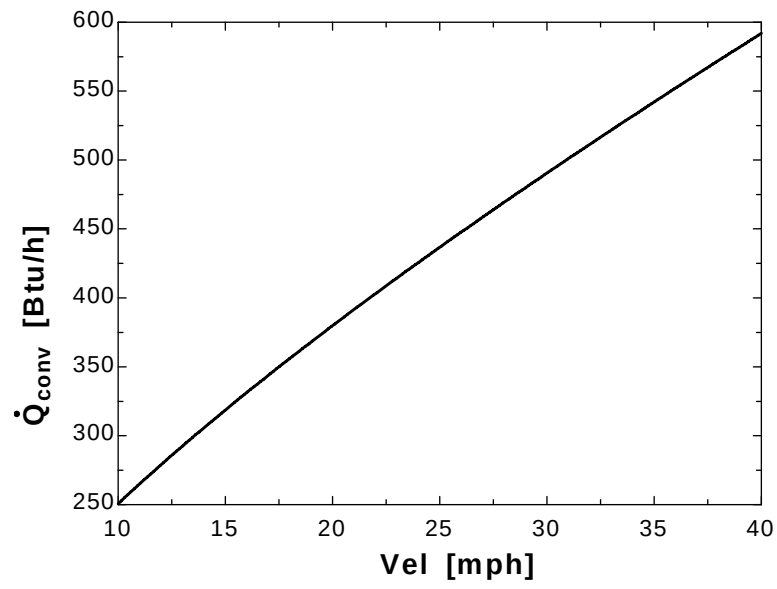
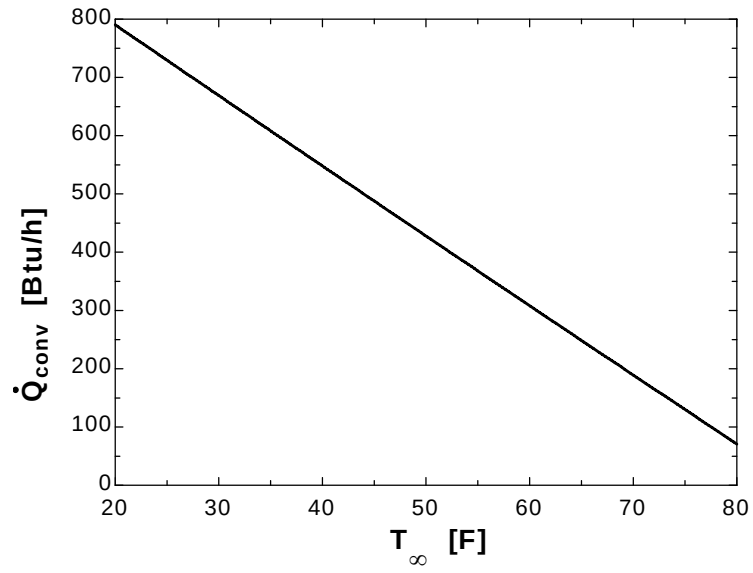
$A=\pi*D*L$

$\dot{Q}_{\text{conv}}=h*A*(T_s-T_{\infty})$

| $T_{\infty}$ [F] | $\dot{Q}_{\text{conv}}$ [Btu/h] |
|------------------|---------------------------------|
| 20               | 790.2                           |
| 25               | 729.4                           |
| 30               | 668.7                           |
| 35               | 608.2                           |
| 40               | 547.9                           |
| 45               | 487.7                           |
| 50               | 427.7                           |
| 55               | 367.9                           |
| 60               | 308.2                           |
| 65               | 248.6                           |
| 70               | 189.2                           |
| 75               | 129.9                           |
| 80               | 70.77                           |

| Vel [mph] | $\dot{Q}_{\text{conv}}$ [Btu/h] |
|-----------|---------------------------------|
| 10        | 250.6                           |
| 12        | 278.9                           |
| 14        | 305.7                           |
| 16        | 331.3                           |
| 18        | 356                             |
| 20        | 379.8                           |
| 22        | 403                             |
| 24        | 425.6                           |
| 26        | 447.7                           |
| 28        | 469.3                           |
| 30        | 490.5                           |
| 32        | 511.4                           |
| 34        | 532                             |
| 36        | 552.2                           |
| 38        | 572.2                           |
| 40        | 591.9                           |





**7-44** The average surface temperature of the head of a person when it is not covered and is subjected to winds is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 One-quarter of the heat the person generates is lost from the head. 5 The head can be approximated as a 30-cm-diameter sphere. 6 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 10°C are (Table A-15)

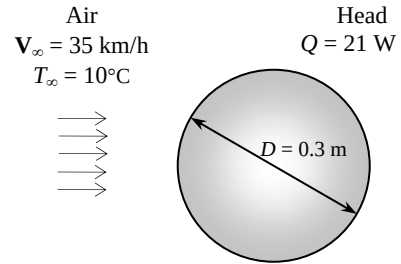
$$k = 0.02439 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_\infty = 1.778 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\mu_{s, @15^\circ\text{C}} = 1.802 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\text{Pr} = 0.7336$$



**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(35 \times 1000/3600) \text{ m/s}](0.3 \text{ m})}{1.426 \times 10^{-5} \text{ m}^2/\text{s}} = 2.045 \times 10^5$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(2.045 \times 10^5)^{0.5} + 0.06(2.045 \times 10^5)^{2/3} \right] (0.7336)^{0.4} \left( \frac{1.778 \times 10^{-5}}{1.802 \times 10^{-5}} \right)^{1/4} = 344.7 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02439 \text{ W/m} \cdot ^\circ\text{C}}{0.3 \text{ m}} (344.7) = 28.02 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Then the surface temperature of the head is determined to be

$$A_s = \pi D^2 = \pi (0.3 \text{ m})^2 = 0.2827 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 10^\circ\text{C} + \frac{(84/4) \text{ W}}{(28.02 \text{ W/m}^2 \cdot ^\circ\text{C})(0.2827 \text{ m}^2)} = 12.7^\circ\text{C}$$

**7-45** The flow of a fluid across an isothermal cylinder is considered. The change in the drag force and the rate of heat transfer when the free-stream velocity of the fluid is doubled is to be determined.

**Analysis** The drag force on a cylinder is given by

$$F_{D1} = C_D A_N \frac{\rho \mathbf{V}_\infty^2}{2}$$

When the free-stream velocity of the fluid is doubled, the drag force becomes

$$F_{D2} = C_D A_N \frac{\rho (2\mathbf{V}_\infty)^2}{2}$$

Taking the ratio of them yields

$$\frac{F_{D2}}{F_{D1}} = \frac{(2\mathbf{V}_\infty)^2}{\mathbf{V}_\infty^2} = \mathbf{4}$$

The rate of heat transfer between the fluid and the cylinder is given by Newton's law of cooling. We assume the Nusselt number is proportional to the  $n$ th power of the Reynolds number with  $0.33 < n < 0.805$ . Then,

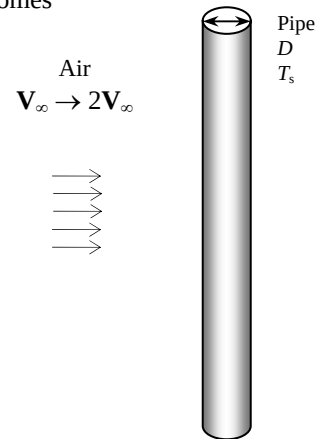
$$\begin{aligned} \dot{Q}_1 &= h A_s (T_s - T_\infty) = \left( \frac{k}{D} Nu \right) A_s (T_s - T_\infty) = \frac{k}{D} (\text{Re})^n A_s (T_s - T_\infty) \\ &= \frac{k}{D} \left( \frac{\mathbf{V}_\infty D}{\nu} \right)^n A_s (T_s - T_\infty) \\ &= \mathbf{V}_\infty^n \frac{k}{D} \left( \frac{D}{\nu} \right)^n A_s (T_s - T_\infty) \end{aligned}$$

When the free-stream velocity of the fluid is doubled, the heat transfer rate becomes

$$\dot{Q}_2 = (2\mathbf{V}_\infty)^n \frac{k}{D} \left( \frac{D}{\nu} \right)^n A_s (T_s - T_\infty)$$

Taking the ratio of them yields

$$\frac{\dot{Q}_2}{\dot{Q}_1} = \frac{(2\mathbf{V}_\infty)^n}{\mathbf{V}_\infty^n} = \mathbf{2^n}$$



**7-46** The wind is blowing across the wire of a transmission line. The surface temperature of the wire is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** We assume the film temperature to be 10°C. The properties of air at this temperature are (Table A-15)

$$\rho = 1.246 \text{ kg/m}^3$$

$$k = 0.02439 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7336$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](0.006 \text{ m})}{1.426 \times 10^{-5} \text{ m}^2/\text{s}} = 4674$$

The Nusselt number corresponding this Reynolds number is determined to be

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(4674)^{0.5} (0.7336)^{1/3}}{\left[1 + (0.4/0.7336)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{4674}{282,000}\right)^{5/8}\right]^{4/5} = 36.0 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02439 \text{ W/m} \cdot ^\circ\text{C}}{0.006 \text{ m}} (36.0) = 146.3 \text{ W/m}^2 \cdot ^\circ\text{C}$$

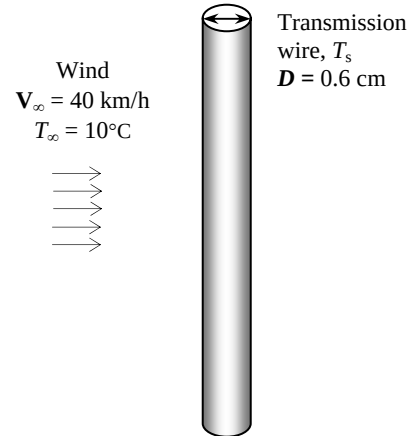
The rate of heat generated in the electrical transmission lines per meter length is

$$\dot{W} = \dot{Q} = I^2 R = (50 \text{ A})^2 (0.002 \text{ Ohm}) = 5.0 \text{ W}$$

The entire heat generated in electrical transmission line has to be transferred to the ambient air. The surface temperature of the wire then becomes

$$A_s = \pi DL = \pi(0.006 \text{ m})(1 \text{ m}) = 0.01885 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 10^\circ\text{C} + \frac{5 \text{ W}}{(146.3 \text{ W/m}^2 \cdot ^\circ\text{C})(0.01885 \text{ m}^2)} = \mathbf{11.8^\circ\text{C}}$$



## 7-47 "PROBLEM 7-47"

**"GIVEN"**

D=0.006 "[m]"

L=1 "[m], unit length is considered"

I=50 "[Ampere]"

R=0.002 "[Ohm]"

T\_infinity=10 "[C]"

"Vel=40 [km/h], parameter to be varied"

**"PROPERTIES"**

Fluid\$='air'

k=Conductivity(Fluid\$, T=T\_film)

Pr=Prandtl(Fluid\$, T=T\_film)

rho=Density(Fluid\$, T=T\_film, P=101.3)

mu=Viscosity(Fluid\$, T=T\_film)

nu=mu/rho

T\_film=1/2\*(T\_s+T\_infinity)

**"ANALYSIS"**

Re=(Vel\*Convert(km/h, m/s)\*D)/nu

Nusselt=0.3+(0.62\*Re<sup>0.5</sup>\*Pr<sup>(1/3)</sup>)/(1+(0.4/Pr)<sup>(2/3)</sup>)<sup>0.25</sup>\*(1+(Re/282000)<sup>(5/8)</sup>)<sup>(4/5)</sup>

h=k/D\*Nusselt

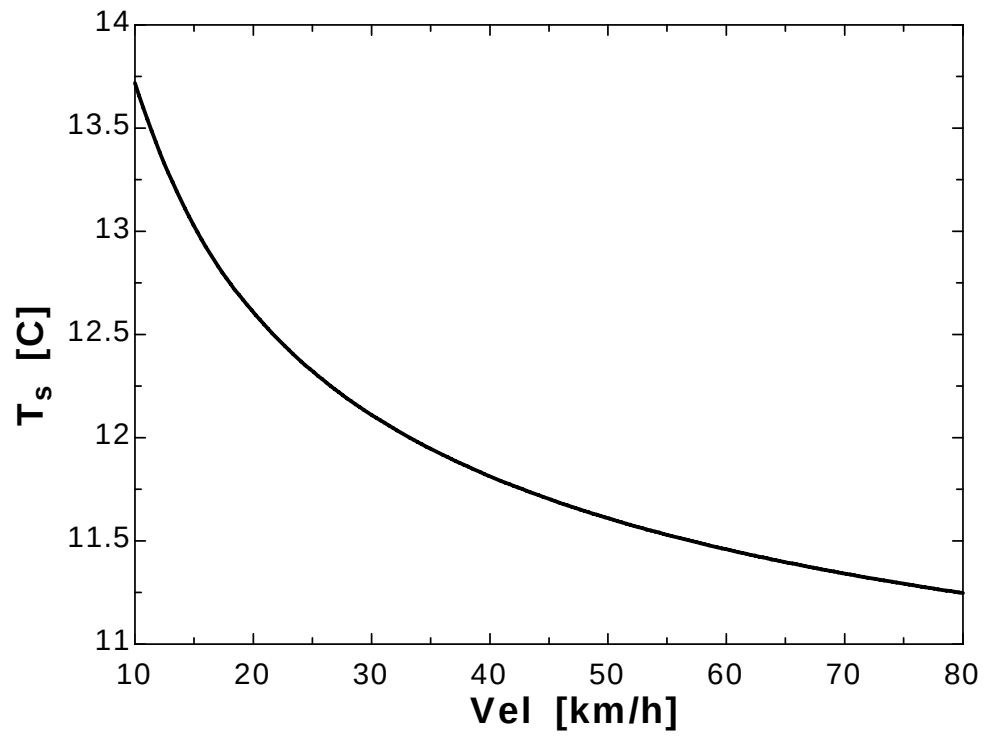
W\_dot=I<sup>2</sup>\*R

Q\_dot=W\_dot

A=pi\*D\*L

Q\_dot=h\*A\*(T\_s-T\_infinity)

| Vel [km/h] | T <sub>s</sub> [C] |
|------------|--------------------|
| 10         | 13.72              |
| 15         | 13.02              |
| 20         | 12.61              |
| 25         | 12.32              |
| 30         | 12.11              |
| 35         | 11.95              |
| 40         | 11.81              |
| 45         | 11.7               |
| 50         | 11.61              |
| 55         | 11.53              |
| 60         | 11.46              |
| 65         | 11.4               |
| 70         | 11.34              |
| 75         | 11.29              |
| 80         | 11.25              |



**7-48** An aircraft is cruising at 900 km/h. A heating system keeps the wings above freezing temperatures. The average convection heat transfer coefficient on the wing surface and the average rate of heat transfer per unit surface area are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The wing is approximated as a cylinder of elliptical cross section whose minor axis is 30 cm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (0 - 55.4)/2 = -27.7^\circ\text{C}$  are (Table A-15)

$$k = 0.02152 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.106 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7422$$

Note that the atmospheric pressure will only affect the kinematic viscosity. The atmospheric pressure in atm unit is

$$P = (18.8 \text{ kPa}) \frac{1 \text{ atm}}{101.325 \text{ kPa}} = 0.1855 \text{ atm}$$

The kinematic viscosity at this atmospheric pressure is

$$\nu = (1.106 \times 10^{-5} \text{ m}^2/\text{s})/0.1855 = 5.961 \times 10^{-5} \text{ m}^2/\text{s}$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(900 \times 1000/3600) \text{ m/s}](0.3 \text{ m})}{5.961 \times 10^{-5} \text{ m}^2/\text{s}} = 1.258 \times 10^6$$

The Nusselt number relation for a cylinder of elliptical cross-section is limited to  $\text{Re} < 15,000$ , and the relation below is not really applicable in this case. However, this relation is all we have for elliptical shapes, and we will use it with the understanding that the results may not be accurate.

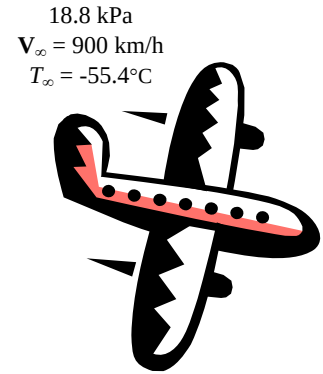
$$\text{Nu} = \frac{hD}{k} = 0.248 \text{Re}^{0.612} \text{Pr}^{1/3} = 0.248(1.258 \times 10^6)^{0.612} (0.724)^{1/3} = 1204$$

The average heat transfer coefficient on the wing surface is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02152 \text{ W/m}\cdot^\circ\text{C}}{0.3 \text{ m}} (1204) = 86.39 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the average rate of heat transfer per unit surface area becomes

$$\dot{q} = h(T_s - T_\infty) = (86.39 \text{ W/m}^2\cdot^\circ\text{C})[0 - (-55.4)]^\circ\text{C} = \mathbf{4786 \text{ W/m}^2}$$



**7-49** A long aluminum wire is cooled by cross air flowing over it. The rate of heat transfer from the wire per meter length when it is first exposed to the air is to be determined.

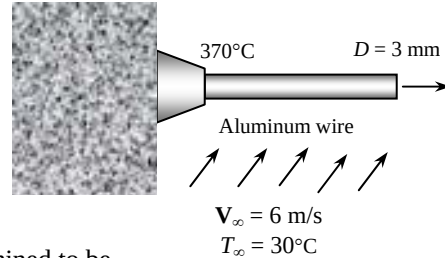
**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (370 + 30)/2 = 200^\circ\text{C}$  are (Table A-15)

$$k = 0.03779 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 3.455 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.6974$$



**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(6 \text{ m/s})(0.003 \text{ m})}{3.455 \times 10^{-5} \text{ m}^2/\text{s}} = 521.0$$

The Nusselt number corresponding this Reynolds number is determined to be

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(521.0)^{0.5} (0.6974)^{1/3}}{\left[1 + (0.4/0.6974)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{521.0}{282,000}\right)^{5/8}\right]^{4/5} = 11.48 \end{aligned}$$

Then the heat transfer coefficient and the heat transfer rate from the wire per meter length become

$$h = \frac{k}{D} \text{Nu} = \frac{0.03779 \text{ W/m}\cdot^\circ\text{C}}{0.003 \text{ m}} (11.48) = 144.6 \text{ W/m}^2\cdot^\circ\text{C}$$

$$A_s = \pi DL = \pi(0.003 \text{ m})(1 \text{ m}) = 0.009425 \text{ m}^2$$

$$\dot{Q}_{\text{conv}} = hA_s(T_s - T_\infty) = (144.6 \text{ W/m}^2\cdot^\circ\text{C})(0.009425 \text{ m}^2)(370 - 30)^\circ\text{C} = \mathbf{463.4 \text{ W}}$$



**7-50E** A fan is blowing air over the entire body of a person. The average temperature of the outer surface of the person is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The average human body can be treated as a 1-ft-diameter cylinder with an exposed surface area of 18 ft<sup>2</sup>. 5 The local atmospheric pressure is 1 atm.

**Properties** We assume the film temperature to be 100 °F. The properties of air at this temperature are (Table A-15E)

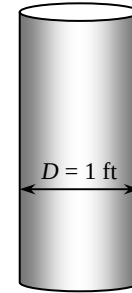
$$k = 0.01529 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.1809 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$\text{Pr} = 0.7260$$

$$\mathbf{V}_\infty = 6 \text{ ft/s}$$

$$T_\infty = 85^\circ\text{F}$$



Person,  $T_s$   
300 Btu/h

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(6 \text{ ft/s})(1 \text{ ft})}{0.1809 \times 10^{-3} \text{ ft}^2/\text{s}} = 3.317 \times 10^4$$

The proper relation for Nusselt number corresponding this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(3.317 \times 10^4)^{0.5} (0.7260)^{1/3}}{\left[1 + (0.4/0.7260)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{3.317 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 107.84 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.01529 \text{ Btu/h.ft.}^\circ\text{F}}{(1 \text{ ft})} (107.84) = 1.649 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

Then the average temperature of the outer surface of the person becomes

$$\dot{Q} = hA_s(T_s - T_\infty) \rightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 85^\circ\text{F} + \frac{300 \text{ Btu/h}}{(1.649 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(18 \text{ ft}^2)} = \mathbf{95.1^\circ\text{F}}$$

If the air velocity were doubled, the Reynolds number would be

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(12 \text{ ft/s})(1 \text{ ft})}{0.1809 \times 10^{-3} \text{ ft}^2/\text{s}} = 6.633 \times 10^4$$

The proper relation for Nusselt number corresponding this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(6.633 \times 10^4)^{0.5} (0.7260)^{1/3}}{\left[1 + (0.4/0.7260)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{6.633 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 165.95 \end{aligned}$$

Heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.01529 \text{ Btu/h.ft.}^\circ\text{F}}{(1 \text{ ft})} (165.95) = 2.537 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

Then the average temperature of the outer surface of the person becomes

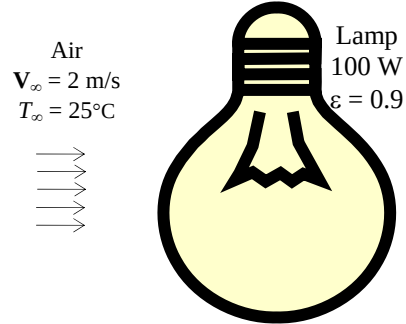
$$\dot{Q} = hA_s(T_s - T_\infty) \rightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 85^\circ\text{F} + \frac{300 \text{ Btu/h}}{(2.537 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(18 \text{ ft}^2)} = \mathbf{91.6^\circ\text{F}}$$

**7-51** A light bulb is cooled by a fan. The equilibrium temperature of the glass bulb is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The light bulb is in spherical shape. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 25°C are (Table A-15)

$$\begin{aligned}k &= 0.02551 \text{ W/m} \cdot ^\circ\text{C} \\ \nu &= 1.562 \times 10^{-5} \text{ m}^2/\text{s} \\ \mu_\infty &= 1.849 \times 10^{-5} \text{ kg/m} \cdot \text{s} \\ \mu_{s, @ 100^\circ\text{C}} &= 2.181 \times 10^{-5} \text{ kg/m} \cdot \text{s} \\ \text{Pr} &= 0.7296\end{aligned}$$



**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{(2 \text{ m/s})(0.1 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2/\text{s}} = 1.280 \times 10^4$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$\begin{aligned}Nu &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(1.280 \times 10^4)^{0.5} + 0.06(1.280 \times 10^4)^{2/3} \right] (0.7296)^{0.4} \left( \frac{1.849 \times 10^{-5}}{2.181 \times 10^{-5}} \right)^{1/4} = 68.06\end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} Nu = \frac{0.02551 \text{ W/m} \cdot ^\circ\text{C}}{0.1 \text{ m}} (68.06) = 17.36 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Noting that 90 % of electrical energy is converted to heat,

$$\dot{Q} = (0.90)(100 \text{ W}) = 90 \text{ W}$$

The bulb loses heat by both convection and radiation. The equilibrium temperature of the glass bulb can be determined by iteration,

$$A_s = \pi D^2 = \pi (0.1 \text{ m})^2 = 0.0314 \text{ m}^2$$

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = hA_s(T_s - T_\infty) + \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4)$$

$$\begin{aligned}90 \text{ W} &= (17.36 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0314 \text{ m}^2)[T_s - (25 + 273)\text{K}] \\ &\quad + (0.9)(0.0314 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[T_s^4 - (25 + 273 \text{ K})^4]\end{aligned}$$

$$T_s = 406.2 \text{ K} = \mathbf{133.2^\circ\text{C}}$$

**7-52** A steam pipe is exposed to a light winds in the atmosphere. The amount of heat loss from the steam during a certain period and the money the facility will save a year as a result of insulating the steam pipe are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The plant operates every day of the year for 10 h a day. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (75+5)/2 = 40^\circ\text{C}$  are (Table A-15)

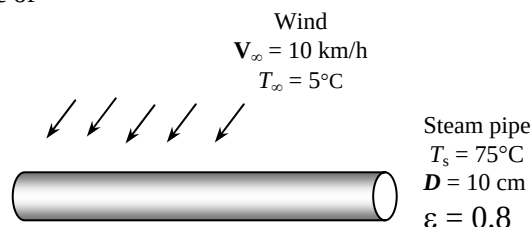
$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(10 \times 1000/3600) \text{ m/s}](0.1 \text{ m})}{1.702 \times 10^{-5} \text{ m}^2/\text{s}} = 1.632 \times 10^4$$



The Nusselt number corresponding this Reynolds number is determined to be

$$\begin{aligned} \text{Nu} &= \frac{hD}{k} = 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(1.632 \times 10^4)^{0.5} (0.7255)^{1/3}}{\left[1 + (0.4/0.7255)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{1.632 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 71.19 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.1 \text{ m}} (71.19) = 18.95 \text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat loss by convection is

$$A_s = \pi DL = \pi(0.1 \text{ m})(12 \text{ m}) = 3.77 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (18.95 \text{ W/m}^2\cdot^\circ\text{C})(3.77 \text{ m}^2)(75 - 5)^\circ\text{C} = 5001 \text{ W}$$

The rate of heat loss by radiation is

$$\begin{aligned} \dot{Q}_{\text{rad}} &= \varepsilon A_s \sigma (T_s^4 - T_{\text{surr}}^4) \\ &= (0.8)(3.77 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(75 + 273 \text{ K})^4 - (0 + 273 \text{ K})^4] = 1558 \text{ W} \end{aligned}$$

The total rate of heat loss then becomes

$$\dot{Q}_{\text{total}} = \dot{Q}_{\text{conv}} + \dot{Q}_{\text{rad}} = 5001 + 1558 = 6559 \text{ W}$$

The amount of heat loss from the steam during a 10-hour work day is

$$Q = \dot{Q}_{\text{total}} \Delta t = (6.559 \text{ kJ/s})(10 \text{ h/day} \times 3600 \text{ s/h}) = \mathbf{2.361 \times 10^5 \text{ kJ/day}}$$

The total amount of heat loss from the steam per year is

$$Q_{\text{total}} = \dot{Q}_{\text{day}} (\text{no. of days}) = (2.361 \times 10^5 \text{ kJ/day})(365 \text{ days/yr}) = 8.619 \times 10^7 \text{ kJ/yr}$$

Noting that the steam generator has an efficiency of 80%, the amount of gas used is

$$Q_{\text{gas}} = \frac{Q_{\text{total}}}{0.80} = \frac{8.619 \times 10^7 \text{ kJ/yr}}{0.80} \left( \frac{1 \text{ therm}}{105,500 \text{ kJ}} \right) = 1021 \text{ therms/yr}$$

Insulation reduces this amount by 90 %. The amount of energy and money saved becomes

$$\text{Energy saved} = (0.90)Q_{\text{gas}} = (0.90)(1021 \text{ therms/yr}) = 919 \text{ therms/yr}$$

$$\text{Money saved} = (\text{Energy saved})(\text{Unit cost of energy}) = (919 \text{ therms/yr})(\$0.54/\text{therm}) = \mathbf{\$496}$$

7-53 A steam pipe is exposed to light winds in the atmosphere. The amount of heat loss from the steam during a certain period and the money the facility will save a year as a result of insulating the steam pipes are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The plant operates every day of the year for 10 h. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (75+5)/2 = 40^\circ\text{C}$  are (Table A-15)

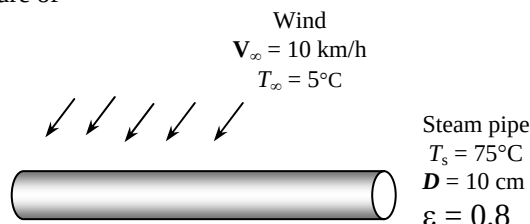
$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7255$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(10 \times 1000/3600) \text{ m/s}](0.1 \text{ m})}{1.702 \times 10^{-5} \text{ m}^2/\text{s}} = 1.632 \times 10^4$$



The Nusselt number corresponding this Reynolds number is determined to be

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{[1 + (0.4/\text{Pr})^{2/3}]^{1/4}} \left[ 1 + \left( \frac{\text{Re}}{282,000} \right)^{5/8} \right]^{4/5} \\ &= 0.3 + \frac{0.62(1.632 \times 10^4)^{0.5} (0.7255)^{1/3}}{[1 + (0.4/0.7255)^{2/3}]^{1/4}} \left[ 1 + \left( \frac{1.632 \times 10^4}{282,000} \right)^{5/8} \right]^{4/5} = 71.19 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.1 \text{ m}} (71.19) = 18.95 \text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat loss by convection is

$$A_s = \pi DL = \pi(0.1 \text{ m})(12 \text{ m}) = 3.77 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (18.95 \text{ W/m}^2\cdot^\circ\text{C})(3.77 \text{ m}^2)(75 - 5)^\circ\text{C} = 5001 \text{ W}$$

For an average surrounding temperature of  $0^\circ\text{C}$ , the rate of heat loss by radiation and the total rate of heat loss are

$$\begin{aligned} \dot{Q}_{rad} &= \epsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.8)(3.77 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(75 + 273 \text{ K})^4 - (0 + 273 \text{ K})^4] = 1558 \text{ W} \end{aligned}$$

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = 5001 + 1558 = 6559 \text{ W}$$

If the average surrounding temperature is  $-20^\circ\text{C}$ , the rate of heat loss by radiation and the total rate of heat loss become

$$\begin{aligned} \dot{Q}_{rad} &= \epsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.8)(3.77 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(75 + 273 \text{ K})^4 - (-20 + 273 \text{ K})^4] \\ &= 1807 \text{ W} \end{aligned}$$

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = 5001 + 1807 = 6808 \text{ W}$$

which is  $6808 - 6559 = 249 \text{ W}$  more than the value for a surrounding temperature of  $0^\circ\text{C}$ . This corresponds to

$$\% \text{change} = \frac{\dot{Q}_{\text{difference}}}{\dot{Q}_{\text{total}, 0^\circ\text{C}}} \times 100 = \frac{249 \text{ W}}{6559 \text{ W}} \times 100 = \mathbf{3.8\%} \quad (\text{increase})$$

If the average surrounding temperature is  $25^\circ\text{C}$ , the rate of heat loss by radiation and the total rate of heat loss become

$$\begin{aligned}\dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.8)(3.77 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4) \left[ (75 + 273 \text{ K})^4 - (25 + 273 \text{ K})^4 \right] \\ &= 1159 \text{ W}\end{aligned}$$

$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = 5001 + 1159 = 6160 \text{ W}$$

which is  $6559 - 6160 = 399 \text{ W}$  less than the value for a surrounding temperature of  $0^\circ\text{C}$ . This corresponds to

$$\% \text{change} = \frac{\dot{Q}_{\text{difference}}}{\dot{Q}_{\text{total}, 0^\circ\text{C}}} \times 100 = \frac{399 \text{ W}}{6559 \text{ W}} \times 100 = \mathbf{6.1\%} \quad (\text{decrease})$$

Therefore, the effect of the temperature variations of the surrounding surfaces on the total heat transfer is less than 6%.

**7-54E** An electrical resistance wire is cooled by a fan. The surface temperature of the wire is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** We assume the film temperature to be 200 °F. The properties of air at this temperature are (Table A-15E)

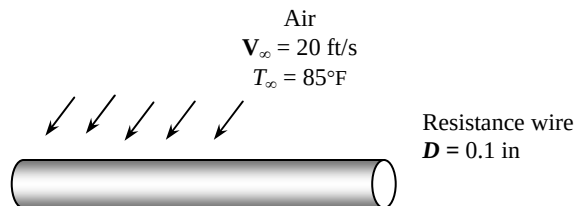
$$k = 0.01761 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.2406 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$\text{Pr} = 0.7124$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(20 \text{ ft/s})(0.1/12 \text{ ft})}{0.2406 \times 10^{-3} \text{ ft}^2/\text{s}} = 692.8$$



The proper relation for Nusselt number corresponding this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(692.8)^{0.5} (0.7124)^{1/3}}{\left[1 + (0.4/0.7124)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{692.8}{282,000}\right)^{5/8}\right]^{4/5} = 13.34 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.01761 \text{ Btu/h.ft.}^\circ\text{F}}{(0.1/12 \text{ ft})} (13.34) = 28.19 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

Then the average temperature of the outer surface of the wire becomes

$$\begin{aligned} A_s &= \pi DL = \pi(0.1/12 \text{ ft})(12 \text{ ft}) = 0.3142 \text{ ft}^2 \\ \dot{Q} &= hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA} = 85^\circ\text{F} + \frac{(1500 \times 3.41214) \text{ Btu/h}}{(28.19 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(0.3142 \text{ ft}^2)} = 662.9^\circ\text{F} \end{aligned}$$

**Discussion** Repeating the calculations at the new film temperature of  $(85+662.9)/2=374^\circ\text{F}$  gives  $T_s=668.3^\circ\text{F}$ .

7-55 The components of an electronic system located in a horizontal duct is cooled by air flowing over the duct. The total power rating of the electronic device is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (65 + 30)/2 = 47.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02717 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.774 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7235$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(200/60) \text{ m/s}](0.2 \text{ m})}{1.774 \times 10^{-5} \text{ m}^2/\text{s}} = 3.758 \times 10^4$$

Using the relation for a square duct from Table 7-1, the Nusselt number is determined to be

$$\text{Nu} = \frac{hD}{k} = 0.102 \text{Re}^{0.675} \text{Pr}^{1/3} = 0.102(3.758 \times 10^4)^{0.675} (0.7235)^{1/3} = 112.2$$

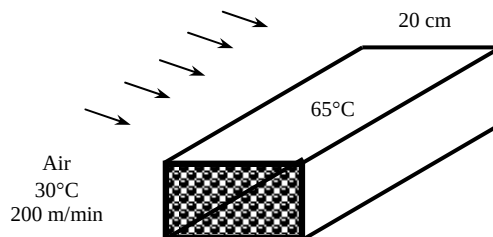
The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02717 \text{ W/m}\cdot^\circ\text{C}}{0.2 \text{ m}} (112.2) = 15.24 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the rate of heat transfer from the duct becomes

$$A_s = (4 \times 0.2 \text{ m})(1.5 \text{ m}) = 1.2 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (15.24 \text{ W/m}^2\cdot^\circ\text{C})(1.2 \text{ m}^2)(65 - 30)^\circ\text{C} = \mathbf{640.0 \text{ W}}$$



**7-56** The components of an electronic system located in a horizontal duct is cooled by air flowing over the duct. The total power rating of the electronic device is to be determined. ✓

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (65+30)/2 = 47.5^\circ\text{C}$  are (Table A-15)

$$k = 0.02717 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.774 \times 10^{-5} \text{ m}^2/\text{s}$$

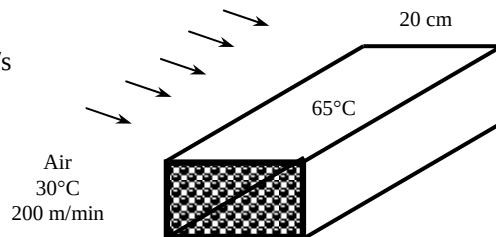
$$\text{Pr} = 0.7235$$

For a location at 4000 m altitude where the atmospheric pressure is 61.66 kPa, only kinematic viscosity of air will be affected. Thus,

$$\nu_{@61.66 \text{ kPa}} = \left( \frac{101.325}{61.66} \right) (1.774 \times 10^{-5}) = 2.915 \times 10^{-5} \text{ m}^2/\text{s}$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{[(200/60) \text{ m/s}](0.2 \text{ m})}{2.915 \times 10^{-5} \text{ m}^2/\text{s}} = 2.287 \times 10^4$$



Using the relation for a square duct from Table 7-1, the Nusselt number is determined to be

$$\text{Nu} = \frac{hD}{k} = 0.102 \text{Re}^{0.675} \text{Pr}^{1/3} = 0.102(2.287)^{0.675} (0.7235)^{1/3} = 80.21$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02717 \text{ W/m}\cdot^\circ\text{C}}{0.2 \text{ m}} (80.21) = 10.90 \text{ W/m}^2\cdot^\circ\text{C}$$

Then the rate of heat transfer from the duct becomes

$$A_s = (4 \times 0.2 \text{ m})(1.5 \text{ m}) = 1.2 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (10.90 \text{ W/m}^2\cdot^\circ\text{C})(1.2 \text{ m}^2)(65 - 30)^\circ\text{C} = \mathbf{457.7 \text{ W}}$$



7-57 A cylindrical electronic component mounted on a circuit board is cooled by air flowing across it. The surface temperature of the component is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** We assume the film temperature to be 50°C. The properties of air at 1 atm and at this temperature are (Table A-15)

$$k = 0.02735 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7228$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(150/60 \text{ m/s})(0.003 \text{ m})}{1.798 \times 10^{-5} \text{ m}^2/\text{s}} = 417.1$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(417.1)^{0.5} (0.7228)^{1/3}}{\left[1 + (0.4/0.7228)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{417.1}{282,000}\right)^{5/8}\right]^{4/5} = 10.43 \end{aligned}$$

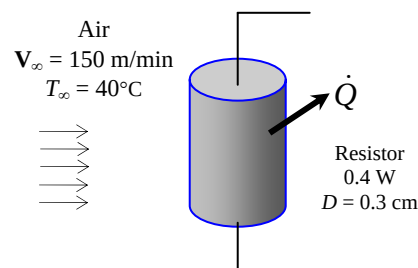
The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02735 \text{ W/m} \cdot ^\circ\text{C}}{0.003 \text{ m}} (10.43) = 95.09 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Then the surface temperature of the component becomes

$$A_s = \pi DL = \pi(0.003 \text{ m})(0.018 \text{ m}) = 0.0001696 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA} = 40^\circ\text{C} + \frac{0.4 \text{ W}}{(95.09 \text{ W/m}^2 \cdot ^\circ\text{C})(0.0001696 \text{ m}^2)} = \mathbf{64.8^\circ\text{C}}$$



**7-58** A cylindrical hot water tank is exposed to windy air. The temperature of the tank after a 45-min cooling period is to be estimated.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The surface of the tank is at the same temperature as the water temperature. 5 The heat transfer coefficient on the top and bottom surfaces is the same as that on the side surfaces.

**Properties** The properties of water at 80°C are (Table A-9)

$$\rho = 971.8 \text{ kg/m}^3$$

$$C_p = 4197 \text{ J/kg} \cdot ^\circ\text{C}$$

The properties of air at 1 atm and at the anticipated film temperature of 50°C are (Table A-15)

$$k = 0.02735 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.798 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7228$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{\left(\frac{40 \times 1000}{3600} \text{ m/s}\right)(0.50 \text{ m})}{1.798 \times 10^{-5} \text{ m}^2/\text{s}} = 309,015$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$\text{Nu} = 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5}$$

$$= 0.3 + \frac{0.62(309,015)^{0.5}(0.7228)^{1/3}}{\left[1 + (0.4/0.7228)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{309,015}{282,000}\right)^{5/8}\right]^{4/5} = 484.9$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02735 \text{ W/m} \cdot ^\circ\text{C}}{0.50 \text{ m}} (484.9) = 26.53 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The surface area of the tank is

$$A_s = \pi DL + 2\pi \frac{D^2}{4} = \pi(0.5)(0.95) + 2\pi(0.5)^2/4 = 1.885 \text{ m}^2$$

The rate of heat transfer is determined from

$$\dot{Q} = hA_s(T_s - T_\infty) = (26.53 \text{ W/m}^2 \cdot ^\circ\text{C})(1.885 \text{ m}^2) \left(\frac{80 + T_2}{2} - 18\right)^\circ\text{C} \quad (\text{Eq. 1})$$

where  $T_2$  is the final temperature of water so that  $(80 + T_2)/2$  gives the average temperature of water during the cooling process. The mass of water in the tank is

$$m = \rho V = \rho \pi \frac{D^2}{4} L = (971.8 \text{ kg/m}^3) \pi (0.50 \text{ m})^2 (0.95 \text{ m}) / 4 = 181.27 \text{ kg}$$

The amount of heat transfer from the water is determined from

$$Q = mC_p(T_2 - T_1) = (181.27 \text{ kg})(4197 \text{ J/kg} \cdot ^\circ\text{C})(80 - T_2)^\circ\text{C}$$

Then average rate of heat transfer is

$$\dot{Q} = \frac{Q}{\Delta t} = \frac{(181.27 \text{ kg})(4197 \text{ J/kg} \cdot ^\circ\text{C})(80 - T_2)^\circ\text{C}}{45 \times 60 \text{ s}} \quad (\text{Eq. 2})$$

Setting Eq. 1 to be equal to Eq. 2 we obtain the final temperature of water

Water tank  
D = 50 cm  
L = 95 cm



Air  
 $\mathbf{V}_\infty = 40 \text{ km/h}$   
 $T_\infty = 18^\circ\text{C}$

$$\dot{Q} = (26.53 \text{ W/m}^2 \cdot ^\circ\text{C})(1.885 \text{ m}^2) \left( \frac{80 + T_2}{2} - 18 \right) ^\circ\text{C} = \frac{(181.27 \text{ kg})(4197 \text{ J/kg} \cdot ^\circ\text{C})(80 - T_2) ^\circ\text{C}}{45 \times 60 \text{ s}}$$

$$\longrightarrow T_2 = \mathbf{69.9^\circ\text{C}}$$

## 7-59 "PROBLEM 7-59"

## "GIVEN"

D=0.50 "[m]"

L=0.95 "[m]"

T<sub>w1</sub>=80 "[C]"T<sub>infinity</sub>=18 "[C]"

Vel=40 "[km/h]"

"time=45 [min], parameter to be varied"

## "PROPERTIES"

Fluid\$='air'

k=Conductivity(Fluid\$, T=T<sub>film</sub>)Pr=Prandtl(Fluid\$, T=T<sub>film</sub>)rho=Density(Fluid\$, T=T<sub>film</sub>, P=101.3)mu=Viscosity(Fluid\$, T=T<sub>film</sub>)

nu=mu/rho

T<sub>film</sub>=1/2\*(T<sub>w\_ave</sub>+T<sub>infinity</sub>)rho\_w=Density(water, T=T<sub>w\_ave</sub>, P=101.3)C<sub>p\_w</sub>=CP(Water, T=T<sub>w\_ave</sub>, P=101.3)\*Convert(kJ/kg-C, J/kg-C)T<sub>w\_ave</sub>=1/2\*(T<sub>w1</sub>+T<sub>w2</sub>)

## "ANALYSIS"

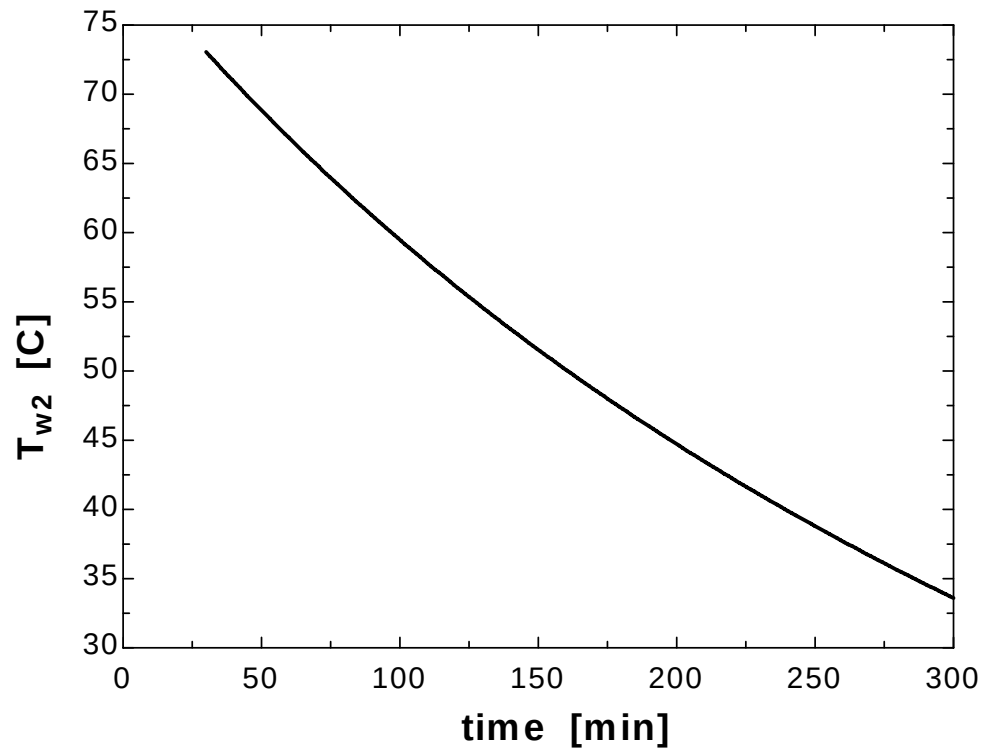
Re=(Vel\*Convert(km/h, m/s)\*D)/nu

Nusselt=0.3+(0.62\*Re<sup>0.5</sup>\*Pr<sup>1/3</sup>)/(1+(0.4/Pr)<sup>2/3</sup>)<sup>0.25</sup>\*(1+(Re/282000)<sup>5/8</sup>)<sup>4/5</sup>

h=k/D\*Nusselt

A=pi\*D\*L+2\*pi\*D<sup>2</sup>/4Q<sub>dot</sub>=h\*A\*(T<sub>w\_ave</sub>-T<sub>infinity</sub>)m<sub>w</sub>=rho\_w\*V<sub>w</sub>V<sub>w</sub>=pi\*D<sup>2</sup>/4\*LQ=m<sub>w</sub>\*C<sub>p\_w</sub>\*(T<sub>w1</sub>-T<sub>w2</sub>)Q<sub>dot</sub>=Q/(time\*Convert(min, s))

| time [min] | T <sub>w2</sub> [C] |
|------------|---------------------|
| 30         | 73.06               |
| 45         | 69.86               |
| 60         | 66.83               |
| 75         | 63.96               |
| 90         | 61.23               |
| 105        | 58.63               |
| 120        | 56.16               |
| 135        | 53.8                |
| 150        | 51.54               |
| 165        | 49.39               |
| 180        | 47.33               |
| 195        | 45.36               |
| 210        | 43.47               |
| 225        | 41.65               |
| 240        | 39.91               |
| 255        | 38.24               |
| 270        | 36.63               |
| 285        | 35.09               |
| 300        | 33.6                |



**7-60** Air flows over a spherical tank containing iced water. The rate of heat transfer to the tank and the rate at which ice melts are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The local atmospheric pressure is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 25°C are (Table A-15)

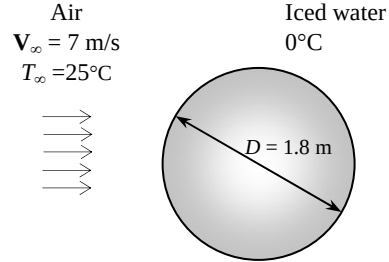
$$k = 0.02551 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.562 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_\infty = 1.849 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\mu_{s, @ 0^\circ\text{C}} = 1.729 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\text{Pr} = 0.7296$$



**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{(7 \text{ m/s})(1.8 \text{ m})}{1.562 \times 10^{-5} \text{ m}^2/\text{s}} = 806,658$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$\begin{aligned} \text{Nu} = \frac{hD}{k} &= 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(806,658)^{0.5} + 0.06(806,658)^{2/3} \right] (0.7296)^{0.4} \left( \frac{1.849 \times 10^{-5}}{1.729 \times 10^{-5}} \right)^{1/4} = 790.1 \end{aligned}$$

The heat transfer coefficient is

$$h = \frac{k}{D} \text{Nu} = \frac{0.02551 \text{ W/m} \cdot ^\circ\text{C}}{1.8 \text{ m}} (790.1) = 11.20 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Then the rate of heat transfer is determined to be

$$A_s = \pi D^2 = \pi (1.8 \text{ m})^2 = 10.18 \text{ m}^2$$

$$\dot{Q} = hA_s(T_s - T_\infty) = (11.20 \text{ W/m}^2 \cdot ^\circ\text{C})(10.18 \text{ m}^2)(25 - 0)^\circ\text{C} = \mathbf{2850 \text{ W}}$$

The rate at which ice melts is

$$\dot{Q} = \dot{m}h_{fg} \longrightarrow = 2.850 \text{ kW} = \dot{m}(333.7 \text{ kJ/kg}) \longrightarrow \dot{m} = 0.00854 \text{ kg/s} = \mathbf{0.512 \text{ kg/min}}$$

**7-61** A cylindrical bottle containing cold water is exposed to windy air. The average wind velocity is to be estimated.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 Heat transfer at the top and bottom surfaces is negligible.

**Properties** The properties of water at the average temperature of  $(T_1 + T_2)/2 = (3+11)/2 = 7^\circ\text{C}$  are (Table A-9)

$$\rho = 999.8 \text{ kg/m}^3$$

$$C_p = 4200 \text{ J/kg}\cdot^\circ\text{C}$$

The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (7+27)/2 = 17^\circ\text{C}$  are (Table A-15)

$$k = 0.02491 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.489 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7317$$

**Analysis** The mass of water in the bottle is

$$m = \rho V = \rho \pi \frac{D^2}{4} L = (999.8 \text{ kg/m}^3) \pi (0.10 \text{ m})^2 (0.30 \text{ m})/4 = 2.356 \text{ kg}$$

Then the amount of heat transfer to the water is

$$Q = m C_p (T_2 - T_1) = (2.356 \text{ kg})(4200 \text{ J/kg}\cdot^\circ\text{C})(11-3)^\circ\text{C} = 79,162 \text{ J}$$

The average rate of heat transfer is

$$\dot{Q} = \frac{Q}{\Delta t} = \frac{79,162 \text{ J}}{45 \times 60 \text{ s}} = 29.32 \text{ W}$$

The heat transfer coefficient is

$$A_s = \pi D L = \pi (0.10 \text{ m})(0.30 \text{ m}) = 0.09425 \text{ m}^2$$

$$\dot{Q}_{\text{conv}} = h A_s (T_s - T_\infty) \longrightarrow 29.32 \text{ W} = h (0.09425 \text{ m}^2) (27-7)^\circ\text{C} \longrightarrow h = 15.55 \text{ W/m}^2\cdot^\circ\text{C}$$

The Nusselt number is

$$Nu = \frac{hD}{k} = \frac{(15.55 \text{ W/m}^2\cdot^\circ\text{C})(0.10 \text{ m})}{0.02491 \text{ W/m}\cdot^\circ\text{C}} = 62.42$$

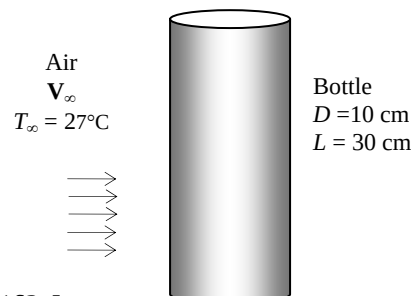
Reynolds number can be obtained from the Nusselt number relation for a flow over the cylinder

$$Nu = 0.3 + \frac{0.62 \text{ Re}^{0.5} \text{ Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5}$$

$$62.42 = 0.3 + \frac{0.62 \text{ Re}^{0.5} (0.7317)^{1/3}}{\left[1 + (0.4/0.7317)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \longrightarrow \text{Re} = 12,856$$

Then using the Reynolds number relation we determine the wind velocity

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} \longrightarrow 12,856 = \frac{\mathbf{V}_\infty (0.10 \text{ m})}{1.489 \times 10^{-5} \text{ m}^2/\text{s}} \longrightarrow \mathbf{V}_\infty = \mathbf{1.91 \text{ m/s}}$$



### Flow Across Tube Banks

**7-62C** In tube banks, the flow characteristics are dominated by the *maximum velocity*  $V_{\max}$  that occurs within the tube bank rather than the approach velocity  $V$ . Therefore, the Reynolds number is defined on the basis of maximum velocity.

**7-63C** The level of turbulence, and thus the heat transfer coefficient, increases with row number because of the combined effects of upstream rows in turbulence caused and the wakes formed. But there is no significant change in turbulence level after the first few rows, and thus the heat transfer coefficient remains constant. There is no change in transverse direction.

**7-64** Combustion air is preheated by hot water in a tube bank. The rate of heat transfer to air and the pressure drop of air are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of hot water.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of 20°C (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02514 \text{ W/m}\cdot\text{K} & \rho &= 1.204 \text{ kg/m}^3 \\ C_p &= 1.007 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7309 \\ \mu &= 1.825 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7132 \end{aligned}$$

Also, the density of air at the inlet temperature of 15°C (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.225 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.021 \text{ m}$ ,  $S_L = S_T = 0.05 \text{ m}$ , and  $V = 3.8 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$\begin{aligned} V_{\max} &= \frac{S_T}{S_T - D} V = \frac{0.05}{0.05 - 0.021} (3.8 \text{ m/s}) = 6.552 \text{ m/s} \\ \text{Re}_D &= \frac{\rho V_{\max} D}{\mu} = \frac{(1.204 \text{ kg/m}^3)(6.552 \text{ m/s})(0.021 \text{ m})}{1.825 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 9075 \end{aligned}$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.27 \text{Re}_D^{0.63} \text{Pr}^{0.36} (\text{Pr}/\text{Pr}_s)^{0.25} \\ &= 0.27(9075)^{0.63} (0.7309)^{0.36} (0.7309/0.7132)^{0.25} = 75.59 \end{aligned}$$

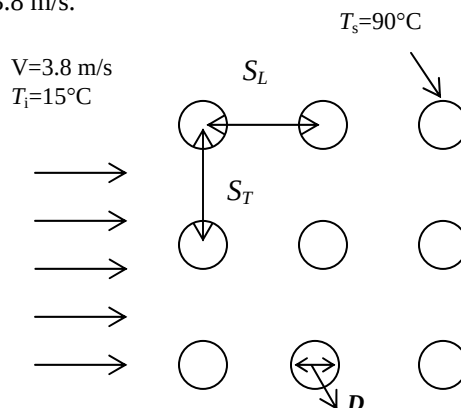
This Nusselt number is applicable to tube banks with  $N_L > 16$ . In our case the number of rows is  $N_L = 8$ , and the corresponding correction factor from Table 7-3 is  $F = 0.967$ . Then the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = F \text{Nu}_D = (0.967)(75.59) = 73.1$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{73.1(0.02514 \text{ W/m}\cdot\text{°C})}{0.021 \text{ m}} = 87.5 \text{ W/m}^2 \cdot \text{°C}$$

The total number of tubes is  $N = N_L \times N_T = 8 \times 8 = 64$ . For a unit tube length ( $L = 1 \text{ m}$ ), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 64\pi(0.021 \text{ m})(1 \text{ m}) = 4.222 \text{ m}^2$$





$$\dot{m} = \dot{m}_i = \rho_i V(N_T S_T L) = (1.225 \text{ kg/m}^3)(3.8 \text{ m/s})(8)(0.05 \text{ m})(1 \text{ m}) = 1.862 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = 90 - (90 - 15) \exp\left(-\frac{(4.222 \text{ m}^2)(87.5 \text{ W/m}^2 \cdot ^\circ\text{C})}{(1.862 \text{ kg/s})(1007 \text{ J/kg} \cdot ^\circ\text{C})}\right) = 28.42^\circ\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(90 - 15) - (90 - 28.42)}{\ln[(90 - 15)/(90 - 28.42)]} = 68.07^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (87.5 \text{ W/m}^2 \cdot ^\circ\text{C})(4.222 \text{ m}^2)(68.07^\circ\text{C}) = \mathbf{25,148 \text{ W}}$$

For this square in-line tube bank, the friction coefficient corresponding to  $\text{Re}_D = 9075$  and  $S_L/D = 5/2.1 = 2.38$  is, from Fig. 7-27a,  $f = 0.22$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 8(0.22)(1) \frac{(1.204 \text{ kg/m}^3)(6.552 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{45.5 \text{ Pa}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (15 + 29.1)/2 = 22.1^\circ\text{C}$ , which is fairly close to the assumed value of  $20^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-65** Combustion air is preheated by hot water in a tube bank. The rate of heat transfer to air and the pressure drop of air are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of hot water.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of 20°C (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02514 \text{ W/m}\cdot\text{K} & \rho &= 1.204 \text{ kg/m}^3 \\ C_p &= 1.007 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7309 \\ \mu &= 1.825 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7132 \end{aligned}$$

Also, the density of air at the inlet temperature of 15°C (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.225 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.021 \text{ m}$ ,  $S_L = S_T = 0.05 \text{ m}$ , and  $V = 3.8 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D} V = \frac{0.05}{0.05 - 0.021} (3.8 \text{ m/s}) = 6.552 \text{ m/s}$$

since  $S_D > (S_T + D)/2$

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{(1.204 \text{ kg/m}^3)(6.552 \text{ m/s})(0.021 \text{ m})}{1.825 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 9075$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.35(S_T / S_L)^{0.2} \text{Re}_D^{0.6} \text{Pr}^{0.36} (\text{Pr} / \text{Pr}_s)^{0.25} \\ &= 0.35(0.05 / 0.05)^{0.2} (9075)^{0.6} (0.7309)^{0.36} (0.7309 / 0.7132)^{0.25} = 74.55 \end{aligned}$$

This Nusselt number is applicable to tube banks with  $N_L > 16$ . In our case the number of rows is  $N_L = 8$ , and the corresponding correction factor from Table 7-3 is  $F = 0.967$ . Then the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = F \text{Nu}_D = (0.967)(74.55) = 72.09$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{72.09(0.02514 \text{ W/m}\cdot\text{K})}{0.021 \text{ m}} = 86.29 \text{ W/m}^2 \cdot \text{K}$$

The total number of tubes is  $N = N_L \times N_T = 8 \times 8 = 64$ . For a unit tube length ( $L = 1 \text{ m}$ ), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 64\pi(0.021 \text{ m})(1 \text{ m}) = 4.222 \text{ m}^2$$

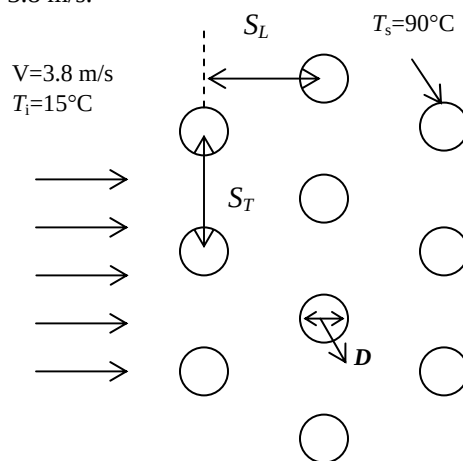
$$\dot{m} = \dot{m}_i = \rho_i V (N_T S_T L) = (1.225 \text{ kg/m}^3)(3.8 \text{ m/s})(8)(0.05 \text{ m})(1 \text{ m}) = 1.862 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = 90 - (90 - 15) \exp\left(-\frac{(4.222 \text{ m}^2)(86.29 \text{ W/m}^2 \cdot \text{K})}{(1.862 \text{ kg/s})(1007 \text{ J/kg}\cdot\text{K})}\right) = 28.25^\circ\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(90 - 15) - (90 - 28.25)}{\ln[(90 - 15)/(90 - 28.25)]} = 68.16^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (86.29 \text{ W/m}^2 \cdot \text{K})(4.222 \text{ m}^2)(68.16^\circ\text{C}) = \mathbf{24,834 \text{ W}}$$



For this staggered tube bank, the friction coefficient corresponding to  $Re_D = 9075$  and  $S_T/D = 5/2.1 = 2.38$  is, from Fig. 7-27ba,  $f = 0.34$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 8(0.34)(1) \frac{(1.204 \text{ kg/m}^3)(6.552 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = 70.3 \text{ Pa}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (15 + 28.3)/2 = 21.7^\circ\text{C}$ , which is fairly close to the assumed value of  $20^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-66** Combustion air is heated by condensing steam in a tube bank. The rate of heat transfer to air, the pressure drop of air, and the rate of condensation of steam are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of steam.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of  $35^\circ\text{C}$  (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02625 \text{ W/m}\cdot\text{K} & \rho &= 1.145 \text{ kg/m}^3 \\ C_p &= 1.007 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7268 \\ \mu &= 1.895 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7111 \end{aligned}$$

Also, the density of air at the inlet temperature of  $20^\circ\text{C}$  (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.204 \text{ kg/m}^3$ . The enthalpy of vaporization of water at  $100^\circ\text{C}$  is  $h_{fg} = 2257 \text{ kJ/kg}$  (Table A-9).

**Analysis** (a) It is given that  $D = 0.016 \text{ m}$ ,  $S_L = S_T = 0.04 \text{ m}$ , and  $V = 5.2 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D} V = \frac{0.04}{0.04 - 0.016} (5.2 \text{ m/s}) = 8.667 \text{ m/s}$$

since  $S_D > (S_T + D)/2$

$$Re_D = \frac{\rho V_{\max} D}{\mu} = \frac{(1.145 \text{ kg/m}^3)(8.667 \text{ m/s})(0.016 \text{ m})}{1.895 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 8380$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} Nu_D &= 0.35(S_T / S_L)^{0.2} Re_D^{0.6} \text{Pr}^{0.36} (\text{Pr} / \text{Pr}_s)^{0.25} \\ &= 0.35(0.04 / 0.04)^{0.2} (8380)^{0.6} (0.7268)^{0.36} (0.7268 / 0.7111)^{0.25} = 70.88 \end{aligned}$$

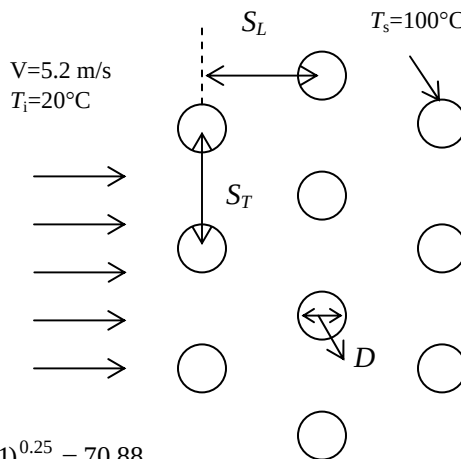
Since  $N_L = 20$ , which is greater than 16, the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$Nu_{D,N_L} = Nu_D = 70.88$$

$$h = \frac{Nu_{D,N_L} k}{D} = \frac{70.88(0.02625 \text{ W/m}\cdot^\circ\text{C})}{0.016 \text{ m}} = 116.3 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The total number of tubes is  $N = N_L \times N_T = 20 \times 10 = 200$ . For a unit tube length ( $L = 1 \text{ m}$ ), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 200\pi(0.016 \text{ m})(1 \text{ m}) = 10.05 \text{ m}^2$$



$$\dot{m} = \dot{m}_i = \rho_i V(N_T S_T L) = (1.204 \text{ kg/m}^3)(5.2 \text{ m/s})(10)(0.04 \text{ m})(1 \text{ m}) = 2.504 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = 100 - (100 - 20) \exp\left(-\frac{(10.05 \text{ m}^2)(116.3 \text{ W/m}^2 \cdot ^\circ\text{C})}{(2.504 \text{ kg/s})(1007 \text{ J/kg} \cdot ^\circ\text{C})}\right) = 49.68^\circ\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(100 - 20) - (100 - 49.68)}{\ln[(100 - 20)/(100 - 49.68)]} = 64.01^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (116.3 \text{ W/m}^2 \cdot ^\circ\text{C})(10.05 \text{ m}^2)(64.01^\circ\text{C}) = \mathbf{74,836 \text{ W}}$$

(b) For this staggered tube bank, the friction coefficient corresponding to  $\text{Re}_D = 7713$  and  $S_T/D = 4/1.6 = 2.5$  is, from Fig. 7-27b,  $f = 0.33$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 20(0.33)(1) \frac{(1.145 \text{ kg/m}^3)(8.667 \text{ m/s})^2}{2} \left(\frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2}\right) = \mathbf{283.9 \text{ Pa}}$$

(c) The rate of condensation of steam is

$$\dot{Q} = \dot{m}_{\text{cond}} h_{fg @ 100^\circ\text{C}} \longrightarrow \dot{m}_{\text{cond}} = \frac{\dot{Q}}{h_{fg @ 100^\circ\text{C}}} = \frac{74.836 \text{ kW}}{2257 \text{ kJ/kg} \cdot ^\circ\text{C}} = \mathbf{0.03316 \text{ kg/s}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (20 + 49.7)/2 = 34.9^\circ\text{C}$ , which is very close to the assumed value of  $35^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-67** Combustion air is heated by condensing steam in a tube bank. The rate of heat transfer to air, the pressure drop of air, and the rate of condensation of steam are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of steam.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of 35°C (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02625 \text{ W/m}\cdot\text{K} & \rho &= 1.145 \text{ kg/m}^3 \\ C_p &= 1.007 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7268 \\ \mu &= 1.895 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7111 \end{aligned}$$

Also, the density of air at the inlet temperature of 20°C (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.204 \text{ kg/m}^3$ . The enthalpy of vaporization of water at 100°C is  $h_{fg} = 2257 \text{ kJ/kg}\cdot\text{K}$  (Table A-9).

**Analysis** (a) It is given that  $D = 0.016 \text{ m}$ ,  $S_L = S_T = 0.05 \text{ m}$ , and  $V = 5.2 \text{ m/s}$ .

Then the maximum velocity and the Reynolds number

based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D} V = \frac{0.05}{0.05 - 0.016} (5.2 \text{ m/s}) = 7.647 \text{ m/s}$$

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{(1.145 \text{ kg/m}^3)(7.647 \text{ m/s})(0.016 \text{ m})}{1.895 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 7394$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.27 \text{Re}_D^{0.63} \text{Pr}^{0.36} (\text{Pr}/\text{Pr}_s)^{0.25} \\ &= 0.27(7394)^{0.63} (0.7268)^{0.36} (0.7268/0.7111)^{0.25} = 66.26 \end{aligned}$$

Since  $N_L = 20$ , which is greater than 16, the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = \text{Nu}_D = 66.26$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{66.26(0.02625 \text{ W/m}\cdot\text{°C})}{0.016 \text{ m}} = 108.7 \text{ W/m}^2 \cdot \text{°C}$$

The total number of tubes is  $N = N_L \times N_T = 20 \times 10 = 200$ . For a unit tube length ( $L = 1 \text{ m}$ ), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 200\pi(0.016 \text{ m})(1 \text{ m}) = 10.05 \text{ m}^2$$

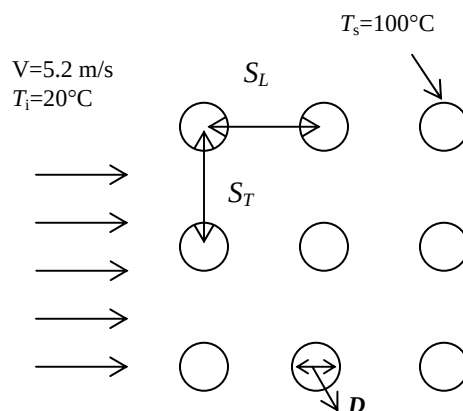
$$\dot{m} = \dot{m}_i = \rho_i V(N_T S_T L) = (1.204 \text{ kg/m}^3)(5.2 \text{ m/s})(10)(0.05 \text{ m})(1 \text{ m}) = 3.130 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = 100 - (100 - 20) \exp\left(-\frac{(10.05 \text{ m}^2)(108.7 \text{ W/m}^2 \cdot \text{°C})}{(3.130 \text{ kg/s})(1007 \text{ J/kg}\cdot\text{°C})}\right) = 43.44^\circ\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(100 - 20) - (100 - 43.44)}{\ln[(100 - 20)/(100 - 43.44)]} = 67.6^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (108.7 \text{ W/m}^2 \cdot \text{°C})(10.05 \text{ m}^2)(67.6^\circ\text{C}) = \mathbf{73,882 \text{ W}}$$



(b) For this in-line arrangement tube bank, the friction coefficient corresponding to  $Re_D = 6806$  and  $S_L/D = 5/1.6 = 3.125$  is, from Fig. 7-27a,  $f = 0.20$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 20(0.20)(1) \frac{(1.145 \text{ kg/m}^3)(7.647 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{133.9 \text{ Pa}}$$

(c) The rate of condensation of steam is

$$\dot{Q} = \dot{m}_{\text{cond}} h_{fg @ 100^\circ\text{C}} \longrightarrow \dot{m}_{\text{cond}} = \frac{\dot{Q}}{h_{fg @ 100^\circ\text{C}}} = \frac{73.882 \text{ kW}}{2257 \text{ kJ/kg} \cdot ^\circ\text{C}} = \mathbf{0.03273 \text{ kg/s}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (20 + 43.4)/2 = 31.7^\circ\text{C}$ , which is fairly close to the assumed value of  $35^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-68** Water is preheated by exhaust gases in a tube bank. The rate of heat transfer, the pressure drop of exhaust gases, and the temperature rise of water are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of steam. 3 For exhaust gases, air properties are used.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of 250°C (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.04104 \text{ W/m}\cdot\text{K} & \rho &= 0.6746 \text{ kg/m}^3 \\ C_p &= 1.033 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.6946 \\ \mu &= 2.76 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7154 \end{aligned}$$

Also, the density of air at the inlet temperature of 300°C (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 0.6158 \text{ kg/m}^3$ .

**Analysis** (a) It is given that  $D = 0.021 \text{ m}$ ,  $S_L = S_T = 0.08 \text{ m}$ , and  $V = 4.5 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D} V = \frac{0.08}{0.08 - 0.021} (4.5 \text{ m/s}) = 6.102 \text{ m/s}$$

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{(0.6746 \text{ kg/m}^3)(6.102 \text{ m/s})(0.021 \text{ m})}{2.76 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 3132$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.27 \text{Re}_D^{0.63} \text{Pr}^{0.36} (\text{Pr}/\text{Pr}_s)^{0.25} \\ &= 0.27(3132)^{0.63} (0.6946)^{0.36} (0.6946/0.7154)^{0.25} = 37.46 \end{aligned}$$

Since  $N_L = 16$ , the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = \text{Nu}_D = 37.46$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{37.46(0.04104 \text{ W/m}\cdot\text{K})}{0.021 \text{ m}} = 73.2 \text{ W/m}^2 \cdot \text{K}$$

The total number of tubes is  $N = N_L \times N_T = 16 \times 8 = 128$ . For a unit tube length ( $L = 1 \text{ m}$ ), the heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 128\pi(0.021 \text{ m})(1 \text{ m}) = 8.445 \text{ m}^2$$

$$\dot{m} = \dot{m}_i = \rho_i V(N_T S_T L) = (0.6158 \text{ kg/m}^3)(4.5 \text{ m/s})(8)(0.08 \text{ m})(1 \text{ m}) = 1.773 \text{ kg/s}$$

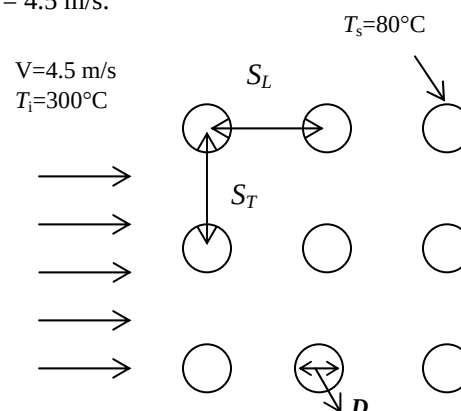
Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = 80 - (80 - 300) \exp\left(-\frac{(8.445 \text{ m}^2)(73.2 \text{ W/m}^2 \cdot \text{K})}{(1.773 \text{ kg/s})(1033 \text{ J/kg}\cdot\text{K})}\right) = 237.0^\circ\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(80 - 300) - (80 - 237)}{\ln[(80 - 300)/(80 - 237)]} = 186.7^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (73.2 \text{ W/m}^2 \cdot \text{K})(8.445 \text{ m}^2)(186.7^\circ\text{C}) = 115,425 \text{ W}$$

(b) For this in-line arrangement tube bank, the friction coefficient corresponding to  $\text{Re}_D = 3132$  and  $S_L/D = 8/2.1 = 3.81$  is, from Fig. 7-27a,  $f = 0.18$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes



$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 16(0.18)(1) \frac{(0.6746 \text{ kg/m}^3)(6.102 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{36.2 \text{ Pa}}$$

(c) The temperature rise of water is

$$\dot{Q} = \dot{m}_{\text{water}} C_{p,\text{water}} \Delta T_{\text{water}} \longrightarrow \Delta T_{\text{water}} = \frac{\dot{Q}}{\dot{m}_{\text{water}} C_{p,\text{water}}} = \frac{115.425 \text{ kW}}{(6 \text{ kg/s})(4.18 \text{ kJ/kg} \cdot ^\circ\text{C})} = \mathbf{4.6^\circ\text{C}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (300 + 237)/2 = 269^\circ\text{C}$ , which is sufficiently close to the assumed value of  $250^\circ\text{C}$ . Therefore, there is no need to repeat calculations.



**7-69** Water is heated by a bundle of resistance heater rods. The number of tube rows is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the rods is constant.

**Properties** The properties of water at the mean temperature of  $(15^\circ\text{C} + 65^\circ\text{C})/2 = 40^\circ\text{C}$  are (Table A-9):

$$\begin{aligned} k &= 0.631 \text{ W/m}\cdot\text{K} & \rho &= 992.1 \text{ kg/m}^3 \\ C_p &= 4.179 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 4.32 \\ \mu &= 0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 1.96 \end{aligned}$$

Also, the density of water at the inlet temperature of  $15^\circ\text{C}$  (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 999.1 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.01 \text{ m}$ ,  $S_L = 0.04 \text{ m}$  and  $S_T = 0.03 \text{ m}$ , and  $V = 0.8 \text{ m/s}$ .

Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$V_{\max} = \frac{S_T}{S_T - D} V = \frac{0.03}{0.03 - 0.01} (0.8 \text{ m/s}) = 1.20 \text{ m/s}$$

$$\text{Re}_D = \frac{\rho V_{\max} D}{\mu} = \frac{(992.1 \text{ kg/m}^3)(1.20 \text{ m/s})(0.01 \text{ m})}{0.653 \times 10^{-3} \text{ kg/m}\cdot\text{s}} = 18,232$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.27 \text{Re}_D^{0.63} \text{Pr}^{0.36} (\text{Pr}/\text{Pr}_s)^{0.25} \\ &= 0.27(18,232)^{0.63} (4.32)^{0.36} (4.32/1.96)^{0.25} = 269.3 \end{aligned}$$

Assuming that  $N_L > 16$ , the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = \text{Nu}_D = 269.3$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{269.3(0.631 \text{ W/m}\cdot^\circ\text{C})}{0.01 \text{ m}} = 16,994 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Consider one-row of tubes in the transverse direction (normal to flow), and thus take  $N_T = 1$ . Then the heat transfer surface area becomes

$$A_s = N_{\text{tube}} \pi D L = (1 \times N_L) \pi (0.01 \text{ m})(4 \text{ m}) = 0.1257 N_L$$

Then the log mean temperature difference, and the expression for the rate of heat transfer become

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(90 - 15) - (90 - 65)}{\ln[(90 - 15)/(90 - 65)]} = 45.51^\circ\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (16,994 \text{ W/m}^2 \cdot ^\circ\text{C})(0.1257 N_L)(45.51^\circ\text{C}) = 97,220 N_L$$

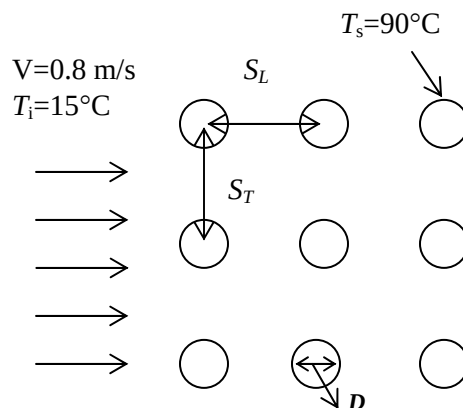
The mass flow rate of water through a cross-section corresponding to  $N_T = 1$  and the rate of heat transfer are

$$\dot{m} = \rho A_c V = (999.1 \text{ kg/m}^3)(4 \times 0.03 \text{ m}^2)(0.8 \text{ m/s}) = 95.91 \text{ kg/s}$$

$$\dot{Q} = \dot{m} C_p (T_e - T_i) = (95.91 \text{ kg/s})(4179 \text{ J/kg}\cdot^\circ\text{C})(65 - 15)^\circ\text{C} = 2.004 \times 10^7 \text{ W}$$

Substituting this result into the heat transfer expression above we find the number of tube rows

$$\dot{Q} = h A_s \Delta T_{\ln} \rightarrow 2.004 \times 10^7 \text{ W} = 97,220 N_L \rightarrow N_L = \mathbf{206}$$



**7-70** Air is cooled by an evaporating refrigerator. The refrigeration capacity and the pressure drop across the tube bank are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of refrigerant.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of  $-5^{\circ}\text{C}$  (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02326 \text{ W/m}\cdot\text{K} & \rho &= 1.316 \text{ kg/m}^3 \\ C_p &= 1.006 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7375 \\ \mu &= 1.705 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7408 \end{aligned}$$

Also, the density of air at the inlet temperature of  $0^{\circ}\text{C}$  (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.292 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.008 \text{ m}$ ,  $S_L = S_T = 0.015 \text{ m}$ , and  $V = 4 \text{ m/s}$ .

Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$\begin{aligned} V_{\max} &= \frac{S_T}{S_T - D} V = \frac{0.015}{0.015 - 0.008} (4 \text{ m/s}) = 8.571 \text{ m/s} \\ \text{Re}_D &= \frac{\rho V_{\max} D}{\mu} = \frac{(1.316 \text{ kg/m}^3)(8.571 \text{ m/s})(0.008 \text{ m})}{1.705 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 5294 \end{aligned}$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.27 \text{Re}_D^{0.63} \text{Pr}^{0.36} (\text{Pr}/\text{Pr}_s)^{0.25} \\ &= 0.27(5294)^{0.63} (0.7375)^{0.36} (0.7375/0.7408)^{0.25} = 53.61 \end{aligned}$$

Since  $N_L > 16$ , the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = F \text{Nu}_D = 53.61$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{53.61(0.02326 \text{ W/m}\cdot^{\circ}\text{C})}{0.008 \text{ m}} = 155.8 \text{ W/m}^2 \cdot ^{\circ}\text{C}$$

The total number of tubes is  $N = N_L \times N_T = 30 \times 15 = 450$ . The heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 300\pi(0.008 \text{ m})(0.4 \text{ m}) = 4.524 \text{ m}^2$$

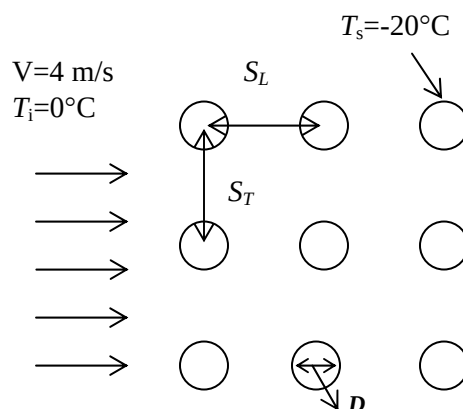
$$\dot{m} = \dot{m}_i = \rho_i V (N_T S_T L) = (1.292 \text{ kg/m}^3)(4 \text{ m/s})(15)(0.015 \text{ m})(0.4 \text{ m}) = 0.4651 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer (refrigeration capacity) become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = -20 - (-20 - 0) \exp\left(-\frac{(4.524 \text{ m}^2)(155.8 \text{ W/m}^2 \cdot ^{\circ}\text{C})}{(0.4651 \text{ kg/s})(1006 \text{ J/kg}\cdot^{\circ}\text{C})}\right) = -15.57^{\circ}\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(-20 - 0) - [-20 - (-15.57)]}{\ln[(-20 - 0)/(-20 + 15.57)]} = 10.33^{\circ}\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (155.8 \text{ W/m}^2 \cdot ^{\circ}\text{C})(4.524 \text{ m}^2)(10.33^{\circ}\text{C}) = \mathbf{7285 \text{ W}}$$



For this square in-line tube bank, the friction coefficient corresponding to  $Re_D = 5294$  and  $S_L/D = 1.5/0.8 = 1.875$  is, from Fig. 7-27a,  $f = 0.27$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 30(0.27)(1) \frac{(1.316 \text{ kg/m}^3)(8.571 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{391.6 \text{ Pa}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (0 - 15.6)/2 = -7.8^\circ\text{C}$ , which is fairly close to the assumed value of  $-5^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-71** Air is cooled by an evaporating refrigerator. The refrigeration capacity and the pressure drop across the tube bank are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is equal to the temperature of refrigerant.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of  $-5^{\circ}\text{C}$  (will be checked later) and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02326 \text{ W/m}\cdot\text{K} & \rho &= 1.316 \text{ kg/m}^3 \\ C_p &= 1.006 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7375 \\ \mu &= 1.705 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7408 \end{aligned}$$

Also, the density of air at the inlet temperature of  $0^{\circ}\text{C}$  (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.292 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.008 \text{ m}$ ,  $S_L = S_T = 0.015 \text{ m}$ , and  $V = 4 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$\begin{aligned} V_{\max} &= \frac{S_T}{S_T - D} V = \frac{0.015}{0.015 - 0.008} (4 \text{ m/s}) = 8.571 \text{ m/s} \\ \text{Re}_D &= \frac{\rho V_{\max} D}{\mu} = \frac{(1.316 \text{ kg/m}^3)(8.571 \text{ m/s})(0.008 \text{ m})}{1.705 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 5294 \end{aligned}$$

The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} \text{Nu}_D &= 0.35(S_T / S_L)^{0.2} \text{Re}_D^{0.6} \text{Pr}^{0.36} (\text{Pr} / \text{Pr}_s)^{0.25} \\ &= 0.35(0.015 / 0.015)^{0.2} (5294)^{0.6} (0.7375)^{0.36} (0.7375 / 0.7408)^{0.25} = 53.73 \end{aligned}$$

Since  $N_L > 16$ , the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$\text{Nu}_{D,N_L} = F \text{Nu}_D = 53.73$$

$$h = \frac{\text{Nu}_{D,N_L} k}{D} = \frac{53.73(0.02326 \text{ W/m}\cdot^{\circ}\text{C})}{0.008 \text{ m}} = 156.2 \text{ W/m}^2 \cdot^{\circ}\text{C}$$

The total number of tubes is  $N = N_L \times N_T = 30 \times 15 = 450$ . The heat transfer surface area and the mass flow rate of air (evaluated at the inlet) are

$$A_s = N\pi DL = 300\pi(0.008 \text{ m})(0.4 \text{ m}) = 4.524 \text{ m}^2$$

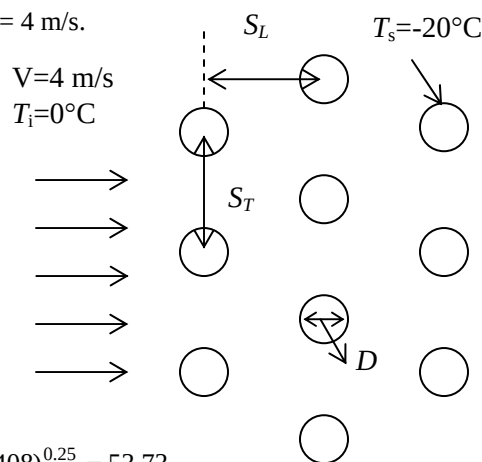
$$\dot{m} = \dot{m}_i = \rho_i V(N_T S_T L) = (1.292 \text{ kg/m}^3)(4 \text{ m/s})(15)(0.015 \text{ m})(0.4 \text{ m}) = 0.4651 \text{ kg/s}$$

Then the fluid exit temperature, the log mean temperature difference, and the rate of heat transfer (refrigeration capacity) become

$$T_e = T_s - (T_s - T_i) \exp\left(-\frac{A_s h}{\dot{m} C_p}\right) = -20 - (-20 - 0) \exp\left(-\frac{(4.524 \text{ m}^2)(156.2 \text{ W/m}^2 \cdot^{\circ}\text{C})}{(0.4651 \text{ kg/s})(1006 \text{ J/kg}\cdot^{\circ}\text{C})}\right) = -15.58^{\circ}\text{C}$$

$$\Delta T_{\ln} = \frac{(T_s - T_i) - (T_s - T_e)}{\ln[(T_s - T_i)/(T_s - T_e)]} = \frac{(-20 - 0) - [-20 - (-15.58)]}{\ln[(-20 - 0)/(-20 + 15.58)]} = 10.32^{\circ}\text{C}$$

$$\dot{Q} = h A_s \Delta T_{\ln} = (156.2 \text{ W/m}^2 \cdot^{\circ}\text{C})(4.524 \text{ m}^2)(10.32^{\circ}\text{C}) = \mathbf{7294 \text{ W}}$$



For this staggered arrangement tube bank, the friction coefficient corresponding to  $Re_D = 5294$  and  $S_L/D = 1.5/0.8 = 1.875$  is, from Fig. 7-27b,  $f = 0.44$ . Also,  $\chi = 1$  for the square arrangements. Then the pressure drop across the tube bank becomes

$$\Delta P = N_L f \chi \frac{\rho V_{\max}^2}{2} = 30(0.44)(1) \frac{(1.316 \text{ kg/m}^3)(8.571 \text{ m/s})^2}{2} \left( \frac{1 \text{ N}}{1 \text{ kg} \cdot \text{m/s}^2} \right) = \mathbf{638.2 \text{ Pa}}$$

**Discussion** The arithmetic mean fluid temperature is  $(T_i + T_e)/2 = (0 - 15.6)/2 = -7.8^\circ\text{C}$ , which is fairly close to the assumed value of  $-5^\circ\text{C}$ . Therefore, there is no need to repeat calculations.

**7-72** Air is heated by hot tubes in a tube bank. The average heat transfer coefficient is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The surface temperature of the tubes is constant.

**Properties** The exit temperature of air, and thus the mean temperature, is not known. We evaluate the air properties at the assumed mean temperature of  $70^\circ\text{C}$  and 1 atm (Table A-15):

$$\begin{aligned} k &= 0.02881 \text{ W/m}\cdot\text{K} & \rho &= 1.028 \text{ kg/m}^3 \\ C_p &= 1.007 \text{ kJ/kg}\cdot\text{K} & \text{Pr} &= 0.7177 \\ \mu &= 2.052 \times 10^{-5} \text{ kg/m}\cdot\text{s} & \text{Pr}_s &= \text{Pr}_{@T_s} = 0.7041 \end{aligned}$$

Also, the density of air at the inlet temperature of  $40^\circ\text{C}$  (for use in the mass flow rate calculation at the inlet) is  $\rho_i = 1.127 \text{ kg/m}^3$ .

**Analysis** It is given that  $D = 0.02 \text{ m}$ ,  $S_L = S_T = 0.06 \text{ m}$ , and  $V = 7 \text{ m/s}$ . Then the maximum velocity and the Reynolds number based on the maximum velocity become

$$\begin{aligned} V_{\max} &= \frac{S_T}{S_T - D} V = \frac{0.06}{0.06 - 0.02} (7 \text{ m/s}) = 10.50 \text{ m/s} \\ Re_D &= \frac{\rho V_{\max} D}{\mu} = \frac{(1.028 \text{ kg/m}^3)(10.50 \text{ m/s})(0.02 \text{ m})}{2.052 \times 10^{-5} \text{ kg/m}\cdot\text{s}} = 10,524 \end{aligned}$$

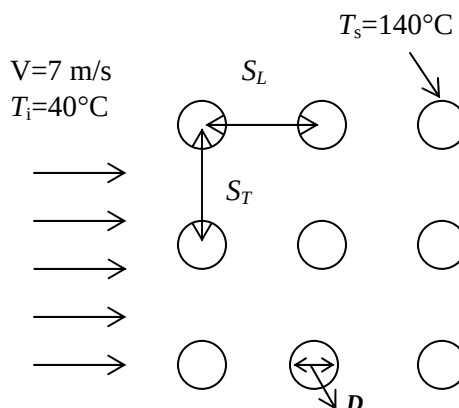
The average Nusselt number is determined using the proper relation from Table 7-2 to be

$$\begin{aligned} Nu_D &= 0.27 Re_D^{0.63} Pr^{0.36} (Pr/Pr_s)^{0.25} \\ &= 0.27(10,524)^{0.63} (0.7177)^{0.36} (0.7177/0.7041)^{0.25} = 82.33 \end{aligned}$$

Since  $N_L > 16$ , the average Nusselt number and heat transfer coefficient for all the tubes in the tube bank become

$$Nu_{D,N_L} = Nu_D = 82.33$$

$$h = \frac{Nu_{D,N_L} k}{D} = \frac{82.33(0.02881 \text{ W/m}\cdot^\circ\text{C})}{0.02 \text{ m}} = \mathbf{118.6 \text{ W/m}^2 \cdot ^\circ\text{C}}$$



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**Special Topic: Thermal Insulation**

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**7-73C** Thermal insulation is a material that is used primarily to provide resistance to heat flow. It differs from other kinds of insulators in that the purpose of an electrical insulator is to halt the flow of electric current, and the purpose of a sound insulator is to slow down the propagation of sound waves.

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**7-74C** In *cold surfaces* such as chilled water lines, refrigerated trucks, and air conditioning ducts, insulation saves energy since the source of “coldness” is *refrigeration* that requires energy input. In this case heat is transferred from the surroundings to the cold surfaces, and the refrigeration unit must now work harder and longer to make up for this heat gain and thus it must consume more electrical energy.

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**7-75C** The  $R$ -value of insulation is the *thermal resistance* of the insulating material *per unit surface area*. For *flat insulation* the  $R$ -value is obtained by simply dividing the thickness of the insulation by its thermal conductivity. That is,  $R\text{-value} = L/k$ . Doubling the thickness  $L$  doubles the  $R$ -value of flat insulation.

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**7-76C** The  $R$ -value of an insulation represents the thermal resistance of insulation *per unit surface area* (or per unit length in the case of pipe insulation).

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**7-77C** Superinsulations are obtained by using layers of highly reflective sheets separated by glass fibers in an evacuated space. Radiation between two surfaces is inversely proportional to the number of sheets used and thus heat loss by radiation will be very low by using this highly reflective sheets. Evacuating the space between the layers forms a vacuum which minimize conduction or convection through the air space.

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**7-78C** Yes, hair or any other cover reduces heat loss from the head, and thus serves as insulation for the head. The insulating ability of hair or feathers is most visible in birds and hairy animals.

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**7-79C** The primary reasons for insulating are energy conservation, personnel protection and comfort, maintaining process temperature, reducing temperature variation and fluctuations, condensation and corrosion prevention, fire protection, freezing protection, and reducing noise and vibration.

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**7-80C** The *optimum* thickness of insulation is the thickness that corresponds to a minimum combined cost of insulation and heat lost. The cost of insulation increases roughly linearly with thickness while the cost of heat lost decreases exponentially. The total cost, which is the sum of the two, decreases first, reaches a minimum, and then increases. The thickness that corresponds to the minimum total cost is the optimum thickness of insulation, and this is the recommended thickness of insulation to be installed.

**7-81** The thickness of flat *R*-8 insulation in SI units is to be determined when the thermal conductivity of the material is known.

**Assumptions** Thermal properties are constant.

**Properties** The thermal conductivity of the insulating material is given to be  $k = 0.04 \text{ W/m}\cdot^\circ\text{C}$ .

**Analysis** The thickness of flat *R*-8 insulation (in  $\text{m}^2\cdot^\circ\text{C/W}$ ) is determined from the definition of *R*-value to be

$$R_{\text{value}} = \frac{L}{k} \rightarrow L = R_{\text{value}} k = (8 \text{ m}^2\cdot^\circ\text{C/W})(0.04 \text{ W/m}\cdot^\circ\text{C}) = \mathbf{0.32 \text{ m}}$$



**7-82E** The thickness of flat *R*-20 insulation in English units is to be determined when the thermal conductivity of the material is known.

**Assumptions** Thermal properties are constant.

**Properties** The thermal conductivity of the insulating material is given to be  $k = 0.02 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$ .

**Analysis** The thickness of flat *R*-20 insulation (in  $\text{h}\cdot\text{ft}^2\cdot^\circ\text{F/Btu}$ ) is determined from the definition of *R*-value to be

$$R_{\text{value}} = \frac{L}{k} \rightarrow L = R_{\text{value}} k = (20 \text{ h}\cdot\text{ft}^2\cdot^\circ\text{F/Btu})(0.02 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}) = \mathbf{0.4 \text{ ft}}$$



**7-83** A steam pipe is to be covered with enough insulation to reduce the exposed surface temperature to 30°C. The thickness of insulation that needs to be installed is to be determined.

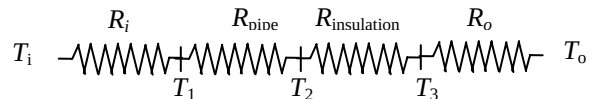
**Assumptions** 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. 3 Thermal properties are constant. 4 The thermal contact resistance at the interface is negligible.

**Properties** The thermal conductivities are given to be  $k = 52 \text{ W/m}\cdot^\circ\text{C}$  for cast iron pipe and  $k = 0.038 \text{ W/m}\cdot^\circ\text{C}$  for fiberglass insulation.

**Analysis** The thermal resistance network for this problem involves 4 resistances in series. The inner radius of the pipe is  $r_1 = 2.0 \text{ cm}$  and the outer radius of the pipe and thus the inner radius of insulation is  $r_2 = 2.3 \text{ cm}$ . Letting  $r_3$  represent the outer radius of insulation, the areas of the surfaces exposed to convection for a  $L = 1 \text{ m}$  long section of the pipe become

$$A_1 = 2\pi r_1 L = 2\pi(0.02 \text{ m})(1 \text{ m}) = 0.1257 \text{ m}^2$$

$$A_3 = 2\pi r_3 L = 2\pi r_3 (1 \text{ m}) = 2\pi r_3 \text{ m}^2 \quad (r_3 \text{ in m})$$



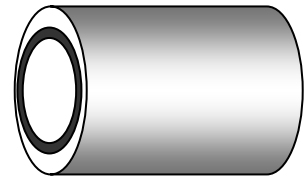
Then the individual thermal resistances are determined to be

$$R_i = R_{\text{conv},1} = \frac{1}{h_i A_1} = \frac{1}{(80 \text{ W/m}^2 \cdot ^\circ\text{C})(0.1257 \text{ m}^2)} = 0.9944 \text{ }^\circ\text{C/W}$$

$$R_1 = R_{\text{pipe}} = \frac{\ln(r_2 / r_1)}{2\pi k_1 L} = \frac{\ln(0.023 / 0.02)}{2\pi(52 \text{ W/m}\cdot^\circ\text{C})(1 \text{ m})} = 0.00043 \text{ }^\circ\text{C/W}$$

$$R_2 = R_{\text{insulation}} = \frac{\ln(r_3 / r_2)}{2\pi k_2 L} = \frac{\ln(r_3 / 0.023)}{2\pi(0.038 \text{ W/m}\cdot^\circ\text{C})(1 \text{ m})} = 4.188 \ln(r_3 / 0.023) \text{ }^\circ\text{C/W}$$

$$R_o = R_{\text{conv},2} = \frac{1}{h_o A_3} = \frac{1}{(22 \text{ W/m}^2 \cdot ^\circ\text{C})(2\pi r_3 \text{ m}^2)} = \frac{1}{138.2 r_3} \text{ }^\circ\text{C/W}$$



Noting that all resistances are in series, the total resistance is

$$R_{\text{total}} = R_i + R_1 + R_2 + R_o = 0.9944 + 0.00043 + 4.188 \ln(r_3 / 0.023) + 1/(138.2 r_3) \text{ }^\circ\text{C/W}$$

Then the steady rate of heat loss from the steam becomes

$$\dot{Q} = \frac{T_i - T_o}{R_{\text{total}}} = \frac{(110 - 22)^\circ\text{C}}{[0.9944 + 0.00043 + 4.188 \ln(r_3 / 0.023) + 1/(138.2 r_3)]^\circ\text{C/W}}$$

Noting that the outer surface temperature of insulation is specified to be 30°C, the rate of heat loss can also be expressed as

$$\dot{Q} = \frac{T_3 - T_o}{R_o} = \frac{(30 - 22)^\circ\text{C}}{1/(138.2 r_3)^\circ\text{C/W}} = 1106 r_3$$

Setting the two relations above equal to each other and solving for  $r_3$  gives  $r_3 = 0.0362 \text{ m}$ . Then the minimum thickness of fiberglass insulation required is

$$t = r_3 - r_2 = 0.0362 - 0.0230 = 0.0132 \text{ m} = \mathbf{1.32 \text{ cm}}$$

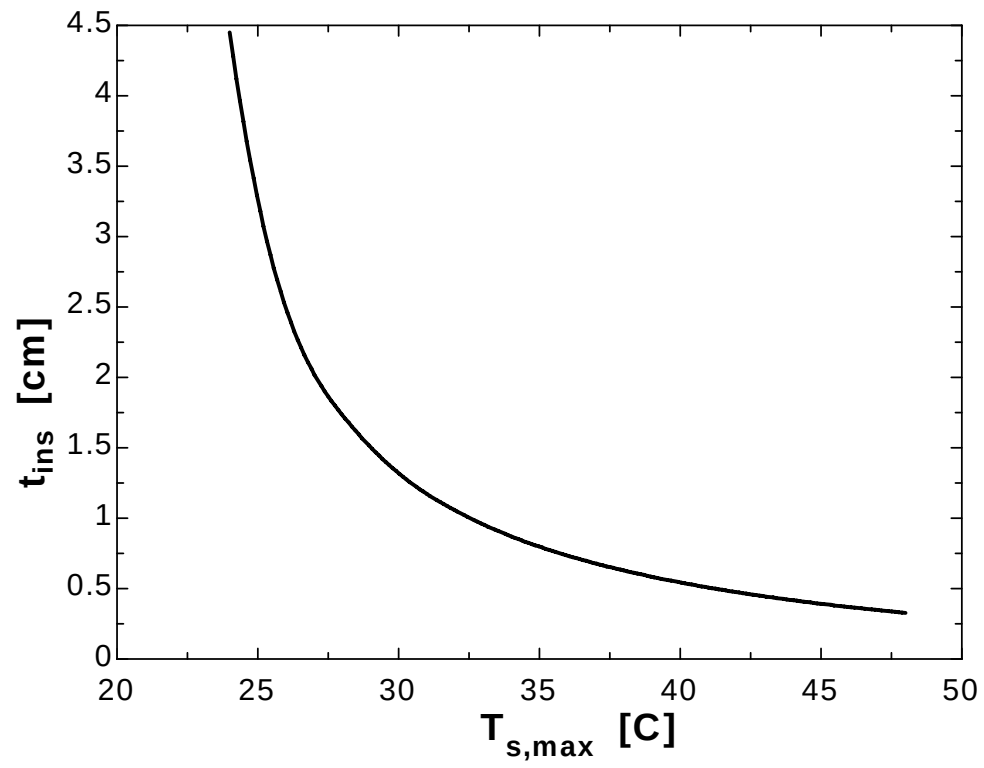
Therefore, insulating the pipe with at least 1.32 cm thick fiberglass insulation will ensure that the outer surface temperature of the pipe will be at 30°C or below.



## 7-84 "PROBLEM 7-84"

**"GIVEN"** $T_i = 110$  "[C]" $T_o = 22$  "[C]" $k_{\text{pipe}} = 52$  "[W/m-C]" $r_1 = 0.02$  "[m]" $t_{\text{pipe}} = 0.003$  "[m]" $T_{s,\text{max}} = 30$  [C], parameter to be varied $h_i = 80$  "[W/m<sup>2</sup>-C]" $h_o = 22$  "[W/m<sup>2</sup>-C]" $k_{\text{ins}} = 0.038$  "[W/m-C]"**"ANALYSIS"** $L = 1$  "[m], 1 m long section of the pipe is considered" $A_i = 2\pi r_1 L$  $A_o = 2\pi r_3 L$  $r_3 = r_2 + t_{\text{ins}}$  Convert(cm, m) " $t_{\text{ins}}$  is in cm" $r_2 = r_1 + t_{\text{pipe}}$  $R_{\text{conv}_i} = 1/(h_i A_i)$  $R_{\text{pipe}} = \ln(r_2/r_1)/(2\pi k_{\text{pipe}} L)$  $R_{\text{ins}} = \ln(r_3/r_2)/(2\pi k_{\text{ins}} L)$  $R_{\text{conv}_o} = 1/(h_o A_o)$  $R_{\text{total}} = R_{\text{conv}_i} + R_{\text{pipe}} + R_{\text{ins}} + R_{\text{conv}_o}$  $\dot{Q} = (T_i - T_o)/R_{\text{total}}$  $\dot{Q} = (T_{s,\text{max}} - T_o)/R_{\text{conv}_o}$ 

| $T_{s,\text{max}}$ [C] | $t_{\text{ins}}$ [cm] |
|------------------------|-----------------------|
| 24                     | 4.45                  |
| 26                     | 2.489                 |
| 28                     | 1.733                 |
| 30                     | 1.319                 |
| 32                     | 1.055                 |
| 34                     | 0.871                 |
| 36                     | 0.7342                |
| 38                     | 0.6285                |
| 40                     | 0.5441                |
| 42                     | 0.4751                |
| 44                     | 0.4176                |
| 46                     | 0.3688                |
| 48                     | 0.327                 |



**7-85** A cylindrical oven is to be insulated to reduce heat losses. The optimum thickness of insulation and the amount of money saved per year are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the insulation is one-dimensional. 3 Thermal conductivities are constant. 4 The thermal contact resistance at the interface is negligible. 5 The surface temperature of the furnace and the heat transfer coefficient remain constant. 6 The surfaces of the cylindrical oven can be treated as plain surfaces since its diameter is greater than 1 m.

**Properties** The thermal conductivity of insulation is given to be  $k = 0.038 \text{ W/m}\cdot^\circ\text{C}$ .

**Analysis** We treat the surfaces of this cylindrical furnace as plain surfaces since its diameter is greater than 1 m, and disregard the curvature effects. The exposed surface area of the furnace is

$$A_o = 2A_{\text{base}} + A_{\text{side}} = 2\pi r^2 + 2\pi rL = 2\pi(1.5 \text{ m})^2 + 2\pi(1.5 \text{ m})(6 \text{ m}) = 70.69 \text{ m}^2$$

The rate of heat loss from the furnace before the insulation is installed is

$$\dot{Q} = h_o A_o (T_s - T_\infty) = (30 \text{ W/m}^2 \cdot ^\circ\text{C})(70.69 \text{ m}^2)(90 - 27)^\circ\text{C} = 133,600 \text{ W}$$

Noting that the plant operates  $52 \times 80 = 4160 \text{ h/yr}$ , the annual heat lost from the furnace is

$$Q = \dot{Q}\Delta t = (133.6 \text{ kJ/s})(4160 \times 3600 \text{ s/yr}) = 2.001 \times 10^9 \text{ kJ/yr}$$

The efficiency of the furnace is given to be 78 percent. Therefore, to generate this much heat, the furnace must consume energy (in the form of natural gas) at a rate of

$$Q_{\text{in}} = Q / \eta_{\text{oven}} = (2.001 \times 10^9 \text{ kJ/yr}) / 0.78 = 2.565 \times 10^9 \text{ kJ/yr} = 24,314 \text{ therms/yr}$$

since 1 therm = 105,500 kJ. Then the annual fuel cost of this furnace before insulation becomes

$$\text{Annual Cost} = Q_{\text{in}} \times \text{Unit cost} = (24,314 \text{ therm/yr})(\$0.50/\text{therm}) = \$12,157/\text{yr}$$

We expect the surface temperature of the furnace to increase, and the heat transfer coefficient to decrease somewhat when insulation is installed. We assume these two effects to counteract each other. Then the rate of heat loss for 1-cm thick insulation becomes

$$\dot{Q}_{\text{ins}} = \frac{T_s - T_\infty}{R_{\text{total}}} = \frac{T_s - T_\infty}{R_{\text{ins}} + R_{\text{conv}}} = \frac{A_o(T_s - T_\infty)}{\frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_o}} = \frac{(70.69 \text{ m}^2)(90 - 27)^\circ\text{C}}{\frac{0.01 \text{ m}}{0.038 \text{ W/m}\cdot^\circ\text{C}} + \frac{1}{30 \text{ W/m}^2 \cdot ^\circ\text{C}}} = 15,021 \text{ W}$$

Also, the total amount of heat loss from the furnace per year and the amount and cost of energy consumption of the furnace become

$$Q_{\text{ins}} = \dot{Q}_{\text{ins}}\Delta t = (15.021 \text{ kJ/s})(4160 \times 3600 \text{ s/yr}) = 2.249 \times 10^8 \text{ kJ/yr}$$

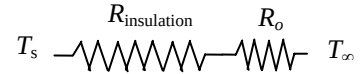
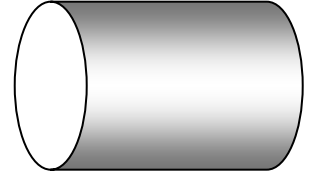
$$Q_{\text{in,ins}} = Q_{\text{ins}} / \eta_{\text{oven}} = (2.249 \times 10^8 \text{ kJ/yr}) / 0.78 = 2.884 \times 10^8 \text{ kJ/yr} = 2734 \text{ therms}$$

$$\text{Annual Cost} = Q_{\text{in,ins}} \times \text{Unit cost} = (2734 \text{ therm/yr})(\$0.50/\text{therm}) = \$1367/\text{yr}$$

$$\text{Cost savings} = \text{Energy cost w/o insulation} - \text{Energy cost w/insulation} = 12,157 - 1367 = \$10,790/\text{yr}$$

The unit cost of insulation is given to be  $\$10/\text{m}^2$  per cm thickness, plus  $\$30/\text{m}^2$  for labor. Then the total cost of insulation becomes

$$\text{Insulation Cost} = (\text{Unit cost})(\text{Surface area}) = [(\$10/\text{cm})(1 \text{ cm}) + \$30/\text{m}^2](70.69 \text{ m}^2) = \$2828$$



To determine the thickness of insulation whose cost is equal to annual energy savings, we repeat the calculations above for 2, 3, . . . 15 cm thick insulations, and list the results in the table below.

| Insulation thickness | Rate of heat loss<br>W | Cost of heat lost<br>\$/yr | Cost savings<br>\$/yr | Insulation cost<br>\$ |
|----------------------|------------------------|----------------------------|-----------------------|-----------------------|
| 0 cm                 | 133,600                | 12,157                     | 0                     | 0                     |
| 1 cm                 | 15,021                 | 1367                       | 10,790                | 2828                  |
| 5 cm                 | 3301                   | 300                        | 11,850                | 3535                  |
| 10 cm                | 1671                   | 152                        | 12,005                | 9189                  |
| 11 cm                | 1521                   | 138                        | 12,019                | 9897                  |
| 12 cm                | 1396                   | 127                        | 12,030                | 10,604                |
| 13 cm                | 1289                   | 117                        | 12,040                | 11,310                |
| 14 cm                | 1198                   | 109                        | 12,048                | 12,017                |
| 15 cm                | 1119                   | 102                        | 12,055                | 12,724                |

Therefore, the thickest insulation that will pay for itself in one year is the one whose thickness is **14 cm**.

**7-86** A cylindrical oven is to be insulated to reduce heat losses. The optimum thickness of insulation and the amount of money saved per year are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Heat transfer through the insulation is one-dimensional. 3 Thermal conductivities are constant. 4 The thermal contact resistance at the interface is negligible. 5 The surface temperature of the furnace and the heat transfer coefficient remain constant. 6 The surfaces of the cylindrical oven can be treated as plain surfaces since its diameter is greater than 1 m.

**Properties** The thermal conductivity of insulation is given to be  $k = 0.038 \text{ W/m}\cdot^\circ\text{C}$ .

**Analysis** We treat the surfaces of this cylindrical furnace as plain surfaces since its diameter is greater than 1 m, and disregard the curvature effects. The exposed surface area of the furnace is

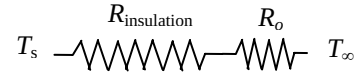
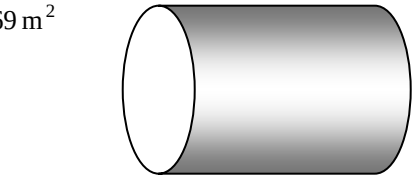
$$A_o = 2A_{\text{base}} + A_{\text{side}} = 2\pi r^2 + 2\pi rL = 2\pi(1.5 \text{ m})^2 + 2\pi(1.5 \text{ m})(6 \text{ m}) = 70.69 \text{ m}^2$$

The rate of heat loss from the furnace before the insulation is installed is

$$\dot{Q} = h_o A_o (T_s - T_\infty) = (30 \text{ W/m}^2 \cdot ^\circ\text{C})(70.69 \text{ m}^2)(75 - 27)^\circ\text{C} = 101,794 \text{ W}$$

Noting that the plant operates  $52 \times 80 = 4160 \text{ h/yr}$ , the annual heat lost from the furnace is

$$Q = \dot{Q}\Delta t = (101,794 \text{ kJ/s})(4160 \times 3600 \text{ s/yr}) = 1.524 \times 10^9 \text{ kJ/yr}$$



The efficiency of the furnace is given to be 78 percent. Therefore, to generate this much heat, the furnace must consume energy (in the form of natural gas) at a rate of

$$Q_{\text{in}} = Q / \eta_{\text{oven}} = (1.524 \times 10^9 \text{ kJ/yr}) / 0.78 = 1.954 \times 10^9 \text{ kJ/yr} = 18,526 \text{ therms/yr}$$

since 1 therm = 105,500 kJ. Then the annual fuel cost of this furnace before insulation becomes

$$\text{Annual Cost} = Q_{\text{in}} \times \text{Unit cost} = (18,526 \text{ therm/yr})(\$0.50/\text{therm}) = \$9,263/\text{yr}$$

We expect the surface temperature of the furnace to increase, and the heat transfer coefficient to decrease somewhat when insulation is installed. We assume these two effects to counteract each other. Then the rate of heat loss for 1-cm thick insulation becomes

$$\dot{Q}_{\text{ins}} = \frac{T_s - T_\infty}{R_{\text{total}}} = \frac{T_s - T_\infty}{R_{\text{ins}} + R_{\text{conv}}} = \frac{A_o(T_s - T_\infty)}{\frac{t_{\text{ins}}}{k_{\text{ins}}} + \frac{1}{h_o}} = \frac{(70.69 \text{ m}^2)(75 - 27)^\circ\text{C}}{\frac{0.01 \text{ m}}{0.038 \text{ W/m}\cdot^\circ\text{C}} + \frac{1}{30 \text{ W/m}^2 \cdot ^\circ\text{C}}} = 11,445 \text{ W}$$

Also, the total amount of heat loss from the furnace per year and the amount and cost of energy consumption of the furnace become

$$Q_{\text{ins}} = \dot{Q}_{\text{ins}}\Delta t = (11,445 \text{ kJ/s})(4160 \times 3600 \text{ s/yr}) = 1.714 \times 10^8 \text{ kJ/yr}$$

$$Q_{\text{in,ins}} = Q_{\text{ins}} / \eta_{\text{oven}} = (1.714 \times 10^8 \text{ kJ/yr}) / 0.78 = 2.197 \times 10^8 \text{ kJ/yr} = 2082 \text{ therms}$$

$$\text{Annual Cost} = Q_{\text{in,ins}} \times \text{Unit cost} = (2082 \text{ therm/yr})(\$0.50/\text{therm}) = \$1041/\text{yr}$$

$$\text{Cost savings} = \text{Energy cost w/o insulation} - \text{Energy cost w/insulation} = 9263 - 1041 = \$8222/\text{yr}$$

The unit cost of insulation is given to be  $\$10/\text{m}^2$  per cm thickness, plus  $\$30/\text{m}^2$  for labor. Then the total cost of insulation becomes

$$\text{Insulation Cost} = (\text{Unit cost})(\text{Surface area}) = [(\$10/\text{cm})(1 \text{ cm}) + \$30/\text{m}^2](70.69 \text{ m}^2) = \$2828$$

To determine the thickness of insulation whose cost is equal to annual energy savings, we repeat the calculations above for 2, 3, . . . . 15 cm thick insulations, and list the results in the table below.

| Insulation Thickness | Rate of heat loss W | Cost of heat lost \$/yr | Cost savings \$/yr | Insulation cost \$ |
|----------------------|---------------------|-------------------------|--------------------|--------------------|
| 0 cm                 | 101,794             | 9263                    | 0                  | 0                  |
| 1 cm                 | 11,445              | 1041                    | 8222               | 2828               |
| 5 cm                 | 2515                | 228                     | 9035               | 3535               |
| 9 cm                 | 1413                | 129                     | 9134               | 8483               |
| 10 cm                | 1273                | 116                     | 9147               | 9189               |
| 11 cm                | 1159                | 105                     | 9158               | 9897               |
| 12 cm                | 1064                | 97                      | 9166               | 10,604             |

Therefore, the thickest insulation that will pay for itself in one year is the one whose thickness is **9 cm**. The 10-cm thick insulation will come very close to paying for itself in one year.

**7-87E** Steam is flowing through an insulated steel pipe, and it is proposed to add another 1-in thick layer of fiberglass insulation on top of the existing one to reduce the heat losses further and to save energy and money. It is to be determined if the new insulation will pay for itself within 2 years.

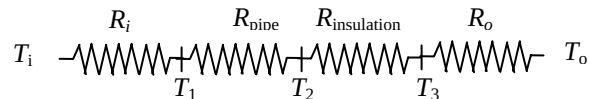
**Assumptions** 1 Heat transfer is steady since there is no indication of any change with time. 2 Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. 3 Thermal properties are constant. 4 The heat transfer coefficients remain constant. 5 The thermal contact resistance at the interface is negligible.

**Properties** The thermal conductivities are given to be  $k = 8.7 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$  for steel pipe and  $k = 0.020 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$  for fiberglass insulation.

**Analysis** The inner radius of the pipe is  $r_1 = 1.75 \text{ in}$ , the outer radius of the pipe is  $r_2 = 2 \text{ in}$ , and the outer radii of the existing and proposed insulation layers are  $r_3 = 3 \text{ in}$  and  $4 \text{ in}$ , respectively. Considering a unit pipe length of  $L = 1 \text{ ft}$ , the individual thermal resistances are determined to be

$$R_i = R_{\text{conv},1} = \frac{1}{h_i A_1} = \frac{1}{h_i (2\pi r_1 L)} = \frac{1}{(30 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})[2\pi(1.75/12 \text{ ft})(1 \text{ ft})]} = 0.0364 \text{ h}\cdot^\circ\text{F/Btu}$$

$$R_1 = R_{\text{pipe}} = \frac{\ln(r_2/r_1)}{2\pi k_1 L} = \frac{\ln(2/1.75)}{2\pi(8.7 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(1 \text{ ft})} = 0.00244 \text{ h}\cdot^\circ\text{F/Btu}$$



**Current Case:**

$$R_{\text{insulation}} = \frac{\ln(r_3/r_2)}{2\pi k_{\text{ins}} L} = \frac{\ln(3/2)}{2\pi(0.020 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(1 \text{ ft})} = 3.227 \text{ h}\cdot^\circ\text{F/Btu}$$

$$R_o = R_{\text{conv},2} = \frac{1}{h_o A_3} = \frac{1}{h_o (2\pi r_3)} = \frac{1}{(5 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})[2\pi(3/12 \text{ ft})(1 \text{ ft})]} = 0.1273 \text{ h}\cdot^\circ\text{F/Btu}$$

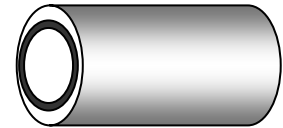
Then the steady rate of heat loss from the steam becomes

$$\dot{Q}_{\text{current}} = \frac{\Delta T}{R_{\text{total}}} = \frac{T_i - T_o}{R_i + R_{\text{pipe}} + R_{\text{ins}} + R_o} = \frac{(400 - 60)^\circ\text{F}}{(0.0364 + 0.00244 + 3.227 + 0.1273) \text{ h}\cdot^\circ\text{F/Btu}} = 100.2 \text{ Btu/h}$$

**Proposed Case:**

$$R_{\text{insulation}} = \frac{\ln(r_3/r_2)}{2\pi k_{\text{ins}} L} = \frac{\ln(4/2)}{2\pi(0.020 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F})(1 \text{ ft})} = 5.516 \text{ h}\cdot^\circ\text{F/Btu}$$

$$R_o = R_{\text{conv},2} = \frac{1}{h_o A_3} = \frac{1}{h_o (2\pi r_3)} = \frac{1}{(5 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})[2\pi(4/12 \text{ ft})(1 \text{ ft})]} = 0.0955 \text{ h}\cdot^\circ\text{F/Btu}$$



Then the steady rate of heat loss from the steam becomes

$$\dot{Q}_{\text{prop}} = \frac{\Delta T}{R_{\text{total}}} = \frac{T_i - T_o}{R_i + R_{\text{pipe}} + R_{\text{ins}} + R_o} = \frac{(400 - 60)^\circ\text{F}}{(0.0364 + 0.00244 + 5.516 + 0.0955) \text{ h}\cdot^\circ\text{F/Btu}} = 60.2 \text{ Btu/h}$$

Therefore, the amount of energy and money saved by the additional insulation per year are

$$\dot{Q}_{\text{saved}} = \dot{Q}_{\text{prop}} - \dot{Q}_{\text{current}} = 100.2 - 60.2 = 40.0 \text{ Btu/h}$$

$$Q_{\text{saved}} = \dot{Q}_{\text{saved}} \Delta t = (40.0 \text{ Btu/h})(8760 \text{ h/yr}) = 350,400 \text{ Btu/yr}$$

$$\text{Money saved} = Q_{\text{saved}} \times (\text{Unit cost}) = (350,400 \text{ Btu/yr})(\$0.01/1000 \text{ Btu}) = \$3.504/\text{yr}$$

or \$7.01 per 2 years, which is barely more than the \$7.0 minimum required. But the criteria is satisfied, and the proposed additional insulation is **justified**.

**7-88** The plumbing system of a plant involves some section of a plastic pipe exposed to the ambient air. The pipe is to be insulated with adequate fiber glass insulation to prevent freezing of water in the pipe. The thickness of insulation that will protect the water from freezing under worst conditions is to be determined.

**Assumptions** 1 Heat transfer is transient, but can be treated as steady at average conditions. 2 Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. 3 Thermal properties are constant. 4 The water in the pipe is stationary, and its initial temperature is 15°C. 5 The thermal contact resistance at the interface is negligible. 6 The convection resistance inside the pipe is negligible.

**Properties** The thermal conductivities are given to be  $k = 0.16 \text{ W/m}\cdot^\circ\text{C}$  for plastic pipe and  $k = 0.035 \text{ W/m}\cdot^\circ\text{C}$  for fiberglass insulation. The density and specific heat of water are to be  $\rho = 1000 \text{ kg/m}^3$  and  $C_p = 4.18 \text{ kJ/kg}\cdot^\circ\text{C}$  (Table A-15).

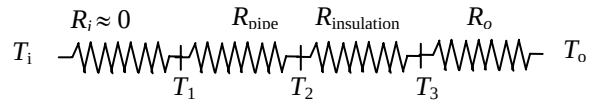
**Analysis** The inner radius of the pipe is  $r_1 = 3.0 \text{ cm}$  and the outer radius of the pipe and thus the inner radius of insulation is  $r_2 = 3.3 \text{ cm}$ . We let  $r_3$  represent the outer radius of insulation. Considering a 1-m section of the pipe, the amount of heat that must be transferred from the water as it cools from 15 to 0°C is determined to be

$$m = \rho V = \rho(\pi r_1^2 L) = (1000 \text{ kg/m}^3)[\pi(0.03 \text{ m})^2(1 \text{ m})] = 2.827 \text{ kg}$$

$$Q_{\text{total}} = m C_p \Delta T = (2.827 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(15 - 0)^\circ\text{C} = 177.3 \text{ kJ}$$

Then the average rate of heat transfer during 60 h becomes

$$\dot{Q}_{\text{ave}} = \frac{Q_{\text{total}}}{\Delta t} = \frac{177,300 \text{ J}}{(60 \times 3600 \text{ s})} = 0.821 \text{ W}$$

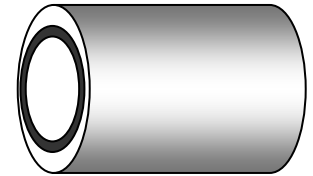


The individual thermal resistances are

$$R_1 = R_{\text{pipe}} = \frac{\ln(r_2 / r_1)}{2\pi k_{\text{pipe}} L} = \frac{\ln(0.033 / 0.03)}{2\pi(0.16 \text{ W / m}\cdot^\circ\text{C})(1 \text{ m})} = 0.0948 \text{ }^\circ\text{C / W}$$

$$R_{\text{insulation}} = \frac{\ln(r_3 / r_2)}{2\pi k_2 L} = \frac{\ln(r_3 / 0.033)}{2\pi(0.035 \text{ W / m}\cdot^\circ\text{C})(1 \text{ m})} = 4.55 \ln(r_3 / 0.033) \text{ }^\circ\text{C / W}$$

$$R_o = R_{\text{conv}} = \frac{1}{h_o A_3} = \frac{1}{(30 \text{ W / m}^2\cdot^\circ\text{C})(2\pi r_3 \text{ m}^2)} = \frac{1}{188.5 r_3} \text{ }^\circ\text{C / W}$$



Then the rate of average heat transfer from the water can be expressed as

$$\dot{Q} = \frac{T_{i,\text{ave}} - T_o}{R_{\text{total}}} \rightarrow 0.821 \text{ W} = \frac{[7.5 - (-10)]^\circ\text{C}}{[0.0948 + 4.55 \ln(r_3 / 0.033) + 1 / (188.5 r_3)]^\circ\text{C / W}} \rightarrow r_3 = 3.50 \text{ m}$$

Therefore, the minimum thickness of fiberglass needed to protect the pipe from freezing is

$$t = r_3 - r_2 = 3.50 - 0.033 = \mathbf{3.467 \text{ m}}$$

which is too large. Installing such a thick insulation is not practical, however, and thus other freeze protection methods should be considered.



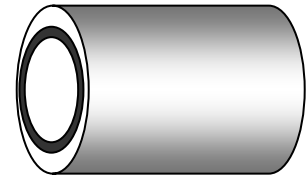
**7-89** The plumbing system of a plant involves some section of a plastic pipe exposed to the ambient air. The pipe is to be insulated with adequate fiber glass insulation to prevent freezing of water in the pipe. The thickness of insulation that will protect the water from freezing more than 20% under worst conditions is to be determined.

**Assumptions** 1 Heat transfer is transient, but can be treated as steady at average conditions. 2 Heat transfer is one-dimensional since there is thermal symmetry about the centerline and no variation in the axial direction. 3 Thermal properties are constant. 4 The water in the pipe is stationary, and its initial temperature is 15°C. 5 The thermal contact resistance at the interface is negligible. 6 The convection resistance inside the pipe is negligible.

**Properties** The thermal conductivities are given to be  $k = 0.16 \text{ W/m}\cdot^\circ\text{C}$  for plastic pipe and  $k = 0.035 \text{ W/m}\cdot^\circ\text{C}$  for fiberglass insulation. The density and specific heat of water are to be  $\rho = 1000 \text{ kg/m}^3$  and  $C_p = 4.18 \text{ kJ/kg}\cdot^\circ\text{C}$  (Table A-15).

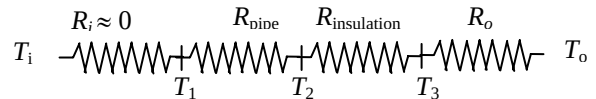
**Analysis** The inner radius of the pipe is  $r_1 = 3.0 \text{ cm}$  and the outer radius of the pipe and thus the inner radius of insulation is  $r_2 = 3.3 \text{ cm}$ . We let  $r_3$  represent the outer radius of insulation. The latent heat of freezing of water is  $33.7 \text{ kJ/kg}$ . Considering a 1-m section of the pipe, the amount of heat that must be transferred from the water as it cools from 15 to 0°C is determined to be

$$\begin{aligned} m &= \rho V = \rho(\pi r_1^2 L) = (1000 \text{ kg/m}^3)[\pi(0.03 \text{ m})^2 (1 \text{ m})] = 2.827 \text{ kg} \\ Q_{\text{total}} &= m C_p \Delta T = (2.827 \text{ kg})(4.18 \text{ kJ/kg}\cdot^\circ\text{C})(15 - 0)^\circ\text{C} = 177.3 \text{ kJ} \\ Q_{\text{freezing}} &= 0.2 \times m h_{if} = 0.2 \times (2.827 \text{ kg})(333.7 \text{ kJ/kg}) = 188.7 \text{ kJ} \\ Q_{\text{total}} &= Q_{\text{cooling}} + Q_{\text{freezing}} = 177.3 + 188.7 = 366.0 \text{ kJ} \end{aligned}$$



Then the average rate of heat transfer during 60 h becomes

$$\dot{Q}_{\text{ave}} = \frac{Q_{\text{total}}}{\Delta t} = \frac{366,000 \text{ J}}{(60 \times 3600 \text{ s})} = 1694 \text{ W}$$



The individual thermal resistances are

$$\begin{aligned} R_1 = R_{\text{pipe}} &= \frac{\ln(r_2 / r_1)}{2\pi k_{\text{pipe}} L} = \frac{\ln(0.033 / 0.03)}{2\pi(0.16 \text{ W/m}\cdot^\circ\text{C})(1 \text{ m})} = 0.0948 \text{ }^\circ\text{C/W} \\ R_{\text{insulation}} &= \frac{\ln(r_3 / r_2)}{2\pi k_2 L} = \frac{\ln(r_3 / 0.033)}{2\pi(0.035 \text{ W/m}\cdot^\circ\text{C})(1 \text{ m})} = 4.55 \ln(r_3 / 0.033) \text{ }^\circ\text{C/W} \\ R_o = R_{\text{conv}} &= \frac{1}{h_o A_3} = \frac{1}{(30 \text{ W/m}^2\cdot^\circ\text{C})(2\pi r_3 \text{ m}^2)} = \frac{1}{188.5 r_3} \text{ }^\circ\text{C/W} \end{aligned}$$

Then the rate of average heat transfer from the water can be expressed as

$$\dot{Q} = \frac{T_{i,\text{ave}} - T_o}{R_{\text{total}}} \rightarrow 1694 \text{ W} = \frac{[7.5 - (-10)]^\circ\text{C}}{[0.0948 + 4.55 \ln(r_3 / 0.033) + 1 / (188.5 r_3)]^\circ\text{C/W}} \rightarrow r_3 = 0.312 \text{ m}$$

Therefore, the minimum thickness of fiberglass needed to protect the pipe from freezing is

$$t = r_3 - r_2 = 0.312 - 0.033 = \mathbf{0.279 \text{ m}}$$

which is too large. Installing such a thick insulation is not practical, however, and thus other freeze protection methods should be considered.

## Review Problems

**7-90** Wind is blowing parallel to the walls of a house. The rate of heat loss from the wall is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties. 5 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of  $T_f = 10^\circ\text{C}$  for the outdoors, the properties of air are evaluated to be (Table A-15)

$$k = 0.02439 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7336$$

**Analysis** Air flows along 8-m side. The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(50 \times 1000 / 3600) \text{ m/s}](8 \text{ m})}{1.426 \times 10^{-5} \text{ m}^2/\text{s}} = 7.792 \times 10^6$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, heat transfer coefficient is determined to be

$$Nu = \frac{h_0 L}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(7.792 \times 10^6)^{0.8} - 871](0.7336)^{1/3} = 10,096$$

$$h_o = \frac{k}{L} Nu = \frac{0.02439 \text{ W/m} \cdot ^\circ\text{C}}{8 \text{ m}} (10,096) = 30.78 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The thermal resistances are

$$A_s = wL = (3 \text{ m})(8 \text{ m}) = 24 \text{ m}^2$$

$$R_i = \frac{1}{h_i A_s} = \frac{1}{(8 \text{ W/m}^2 \cdot ^\circ\text{C})(24 \text{ m}^2)} = 0.0052 \text{ } ^\circ\text{C/W}$$

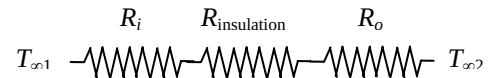
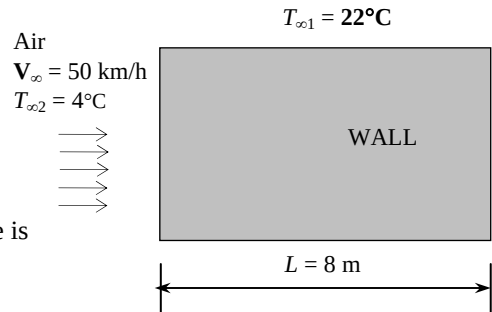
$$R_{insulation} = \frac{(R - 3.38)_{value}}{A_s} = \frac{3.38 \text{ m}^2 \cdot ^\circ\text{C/W}}{24 \text{ m}^2} = 0.1408 \text{ } ^\circ\text{C/W}$$

$$R_o = \frac{1}{h_o A_s} = \frac{1}{(30.78 \text{ W/m}^2 \cdot ^\circ\text{C})(24 \text{ m}^2)} = 0.0014 \text{ } ^\circ\text{C/W}$$

Then the total thermal resistance and the heat transfer rate through the wall are determined from

$$R_{total} = R_i + R_{insulation} + R_o = 0.0052 + 0.1408 + 0.0014 = 0.1474 \text{ } ^\circ\text{C/W}$$

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{total}} = \frac{(22 - 4)^\circ\text{C}}{0.1474 \text{ } ^\circ\text{C/W}} = \mathbf{122.1 \text{ W}}$$



**7-91** A car travels at a velocity of 60 km/h. The rate of heat transfer from the bottom surface of the hot automotive engine block is to be determined for two cases.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Air is an ideal gas with constant properties. 4 The pressure of air is 1 atm. 5 The flow is turbulent over the entire surface because of the constant agitation of the engine block. 6 The bottom surface of the engine is a flat surface.

**Properties** The properties of air at 1 atm and the film temperature of  $(T_s + T_\infty)/2 = (75+5)/2 = 40^\circ\text{C}$  are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7255$$

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(60 \times 1000 / 3600) \text{ m/s}](0.7 \text{ m})}{1.702 \times 10^{-5} \text{ m}^2/\text{s}} = 6.855 \times 10^5$$

which is less than the critical Reynolds number. But we will assume turbulent flow because of the constant agitation of the engine block.

$$Nu = \frac{hL}{k} = 0.037 Re_L^{0.8} Pr^{1/3} = 0.037(6.855 \times 10^5)^{0.8} (0.7255)^{1/3} = 1551$$

$$h = \frac{k}{L} Nu = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.7 \text{ m}} (1551) = 58.97 \text{ W/m}^2\cdot^\circ\text{C}$$

$$\dot{Q}_{conv} = hA_s(T_\infty - T_s) = (58.97 \text{ W/m}^2\cdot^\circ\text{C})[(0.6 \text{ m})(0.7 \text{ m})](75 - 5)^\circ\text{C} = 1734 \text{ W}$$

The heat loss by radiation is then determined from Stefan-Boltzman law to be

$$\begin{aligned} \dot{Q}_{rad} &= \varepsilon A_s \sigma (T_s^4 - T_{surr}^4) \\ &= (0.92)(0.6 \text{ m})(0.7 \text{ m})(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4)[(75 + 273 \text{ K})^4 - (10 + 273 \text{ K})^4] = 181 \text{ W} \end{aligned}$$

Then the total rate of heat loss from the bottom surface of the engine block becomes

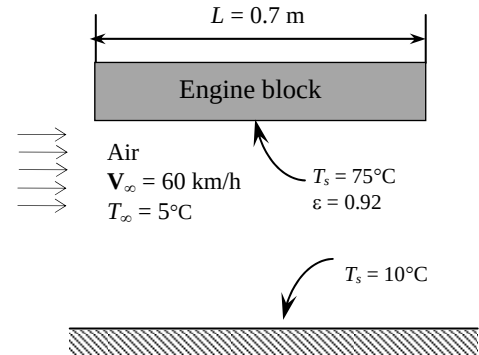
$$\dot{Q}_{total} = \dot{Q}_{conv} + \dot{Q}_{rad} = 1734 + 181 = \mathbf{1915 \text{ W}}$$

The gunk will introduce an additional resistance to heat dissipation from the engine. The total heat transfer rate in this case can be calculated from

$$\dot{Q} = \frac{T_\infty - T_s}{\frac{1}{hA_s} + \frac{L}{kA_s}} = \frac{(75 - 5)^\circ\text{C}}{\frac{1}{(58.97 \text{ W/m}^2\cdot^\circ\text{C})[(0.6 \text{ m})(0.7 \text{ m})]} + \frac{(0.002 \text{ m})}{(3 \text{ W/m}\cdot^\circ\text{C})(0.6 \text{ m} \times 0.7 \text{ m})}} = 1668 \text{ W}$$

The decrease in the heat transfer rate is

$$1734 - 1668 = \mathbf{66 \text{ W}}$$



**7-92E** A minivan is traveling at 60 mph. The rate of heat transfer to the van is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air flow is turbulent because of the intense vibrations involved. 5 Air is an ideal gas with constant properties. 6 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of  $T_f = 80^\circ\text{F}$ , the properties of air are evaluated to be (Table A-15E)

$$k = 0.01481 \text{ Btu/h.ft.}^\circ\text{F}$$

$$\nu = 0.1697 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$Pr = 0.7290$$

**Analysis** Air flows along 11 ft long side. The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(60 \times 5280 / 3600) \text{ ft/s}](11 \text{ ft})}{0.1697 \times 10^{-3} \text{ ft}^2/\text{s}} = 5.704 \times 10^6$$

which is greater than the critical Reynolds number. The air flow is assumed to be entirely turbulent because of the intense vibrations involved. Then the Nusselt number and the heat transfer coefficient are determined to be

$$Nu = \frac{h_o L}{k} = 0.037 Re_L^{0.8} Pr^{1/3} = 0.037 (5.704 \times 10^6)^{0.8} (0.7290)^{1/3} = 8461$$

$$h_o = \frac{k}{L} Nu = \frac{0.01481 \text{ Btu/h.ft.}^\circ\text{F}}{11 \text{ ft}} (8461) = 11.39 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F}$$

The thermal resistances are

$$A_s = 2[(3.2 \text{ ft})(6 \text{ ft}) + (3.2 \text{ ft})(11 \text{ ft}) + (6 \text{ ft})(11 \text{ ft})] = 240.8 \text{ ft}^2$$

$$R_i = \frac{1}{h_i A_s} = \frac{1}{(1.2 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(240.8 \text{ ft}^2)} = 0.0035 \text{ h.}^\circ\text{F/Btu}$$

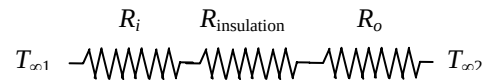
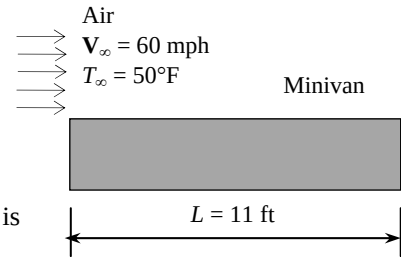
$$R_{insulation} = \frac{(R-3)_{value}}{A_s} = \frac{3 \text{ h.ft}^2 \cdot ^\circ\text{F/Btu}}{(240.8 \text{ ft}^2)} = 0.0125 \text{ h.}^\circ\text{F/Btu}$$

$$R_o = \frac{1}{h_o A_s} = \frac{1}{(11.39 \text{ Btu/h.ft}^2 \cdot ^\circ\text{F})(240.8 \text{ ft}^2)} = 0.0004 \text{ h.}^\circ\text{F/Btu}$$

Then the total thermal resistance and the heat transfer rate into the minivan are determined to be

$$R_{total} = R_i + R_{insulation} + R_o = 0.0035 + 0.0125 + 0.0004 = 0.0164 \text{ h.}^\circ\text{F/Btu}$$

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{total}} = \frac{(90 - 70)^\circ\text{F}}{0.0164 \text{ h.}^\circ\text{F/Btu}} = 1220 \text{ Btu/h}$$



**7-93** Wind is blowing parallel to the walls of a house with windows. The rate of heat loss through the window is to be determined.

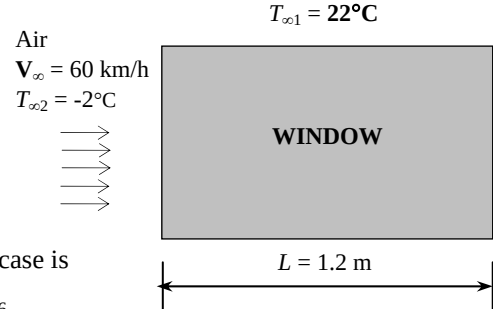
**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air flow is turbulent because of the intense vibrations involved. 5 The minivan is modeled as a rectangular box. 6 Air is an ideal gas with constant properties. 6 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of  $5^\circ\text{C}$ , the properties of air at 1 atm and this temperature are evaluated to be (Table A-15)

$$k = 0.02401 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.382 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7350$$



**Analysis** Air flows along 1.2 m side. The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(60 \times 1000 / 3600) \text{ m/s}](1.2 \text{ m})}{1.382 \times 10^{-5} \text{ m}^2/\text{s}} = 1.447 \times 10^6$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Using the proper relation for Nusselt number, heat transfer coefficient is determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(1.447 \times 10^6)^{0.8} - 871](0.7350)^{1/3} = 2046$$

$$h = \frac{k}{L} Nu = \frac{0.02401 \text{ W/m} \cdot ^\circ\text{C}}{1.2 \text{ m}} (2046) = 40.93 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The thermal resistances are

$$A_s = 3(1.2 \text{ m})(1.5 \text{ m}) = 5.4 \text{ m}^2$$

$$R_{conv,i} = \frac{1}{h_i A_s} = \frac{1}{(8 \text{ W/m}^2 \cdot ^\circ\text{C})(5.4 \text{ m}^2)} = 0.0231 ^\circ\text{C/W}$$

$$R_{cond} = \frac{L}{k A_s} = \frac{0.005 \text{ m}}{(0.78 \text{ W/m} \cdot ^\circ\text{C})(5.4 \text{ m}^2)} = 0.0012 ^\circ\text{C/W}$$

$$R_{conv,o} = \frac{1}{h_o A_s} = \frac{1}{(40.93 \text{ W/m}^2 \cdot ^\circ\text{C})(5.4 \text{ m}^2)} = 0.0045 ^\circ\text{C/W}$$

Then the total thermal resistance and the heat transfer rate through the 3 windows become

$$R_{total} = R_{conv,i} + R_{cond} + R_{conv,o} = 0.0231 + 0.0012 + 0.0045 = 0.0288 ^\circ\text{C/W}$$

$$\dot{Q} = \frac{T_{\infty 1} - T_{\infty 2}}{R_{total}} = \frac{[22 - (-2)]^\circ\text{C}}{0.0288 ^\circ\text{C/W}} = \mathbf{833.3 \text{ W}}$$

**7-94** A fan is blowing air over the entire body of a person. The average temperature of the outer surface of the person is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The pressure of air is 1 atm. 4 The average human body can be treated as a 30-cm-diameter cylinder with an exposed surface area of 1.7 m<sup>2</sup>.

**Properties** We assume the film temperature to be 35°C. The properties of air at 1 atm and this temperature are (Table A-15)

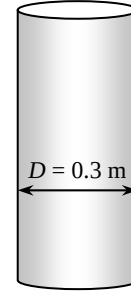
$$k = 0.02625 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.655 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\text{Pr} = 0.7268$$

$$V_\infty = 5 \text{ m/s}$$

$$T_\infty = 32^\circ\text{C}$$



$$\text{Person, } T_s$$

$$90 \text{ W}$$

$$\varepsilon = 0.9$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{(5 \text{ m/s})(0.3 \text{ m})}{1.655 \times 10^{-5} \text{ m}^2/\text{s}} = 9.063 \times 10^4$$

The proper relation for Nusselt number corresponding to this Reynolds number is

$$Nu = \frac{hD}{k} = 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5}$$

$$= 0.3 + \frac{0.62(9.063 \times 10^4)^{0.5} (0.7268)^{1/3}}{\left[1 + (0.4/0.7268)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{9.063 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 203.6$$

Then

$$h = \frac{k}{D} Nu = \frac{0.02655 \text{ W/m} \cdot ^\circ\text{C}}{0.3 \text{ m}} (203.6) = 18.02 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Considering that there is heat generation in that person's body at a rate of 90 W and body gains heat by radiation from the surrounding surfaces, an energy balance can be written as

$$\dot{Q}_{\text{generated}} + \dot{Q}_{\text{radiation}} = \dot{Q}_{\text{convection}}$$

Substituting values with proper units and then application of trial & error method yields the average temperature of the outer surface of the person.

$$90 \text{ W} + \varepsilon A_s \sigma (T_{\text{surr}}^4 - T_s^4) = h A_s (T_s - T_\infty)$$

$$90 + (0.9)(1.7)(5.67 \times 10^{-8})[(40 + 273)^4 - T_s^4] = (18.02)(1.7)[T_s - (32 + 273)]$$

$$T_s = 309.2 \text{ K} = 36.2^\circ\text{C}$$

**7-95** The heat generated by four transistors mounted on a thin vertical plate is dissipated by air blown over the plate on both surfaces. The temperature of the aluminum plate is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 The entire plate is nearly isothermal. 5 The exposed surface area of the transistor is taken to be equal to its base area. 6 Air is an ideal gas with constant properties. 7 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of  $40^\circ\text{C}$ , the properties of air are evaluated to be (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7255$$

**Analysis** The Reynolds number in this case is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(250/60) \text{ m/s}](0.22 \text{ m})}{1.702 \times 10^{-5} \text{ m}^2/\text{s}} = 5.386 \times 10^4$$

which is smaller than the critical Reynolds number. Thus we have laminar flow. Using the proper relation for Nusselt number, heat transfer coefficient is determined to be

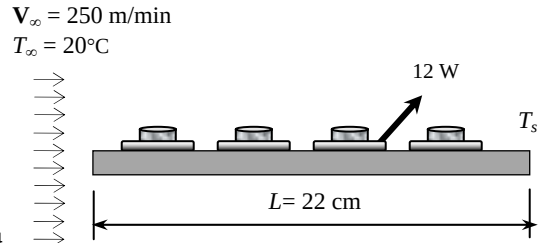
$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664(5.386 \times 10^4)^{0.5} (0.7255)^{1/3} = 138.5$$

$$h = \frac{k}{L} Nu = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.22 \text{ m}} (138.5) = 16.75 \text{ W/m}^2\cdot^\circ\text{C}$$

The temperature of aluminum plate then becomes

$$\dot{Q} = hA_s(T_s - T_\infty) \longrightarrow T_s = T_\infty + \frac{\dot{Q}}{hA_s} = 20^\circ\text{C} + \frac{(4 \times 12) \text{ W}}{(16.75 \text{ W/m}^2\cdot^\circ\text{C})[2(0.22 \text{ m})^2]} = 50.0^\circ\text{C}$$

**Discussion** In reality, the heat transfer coefficient will be higher since the transistors will cause turbulence in the air.



**7-96** A spherical tank used to store iced water is subjected to winds. The rate of heat transfer to the iced water and the amount of ice that melts during a 24-h period are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Thermal resistance of the tank is negligible. 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties. 5 The pressure of air is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 30°C are (Table A-15)

$$k = 0.02588 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_\infty = 1.872 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\mu_{s, @ 0^\circ\text{C}} = 1.729 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

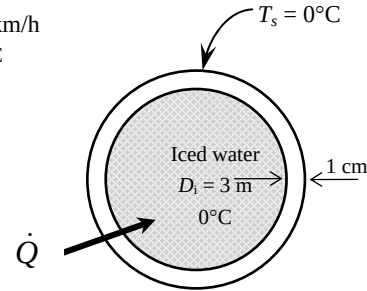
$$\text{Pr} = 0.7282$$

**Analysis** (a) The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(25 \times 1000/3600) \text{ m/s}](3.02 \text{ m})}{1.608 \times 10^{-5} \text{ m}^2/\text{s}} = 1.304 \times 10^6$$

$$V_\infty = 25 \text{ km/h}$$

$$T_\infty = 30^\circ\text{C}$$



The Nusselt number corresponding to this Reynolds number is determined from

$$Nu = \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4}$$

$$= 2 + \left[ 0.4(1.304 \times 10^6)^{0.5} + 0.06(1.304 \times 10^6)^{2/3} \right] (0.7282)^{0.4} \left( \frac{1.872 \times 10^{-5}}{1.729 \times 10^{-5}} \right)^{1/4} = 1056$$

and 
$$h = \frac{k}{D} Nu = \frac{0.02588 \text{ W/m} \cdot ^\circ\text{C}}{3.02 \text{ m}} (1056) = 9.05 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer to the iced water is

$$\dot{Q} = hA_s(T_s - T_\infty) = h(\pi D^2)(T_s - T_\infty) = (9.05 \text{ W/m}^2 \cdot ^\circ\text{C})[\pi(3.02 \text{ m})^2](30 - 0)^\circ\text{C} = \mathbf{7779 \text{ W}}$$

(b) The amount of heat transfer during a 24-hour period is

$$Q = \dot{Q}\Delta t = (7.779 \text{ kJ/s})(24 \times 3600 \text{ s}) = 672,079 \text{ kJ}$$

Then the amount of ice that melts during this period becomes

$$Q = mh_{if} \longrightarrow m = \frac{Q}{h_{if}} = \frac{672,079 \text{ kJ}}{333.7 \text{ kJ/kg}} = \mathbf{2014 \text{ kg}}$$



**7-97** A spherical tank used to store iced water is subjected to winds. The rate of heat transfer to the iced water and the amount of ice that melts during a 24-h period are to be determined.

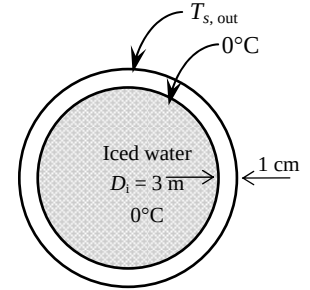
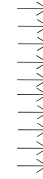
**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 7 The pressure of air is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 30°C are (Table A-15)

|                                                                 |                                                                                 |
|-----------------------------------------------------------------|---------------------------------------------------------------------------------|
| $k = 0.02588 \text{ W/m} \cdot ^\circ\text{C}$                  | $\mu_{s, @ 0^\circ\text{C}} = 1.729 \times 10^{-5} \text{ kg/m} \cdot \text{s}$ |
| $\nu = 1.608 \times 10^{-5} \text{ m}^2/\text{s}$               | $\text{Pr} = 0.7282$                                                            |
| $\mu_\infty = 1.872 \times 10^{-5} \text{ kg/m} \cdot \text{s}$ |                                                                                 |

$$\mathbf{V}_\infty = 25 \text{ km/h}$$

$$T_\infty = 30^\circ\text{C}$$



**Analysis** (a) The Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(25 \times 1000/3600) \text{ m/s}](3.02 \text{ m})}{1.608 \times 10^{-5} \text{ m}^2/\text{s}} = 1.304 \times 10^6$$

The Nusselt number corresponding to this Reynolds number is determined from

$$\begin{aligned} \text{Nu} &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(1.304 \times 10^6)^{0.5} + 0.06(1.304 \times 10^6)^{2/3} \right] (0.7282)^{0.4} \left( \frac{1.872 \times 10^{-5}}{1.729 \times 10^{-5}} \right)^{1/4} = 1056 \end{aligned}$$

and 
$$h = \frac{k}{D} \text{Nu} = \frac{0.02588 \text{ W/m} \cdot ^\circ\text{C}}{3.02 \text{ m}} (1056) = 9.05 \text{ W/m}^2 \cdot ^\circ\text{C}$$

In steady operation, heat transfer through the tank by conduction is equal to the heat transfer from the outer surface of the tank by convection and radiation. Therefore,

$$\begin{aligned} \dot{Q} &= \dot{Q}_{\text{through tank}} = \dot{Q}_{\text{from tank, conv+rad}} \\ \dot{Q} &= \frac{T_{s, \text{out}} - T_{s, \text{in}}}{R_{\text{sphere}}} = h_o A_o (T_{\text{surr}} - T_{s, \text{out}}) + \varepsilon A_o \sigma (T_{\text{surr}}^4 - T_{s, \text{out}}^4) \end{aligned}$$

where 
$$R_{\text{sphere}} = \frac{r_2 - r_1}{4\pi k r_1 r_2} = \frac{(1.51 - 1.50) \text{ m}}{4\pi (15 \text{ W/m} \cdot ^\circ\text{C})(1.51 \text{ m})(1.50 \text{ m})} = 2.342 \times 10^{-5} \text{ } ^\circ\text{C/W}$$

$$A_o = \pi D^2 = \pi (3.02 \text{ m})^2 = 28.65 \text{ m}^2$$

Substituting,

$$\begin{aligned} \dot{Q} &= \frac{T_{s, \text{out}} - 0^\circ\text{C}}{2.34 \times 10^{-5} \text{ } ^\circ\text{C/W}} = (9.05 \text{ W/m}^2 \cdot ^\circ\text{C})(28.65 \text{ m}^2)(30 - T_{s, \text{out}})^\circ\text{C} \\ &\quad + (0.9)(28.65 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(15 + 273 \text{ K})^4 - (T_{s, \text{out}} + 273 \text{ K})^4] \end{aligned}$$

whose solution is

$$T_s = 0.23^\circ\text{C} \text{ and } \dot{Q} = 9630 \text{ W} = \mathbf{9.63 \text{ kW}}$$

(b) The amount of heat transfer during a 24-hour period is

$$Q = \dot{Q} \Delta t = (9.63 \text{ kJ/s})(24 \times 3600 \text{ s}) = 832,032 \text{ kJ}$$

Then the amount of ice that melts during this period becomes

$$Q = m h_{if} \longrightarrow m = \frac{Q}{h_{if}} = \frac{832,032 \text{ kJ}}{333.7 \text{ kJ/kg}} = \mathbf{2493 \text{ kg}}$$

**7-98E** A cylindrical transistor mounted on a circuit board is cooled by air flowing over it. The maximum power rating of the transistor is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The pressure of air is 1 atm.

**Properties** The properties of air at 1 atm and the film temperature of  $T_f = (180 + 120) / 2 = 150^\circ\text{F}$  are (Table A-15)

$$k = 0.01646 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}$$

$$\nu = 0.210 \times 10^{-3} \text{ ft}^2/\text{s}$$

$$\text{Pr} = 0.7188$$

**Analysis** The Reynolds number is

$$\text{Re} = \frac{\mathbf{V}_\infty D}{\nu} = \frac{(500/60 \text{ ft/s})(0.22/12 \text{ ft})}{0.210 \times 10^{-3} \text{ ft}^2/\text{s}} = 727.5$$

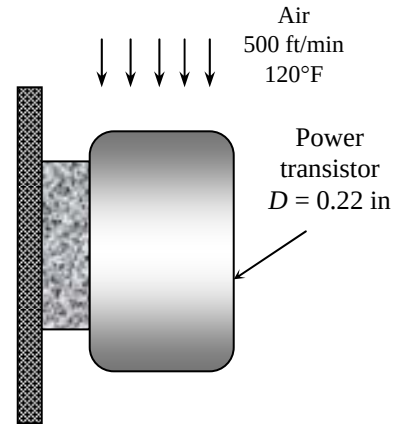
The Nusselt number corresponding to this Reynolds number is

$$\begin{aligned} Nu = \frac{hD}{k} &= 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(727.5)^{0.5} (0.7188)^{1/3}}{\left[1 + (0.4/0.7188)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{727.5}{282,000}\right)^{5/8}\right]^{4/5} = 13.72 \end{aligned}$$

$$\text{and } h = \frac{k}{D} Nu = \frac{0.01646 \text{ Btu/h}\cdot\text{ft}\cdot^\circ\text{F}}{(0.22/12 \text{ ft})} (13.72) = 12.32 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F}$$

Then the amount of power this transistor can dissipate safely becomes

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) = h(\pi DL)(T_s - T_\infty) \\ &= (12.32 \text{ Btu/h}\cdot\text{ft}^2\cdot^\circ\text{F})[\pi(0.22/12 \text{ ft})(0.25/12 \text{ ft})](180 - 120)^\circ\text{C} \\ &= \mathbf{0.887 \text{ Btu/h} = 0.26 \text{ W}} \quad (1 \text{ W} = 3.412 \text{ Btu/h}) \end{aligned}$$



**7-99** Wind is blowing over the roof of a house. The rate of heat transfer through the roof and the cost of this heat loss for 14-h period are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Air is an ideal gas with constant properties. 4 The pressure of air is 1 atm.

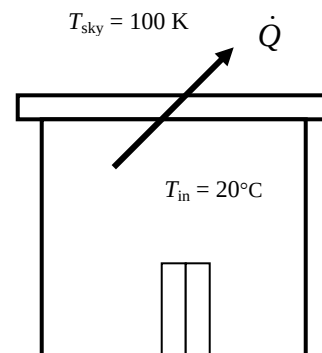
**Properties** Assuming a film temperature of  $10^\circ\text{C}$ , the properties of air are (Table A-15)

$$k = 0.02439 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7336$$

Air  
 $V_\infty = 60 \text{ km/h}$   
 $T_\infty = 10^\circ\text{C}$



**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(60 \times 1000 / 3600) \text{ m/s}](20 \text{ m})}{1.426 \times 10^{-5} \text{ m}^2/\text{s}} = 2.338 \times 10^7$$

which is greater than the critical Reynolds number. Thus we have combined laminar and turbulent flow. Then the Nusselt number and the heat transfer coefficient are determined to be

$$Nu = \frac{hL}{k} = (0.037 Re_L^{0.8} - 871) Pr^{1/3} = [0.037(2.338 \times 10^7)^{0.8} - 871](0.7336)^{1/3} = 2.542 \times 10^4$$

$$h = \frac{k}{L} Nu = \frac{0.02439 \text{ W/m}\cdot^\circ\text{C}}{20 \text{ m}} (2.542 \times 10^4) = 31.0 \text{ W/m}^2\cdot^\circ\text{C}$$

In steady operation, heat transfer from the room to the roof (by convection and radiation) must be equal to the heat transfer from the roof to the surroundings (by convection and radiation), which must be equal to the heat transfer through the roof by conduction. That is,

$$\dot{Q} = \dot{Q}_{\text{room to roof, conv+rad}} = \dot{Q}_{\text{roof, cond}} = \dot{Q}_{\text{roof to surroundings, conv+rad}}$$

Taking the inner and outer surface temperatures of the roof to be  $T_{s,in}$  and  $T_{s,out}$ , respectively, the quantities above can be expressed as

$$\begin{aligned} \dot{Q}_{\text{room to roof, conv+rad}} &= h_i A_s (T_{\text{room}} - T_{s,in}) + \varepsilon A_s \sigma (T_{\text{room}}^4 - T_{s,in}^4) = (5 \text{ W/m}^2\cdot^\circ\text{C})(300 \text{ m}^2)(20 - T_{s,in})^\circ\text{C} \\ &\quad + (0.9)(300 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(20 + 273 \text{ K})^4 - (T_{s,in} + 273 \text{ K})^4] \end{aligned}$$

$$\dot{Q}_{\text{roof, cond}} = k A_s \frac{T_{s,in} - T_{s,out}}{L} = (2 \text{ W/m}\cdot^\circ\text{C})(300 \text{ m}^2) \frac{T_{s,in} - T_{s,out}}{0.15 \text{ m}}$$

$$\begin{aligned} \dot{Q}_{\text{roof to surr, conv+rad}} &= h_o A_s (T_{s,out} - T_{\text{surr}}) + \varepsilon A_s \sigma (T_{s,out}^4 - T_{\text{surr}}^4) = (31.0 \text{ W/m}^2\cdot^\circ\text{C})(300 \text{ m}^2)(T_{s,out} - 10)^\circ\text{C} \\ &\quad + (0.9)(300 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2\cdot\text{K}^4) [(T_{s,out} + 273 \text{ K})^4 - (100 \text{ K})^4] \end{aligned}$$

Solving the equations above simultaneously gives

$$\dot{Q} = 28,025 \text{ W} = \mathbf{28.03 \text{ kW}}, \quad T_{s,in} = 10.6^\circ\text{C}, \text{ and } T_{s,out} = 3.5^\circ\text{C}$$

The total amount of natural gas consumption during a 14-hour period is

$$Q_{\text{gas}} = \frac{Q_{\text{total}}}{0.85} = \frac{\dot{Q} \Delta t}{0.85} = \frac{(28.03 \text{ kJ/s})(14 \times 3600 \text{ s})}{0.85} \left( \frac{1 \text{ therm}}{105,500 \text{ kJ}} \right) = 15.75 \text{ therms}$$

Finally, the money lost through the roof during that period is

$$\text{Money lost} = (15.75 \text{ therms})(\$0.60 / \text{therm}) = \mathbf{\$9.45}$$

**7-100** Steam is flowing in a stainless steel pipe while air is flowing across the pipe. The rate of heat loss from the steam per unit length of the pipe is to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 Air is an ideal gas with constant properties. 3 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of 10°C, the properties of air are (Table A-15)

$$k = 0.02439 \text{ W/m} \cdot ^\circ\text{C}, \quad \nu = 1.426 \times 10^{-5} \text{ m}^2/\text{s}, \quad \text{and} \quad \text{Pr} = 0.7336$$

**Analysis** The outer diameter of insulated pipe is  $D_o = 4.6 + 2 \times 3.5 = 11.6 \text{ cm} = 0.116 \text{ m}$ . The Reynolds number is

$$\text{Re} = \frac{V_\infty D_o}{\nu} = \frac{(4 \text{ m/s})(0.116 \text{ m})}{1.426 \times 10^{-5} \text{ m}^2/\text{s}} = 3.254 \times 10^4$$

The Nusselt number for flow across a cylinder is determined from

$$\begin{aligned} \text{Nu} &= \frac{hD_o}{k} = 0.3 + \frac{0.62 \text{Re}^{0.5} \text{Pr}^{1/3}}{\left[1 + (0.4/\text{Pr})^{2/3}\right]^{1/4}} \left[1 + \left(\frac{\text{Re}}{282,000}\right)^{5/8}\right]^{4/5} \\ &= 0.3 + \frac{0.62(3.254 \times 10^4)^{0.5} (0.7336)^{1/3}}{\left[1 + (0.4/0.7336)^{2/3}\right]^{1/4}} \left[1 + \left(\frac{3.254 \times 10^4}{282,000}\right)^{5/8}\right]^{4/5} = 107.0 \end{aligned}$$

$$\text{and} \quad h_o = \frac{k}{D_o} \text{Nu} = \frac{0.02439 \text{ W/m} \cdot ^\circ\text{C}}{0.116 \text{ m}} (107.0) = 22.50 \text{ W/m}^2 \cdot ^\circ\text{C}$$

Area of the outer surface of the pipe per m length of the pipe is

$$A_o = \pi D_o L = \pi (0.116 \text{ m})(1 \text{ m}) = 0.3644 \text{ m}^2$$

In steady operation, heat transfer from the steam through the pipe and the insulation to the outer surface (by first convection and then conduction) must be equal to the heat transfer from the outer surface to the surroundings (by simultaneous convection and radiation). That is,

$$\dot{Q} = \dot{Q}_{\text{pipe and insulation}} = \dot{Q}_{\text{surface to surroundings}}$$

Using the thermal resistance network, heat transfer from the steam to the outer surface is expressed as

$$R_{\text{conv},i} = \frac{1}{h_i A_i} = \frac{1}{(80 \text{ W/m}^2 \cdot ^\circ\text{C})[\pi(0.04 \text{ m})(1 \text{ m})]} = 0.0995 \text{ } ^\circ\text{C/W}$$

$$R_{\text{pipe}} = \frac{\ln(r_2/r_1)}{2\pi k L} = \frac{\ln(2.3/2)}{2\pi(15 \text{ W/m} \cdot ^\circ\text{C})(1 \text{ m})} = 0.0015 \text{ } ^\circ\text{C/W}$$

$$R_{\text{insulation}} = \frac{\ln(r_3/r_2)}{2\pi k L} = \frac{\ln(5.8/2.3)}{2\pi(0.038 \text{ W/m} \cdot ^\circ\text{C})(1 \text{ m})} = 3.874 \text{ } ^\circ\text{C/W}$$

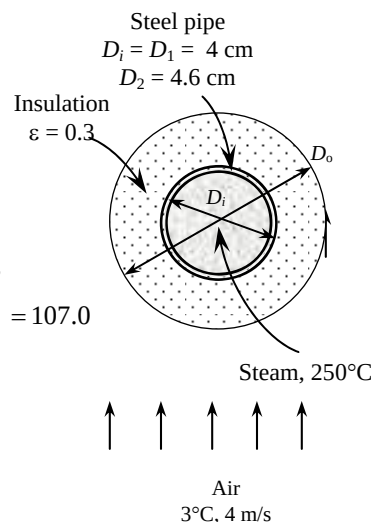
$$\text{and} \quad \dot{Q}_{\text{pipe and ins}} = \frac{T_{\infty 1} - T_s}{R_{\text{conv},i} + R_{\text{pipe}} + R_{\text{insulation}}} = \frac{(250 - T_s)^\circ\text{C}}{(0.0995 + 0.0015 + 3.874)^\circ\text{C/W}}$$

Heat transfer from the outer surface can be expressed as

$$\begin{aligned} \dot{Q}_{\text{surface to surr, conv+rad}} &= h_o A_o (T_s - T_{\text{surr}}) + \varepsilon A_o \sigma (T_s^4 - T_{\text{surr}}^4) = (22.50 \text{ W/m}^2 \cdot ^\circ\text{C})(0.3644 \text{ m}^2)(T_s - 3)^\circ\text{C} \\ &\quad + (0.3)(0.3644 \text{ m}^2)(5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)[(T_s + 273 \text{ K})^4 - (3 + 273 \text{ K})^4] \end{aligned}$$

Solving the two equations above simultaneously, the surface temperature and the heat transfer rate per m length of the pipe are determined to be

$$T_s = 9.9^\circ\text{C} \text{ and } \dot{Q} = \mathbf{60.4 \text{ W}} \text{ (per m length)}$$



**7-101** A spherical tank filled with liquid nitrogen is exposed to winds. The rate of evaporation of the liquid nitrogen due to heat transfer from the air is to be determined for three cases.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 4 The pressure of air is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 20°C are (Table A-15)

$$k = 0.02514 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.516 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_\infty = 1.825 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\mu_{s, @ -196^\circ\text{C}} = 5.023 \times 10^{-6} \text{ kg/m} \cdot \text{s}$$

$$\text{Pr} = 0.7309$$

**Analysis** (a) When there is no insulation,  $D = D_i = 4 \text{ m}$ , and the Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 2.932 \times 10^6$$

The Nusselt number is determined from

$$\begin{aligned} Nu &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(2.932 \times 10^6)^{0.5} + 0.06(2.932 \times 10^6)^{2/3} \right] (0.7309)^{0.4} \left( \frac{1.825 \times 10^{-5}}{5.023 \times 10^{-6}} \right)^{1/4} = 2333 \end{aligned}$$

and 
$$h = \frac{k}{D} Nu = \frac{0.02514 \text{ W/m} \cdot ^\circ\text{C}}{4 \text{ m}} (2333) = 14.66 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer to the liquid nitrogen is

$$\begin{aligned} \dot{Q} &= hA_s(T_s - T_\infty) = h(\pi D^2)(T_s - T_\infty) \\ &= (14.66 \text{ W/m}^2 \cdot ^\circ\text{C})[\pi(4 \text{ m})^2][(20 - (-196))^\circ\text{C}] = 159,200 \text{ W} \end{aligned}$$

The rate of evaporation of liquid nitrogen then becomes

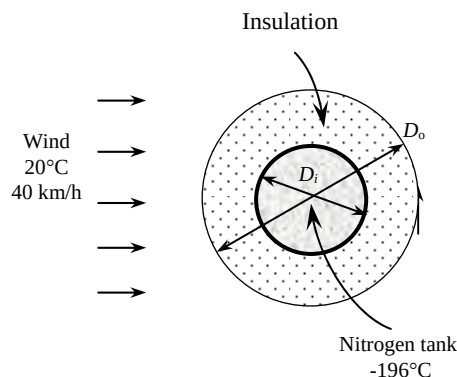
$$\dot{Q} = \dot{m}h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{159.2 \text{ kJ/s}}{198 \text{ kJ/kg}} = \mathbf{0.804 \text{ kg/s}}$$

(b) Note that after insulation the outer surface temperature and diameter will change. Therefore we need to evaluate dynamic viscosity at a new surface temperature which we will assume to be  $-100^\circ\text{C}$ . At  $-100^\circ\text{C}$ ,  $\mu = 1.189 \times 10^{-5} \text{ kg/m} \cdot \text{s}$ . Noting that  $D = D_0 = 4.1 \text{ m}$ , the Nusselt number becomes

$$\begin{aligned} \text{Re} &= \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4.1 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 3.005 \times 10^6 \\ Nu &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(3.005 \times 10^6)^{0.5} + 0.06(3.005 \times 10^6)^{2/3} \right] (0.7309)^{0.4} \left( \frac{1.825 \times 10^{-5}}{1.189 \times 10^{-5}} \right)^{1/4} = 1910 \end{aligned}$$

and 
$$h = \frac{k}{D} Nu = \frac{0.02514 \text{ W/m} \cdot ^\circ\text{C}}{4.1 \text{ m}} (1910) = 11.71 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer to the liquid nitrogen is



$$\begin{aligned}
 A_s &= \pi D^2 = \pi (4.1 \text{ m})^2 = 52.81 \text{ m}^2 \\
 \dot{Q} &= \frac{T_\infty - T_{s, \text{tan } k}}{R_{\text{insulation}} + R_{\text{conv}}} = \frac{T_\infty - T_{s, \text{tan } k}}{\frac{r_2 - r_1}{4\pi k r_1 r_2} + \frac{1}{h A_s}} \\
 &= \frac{[20 - (-196)]^\circ\text{C}}{\frac{(2.05 - 2) \text{ m}}{4\pi (0.035 \text{ W/m}\cdot^\circ\text{C})(2.05 \text{ m})(2 \text{ m})} + \frac{1}{(11.71 \text{ W/m}^2\cdot^\circ\text{C})(52.81 \text{ m}^2)}} = 7361 \text{ W}
 \end{aligned}$$

The rate of evaporation of liquid nitrogen then becomes

$$\dot{Q} = \dot{m} h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{7.361 \text{ kJ/s}}{198 \text{ kJ/kg}} = \mathbf{0.0372 \text{ kg/s}}$$

(c) We use the dynamic viscosity value at the new estimated surface temperature of  $0^\circ\text{C}$  to be  $\mu = 1.729 \times 10^{-5} \text{ kg/m}\cdot\text{s}$ . Noting that  $D = D_0 = 4.04 \text{ m}$  in this case, the Nusselt number becomes

$$\begin{aligned}
 \text{Re} &= \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4.04 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 2.961 \times 10^6 \\
 \text{Nu} &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\
 &= 2 + \left[ 0.4(2.961 \times 10^6)^{0.5} + 0.06(2.961 \times 10^6)^{2/3} \right] (0.7309)^{0.4} \left( \frac{1.825 \times 10^{-5}}{1.729 \times 10^{-5}} \right)^{1/4} = 1724
 \end{aligned}$$

$$\text{and } h = \frac{k}{D} \text{Nu} = \frac{0.02514 \text{ W/m}\cdot^\circ\text{C}}{4.04 \text{ m}} (1724) = 10.73 \text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat transfer to the liquid nitrogen is

$$\begin{aligned}
 A_s &= \pi D^2 = \pi (4.04 \text{ m})^2 = 51.28 \text{ m}^2 \\
 \dot{Q} &= \frac{T_\infty - T_{s, \text{tan } k}}{R_{\text{insulation}} + R_{\text{conv}}} = \frac{T_\infty - T_{s, \text{tan } k}}{\frac{r_2 - r_1}{4\pi k r_1 r_2} + \frac{1}{h A_s}} \\
 &= \frac{[20 - (-196)]^\circ\text{C}}{\frac{(2.02 - 2) \text{ m}}{4\pi (0.00005 \text{ W/m}\cdot^\circ\text{C})(2.02 \text{ m})(2 \text{ m})} + \frac{1}{(10.73 \text{ W/m}^2\cdot^\circ\text{C})(51.28 \text{ m}^2)}} = 27.4 \text{ W}
 \end{aligned}$$

The rate of evaporation of liquid nitrogen then becomes

$$\dot{Q} = \dot{m} h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{0.0274 \text{ kJ/s}}{198 \text{ kJ/kg}} = \mathbf{1.38 \times 10^{-4} \text{ kg/s}}$$

**7-102** A spherical tank filled with liquid oxygen is exposed to ambient winds. The rate of evaporation of the liquid oxygen due to heat transfer from the air is to be determined for three cases.

**Assumptions** 1 Steady operating conditions exist. 2 Radiation effects are negligible. 3 Air is an ideal gas with constant properties. 7 The pressure of air is 1 atm.

**Properties** The properties of air at 1 atm pressure and the free stream temperature of 20°C are (Table A-15)

$$k = 0.02514 \text{ W/m} \cdot ^\circ\text{C}$$

$$\nu = 1.516 \times 10^{-5} \text{ m}^2/\text{s}$$

$$\mu_\infty = 1.825 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\mu_{s, @ -183^\circ\text{C}} = 6.127 \times 10^{-5} \text{ kg/m} \cdot \text{s}$$

$$\text{Pr} = 0.7309$$

**Analysis** (a) When there is no insulation,  $D = D_i = 4 \text{ m}$ , and the Reynolds number is

$$\text{Re} = \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 2.932 \times 10^6$$

The Nusselt number is determined from

$$\begin{aligned} \text{Nu} &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(2.932 \times 10^6)^{0.5} + 0.06(2.932 \times 10^6)^{2/3} \right] (0.7309)^{0.4} \left( \frac{1.825 \times 10^{-5}}{6.127 \times 10^{-5}} \right)^{1/4} = 2220 \end{aligned}$$

$$\text{and } h = \frac{k}{D} \text{Nu} = \frac{0.02514 \text{ W/m} \cdot ^\circ\text{C}}{4 \text{ m}} (2220) = 13.95 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer to the liquid oxygen is

$$\dot{Q} = hA_s(T_s - T_\infty) = h(\pi D^2)(T_s - T_\infty) = (13.95 \text{ W/m}^2 \cdot ^\circ\text{C})(\pi(4 \text{ m})^2)[(20 - (-183))^\circ\text{C}] = 142,372 \text{ W}$$

The rate of evaporation of liquid oxygen then becomes

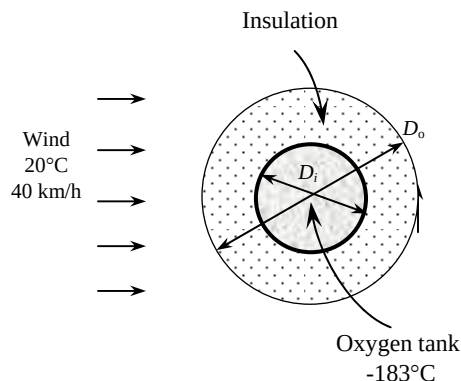
$$\dot{Q} = \dot{m}h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{142.4 \text{ kJ/s}}{213 \text{ kJ/kg}} = \mathbf{0.668 \text{ kg/s}}$$

(b) Note that after insulation the outer surface temperature and diameter will change. Therefore we need to evaluate dynamic viscosity at a new surface temperature which we will assume to be  $-100^\circ\text{C}$ . At  $-100^\circ\text{C}$ ,  $\mu = 1.189 \times 10^{-5} \text{ kg/m} \cdot \text{s}$ . Noting that  $D = D_0 = 4.1 \text{ m}$ , the Nusselt number becomes

$$\begin{aligned} \text{Re} &= \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4.1 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 3.005 \times 10^6 \\ \text{Nu} &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\ &= 2 + \left[ 0.4(3.005 \times 10^6)^{0.5} + 0.06(3.005 \times 10^6)^{2/3} \right] (0.7309)^{0.4} \left( \frac{1.825 \times 10^{-5}}{1.189 \times 10^{-5}} \right)^{1/4} = 1910 \end{aligned}$$

$$\text{and } h = \frac{k}{D} \text{Nu} = \frac{0.02514 \text{ W/m} \cdot ^\circ\text{C}}{4.1 \text{ m}} (1910) = 11.71 \text{ W/m}^2 \cdot ^\circ\text{C}$$

The rate of heat transfer to the liquid nitrogen is



$$\begin{aligned}
 A_s &= \pi D^2 = \pi (4.1 \text{ m})^2 = 52.81 \text{ m}^2 \\
 \dot{Q} &= \frac{T_\infty - T_{s, \text{tan } k}}{R_{\text{insulation}} + R_{\text{conv}}} = \frac{T_\infty - T_{s, \text{tan } k}}{\frac{r_2 - r_1}{4\pi k r_1 r_2} + \frac{1}{h A_s}} \\
 &= \frac{[20 - (-183)]^\circ\text{C}}{\frac{(2.05 - 2) \text{ m}}{4\pi (0.035 \text{ W/m}\cdot^\circ\text{C})(2.05 \text{ m})(2 \text{ m})} + \frac{1}{(11.71 \text{ W/m}^2\cdot^\circ\text{C})(52.81 \text{ m}^2)}} = 6918 \text{ W}
 \end{aligned}$$

The rate of evaporation of liquid nitrogen then becomes

$$\dot{Q} = \dot{m} h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{6.918 \text{ kJ/s}}{213 \text{ kJ/kg}} = \mathbf{0.0325 \text{ kg/s}}$$

(c) Again we use the dynamic viscosity value at the estimated surface temperature of  $0^\circ\text{C}$  to be  $\mu = 1.729 \times 10^{-5} \text{ kg/m}\cdot\text{s}$ . Noting that  $D = D_0 = 4.04 \text{ m}$  in this case, the Nusselt number becomes

$$\begin{aligned}
 \text{Re} &= \frac{V_\infty D}{\nu} = \frac{[(40 \times 1000/3600) \text{ m/s}](4.04 \text{ m})}{1.516 \times 10^{-5} \text{ m}^2/\text{s}} = 2.961 \times 10^6 \\
 \text{Nu} &= \frac{hD}{k} = 2 + \left[ 0.4 \text{Re}^{0.5} + 0.06 \text{Re}^{2/3} \right] \text{Pr}^{0.4} \left( \frac{\mu_\infty}{\mu_s} \right)^{1/4} \\
 &= 2 + \left[ 0.4(2.961 \times 10^6)^{0.5} + 0.06(2.961 \times 10^6)^{2/3} \right] (0.713)^{0.4} \left( \frac{1.825 \times 10^{-5}}{1.729 \times 10^{-5}} \right)^{1/4} = 1724
 \end{aligned}$$

and 
$$h = \frac{k}{D} \text{Nu} = \frac{0.02514 \text{ W/m}\cdot^\circ\text{C}}{4.04 \text{ m}} (1724) = 10.73 \text{ W/m}^2\cdot^\circ\text{C}$$

The rate of heat transfer to the liquid nitrogen is

$$\begin{aligned}
 A_s &= \pi D^2 = \pi (4.04 \text{ m})^2 = 51.28 \text{ m}^2 \\
 \dot{Q} &= \frac{T_\infty - T_{s, \text{tan } k}}{R_{\text{insulation}} + R_{\text{conv}}} = \frac{T_\infty - T_{s, \text{tan } k}}{\frac{r_2 - r_1}{4\pi k r_1 r_2} + \frac{1}{h A_s}} \\
 &= \frac{[20 - (-183)]^\circ\text{C}}{\frac{(2.02 - 2) \text{ m}}{4\pi (0.00005 \text{ W/m}\cdot^\circ\text{C})(2.02 \text{ m})(2 \text{ m})} + \frac{1}{(10.73 \text{ W/m}^2\cdot^\circ\text{C})(51.28 \text{ m}^2)}} = 25.8 \text{ W}
 \end{aligned}$$

The rate of evaporation of liquid oxygen then becomes

$$\dot{Q} = \dot{m} h_{if} \longrightarrow \dot{m} = \frac{\dot{Q}}{h_{if}} = \frac{0.0258 \text{ kJ/s}}{213 \text{ kJ/kg}} = \mathbf{1.21 \times 10^{-4} \text{ kg/s}}$$



**7-103** A circuit board houses 80 closely spaced logic chips on one side. All the heat generated is conducted across the circuit board and is dissipated from the back side of the board to the ambient air, which is forced to flow over the surface by a fan. The temperatures on the two sides of the circuit board are to be determined.

**Assumptions** 1 Steady operating conditions exist. 2 The critical Reynolds number is  $Re_{cr} = 5 \times 10^5$ . 3 Radiation effects are negligible. 4 Air is an ideal gas with constant properties. 7 The pressure of air is 1 atm.

**Properties** Assuming a film temperature of  $40^\circ\text{C}$ , the properties of air are (Table A-15)

$$k = 0.02662 \text{ W/m}\cdot^\circ\text{C}$$

$$\nu = 1.702 \times 10^{-5} \text{ m}^2/\text{s}$$

$$Pr = 0.7255$$

**Analysis** The Reynolds number is

$$Re_L = \frac{V_\infty L}{\nu} = \frac{[(400/60) \text{ m/s}](0.18 \text{ m})}{1.702 \times 10^{-5} \text{ m}^2/\text{s}} = 7.051 \times 10^4$$

which is less than the critical Reynolds number. Therefore, the flow is laminar. Using the proper relation for Nusselt number, heat transfer coefficient is determined to be

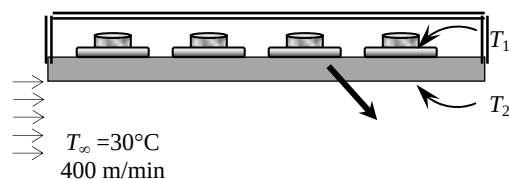
$$Nu = \frac{hL}{k} = 0.664 Re_L^{0.5} Pr^{1/3} = 0.664(7.051 \times 10^4)^{0.5} (0.7255)^{1/3} = 158.4$$

$$h = \frac{k}{L} Nu = \frac{0.02662 \text{ W/m}\cdot^\circ\text{C}}{0.18 \text{ m}} (158.4) = 23.43 \text{ W/m}^2\cdot^\circ\text{C}$$

The temperatures on the two sides of the circuit board are

$$\begin{aligned} \dot{Q} &= hA_s(T_2 - T_\infty) \rightarrow T_2 = T_\infty + \frac{\dot{Q}}{hA_s} \\ &= 30^\circ\text{C} + \frac{(80 \times 0.06) \text{ W}}{(23.43 \text{ W/m}^2\cdot^\circ\text{C})(0.12 \text{ m})(0.18 \text{ m})} = \mathbf{39.48^\circ\text{C}} \end{aligned}$$

$$\begin{aligned} \dot{Q} &= \frac{kA_s}{L}(T_1 - T_2) \rightarrow T_1 = T_2 + \frac{\dot{Q}L}{kA_s} \\ &= 39.48^\circ\text{C} + \frac{(80 \times 0.06 \text{ W})(0.003 \text{ m})}{(16 \text{ W/m}\cdot^\circ\text{C})(0.12 \text{ m})(0.18 \text{ m})} = \mathbf{39.52^\circ\text{C}} \end{aligned}$$



**7-104E** The equivalent wind chill temperature of an environment at 10°F at various winds speeds are

$$\begin{aligned} \mathbf{V} = 10 \text{ mph:} \quad T_{equiv} &= 91.4 - (91.4 - T_{ambient})(0.475 - 0.0203V + 0.304\sqrt{V}) \\ &= 91.4 - [91.4 - (10^\circ \text{F})][0.475 - 0.0203(10 \text{ mph}) + 0.304\sqrt{10 \text{ mph}}] = \mathbf{-9^\circ \text{F}} \end{aligned}$$

$$\mathbf{V} = 20 \text{ mph:} \quad T_{equiv} = 91.4 - [91.4 - (10^\circ \text{F})][0.475 - 0.0203(20 \text{ mph}) + 0.304\sqrt{20 \text{ mph}}] = \mathbf{-24.9^\circ \text{F}}$$

$$\mathbf{V} = 30 \text{ mph:} \quad T_{equiv} = 91.4 - [91.4 - (10^\circ \text{F})][0.475 - 0.0203(30 \text{ mph}) + 0.304\sqrt{30 \text{ mph}}] = \mathbf{-33.2^\circ \text{F}}$$

$$\mathbf{V} = 40 \text{ mph:} \quad T_{equiv} = 91.4 - [91.4 - (10^\circ \text{F})][0.475 - 0.0203(40 \text{ mph}) + 0.304\sqrt{40 \text{ mph}}] = \mathbf{-37.7^\circ \text{F}}$$

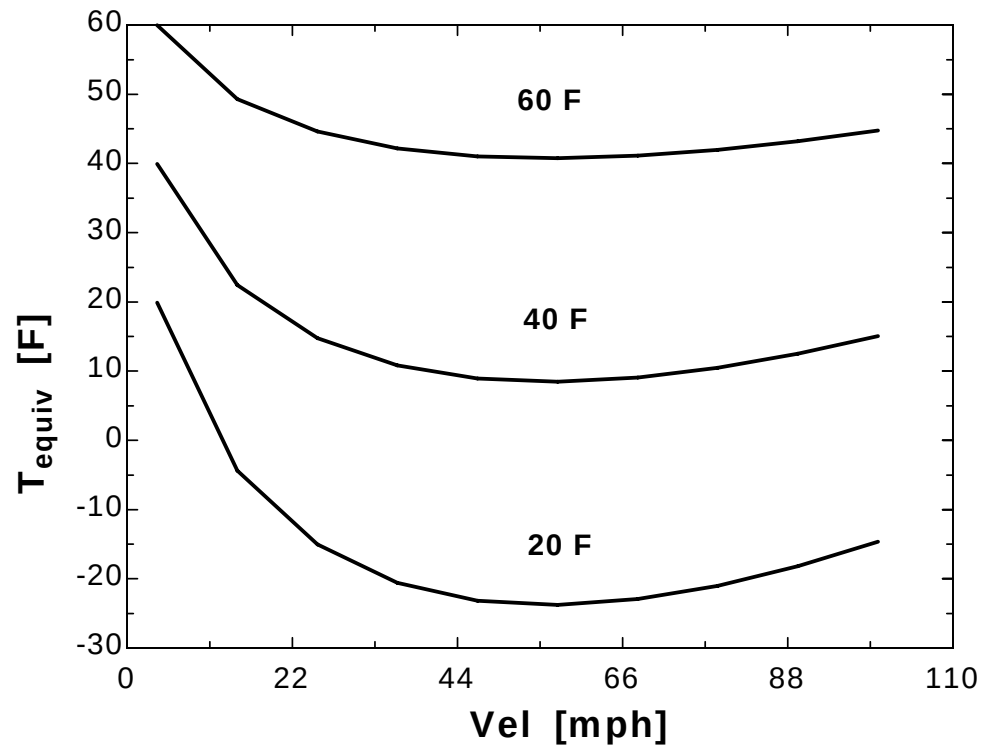
In the last 3 cases, the person needs to be concerned about the possibility of freezing.

## 7-105E "PROBLEM 7-105E"

## "ANALYSIS"

$$T_{\text{equiv}} = 91.4 - (91.4 - T_{\text{ambient}}) * (0.475 - 0.0203 * \text{Vel} + 0.304 * \sqrt{\text{Vel}})$$

| Vel [mph] | T <sub>ambient</sub> [F] | T <sub>equiv</sub> [F] |
|-----------|--------------------------|------------------------|
| 4         | 20                       | 19.87                  |
| 14.67     | 20                       | -4.383                 |
| 25.33     | 20                       | -15.05                 |
| 36        | 20                       | -20.57                 |
| 46.67     | 20                       | -23.15                 |
| 57.33     | 20                       | -23.77                 |
| 68        | 20                       | -22.94                 |
| 78.67     | 20                       | -21.01                 |
| 89.33     | 20                       | -18.19                 |
| 100       | 20                       | -14.63                 |
| 4         | 40                       | 39.91                  |
| 14.67     | 40                       | 22.45                  |
| 25.33     | 40                       | 14.77                  |
| 36        | 40                       | 10.79                  |
| 46.67     | 40                       | 8.935                  |
| 57.33     | 40                       | 8.493                  |
| 68        | 40                       | 9.086                  |
| 78.67     | 40                       | 10.48                  |
| 89.33     | 40                       | 12.51                  |
| 100       | 40                       | 15.07                  |
| 4         | 60                       | 59.94                  |
| 14.67     | 60                       | 49.28                  |
| 25.33     | 60                       | 44.59                  |
| 36        | 60                       | 42.16                  |
| 46.67     | 60                       | 41.02                  |
| 57.33     | 60                       | 40.75                  |
| 68        | 60                       | 41.11                  |
| 78.67     | 60                       | 41.96                  |
| 89.33     | 60                       | 43.21                  |
| 100       | 60                       | 44.77                  |



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7-106 .... 7-110 Design and Essay Problems

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